

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

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ABSTRACT. We relate the geometry of acquired information to its realization by a finite-state observer. For arbitrary Borel dynamics with noisy renewal observations, the quantization error of the weighted posterior future orbit is a lower bound for every randomized, time-dependent causal machine. Conversely, an N -centre orbit code truncated at depth n is realized by $Nn + 1$ states. Under stretched-exponential renewal tails this preserves every finite positive quantization exponent. For noisy expanding torus experiments in every dimension, a shift-closed suffix code yields matching error bounds with all persistent states counted. The resulting profile treats nongeometric renewal laws, separates the Bayes noise floor from memory loss, and gives a uniform crossover at the expansion–renewal threshold. We also derive global and stratified minimax laws for calibrated Gaussian observations of colliding-root moments, proving the singular modulus from the moment map rather than assuming a contact geometry. The earlier checkpoint, filtering, control, and measurable-experiment results are retained with their proofs as complementary quantitative and foundational development.

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1. INTRODUCTION

A finite representation of an acquired state need not be a finite representation of its future. A coordinate which is inexpensive to forget now may become visible after expansion, and a codebook for a whole future may require an additional register to be executed one step at a time. The purpose of this paper is to separate these two questions and then relate them quantitatively. The first concerns the geometry of the posterior future orbit. The second concerns its realization by a finite-state observer.

The resource is a hard cardinality bound M on the observer's persistent state. We do not replace it by mutual information, by an average coding length, or by a communication rate. Updates and outputs may be randomized and time dependent. An absolute clock is permitted, but the time since the last random acquisition is not available unless represented in the persistent state. This distinction will be essential even in a two-point example.

1.1. The main results. Consider a dynamical system T which is renewed at independent random times. A fresh hidden state U is observed through a kernel $K(U, dy)$ at each renewal. Between renewals there are no new data. Let w_k be the stationary age probabilities, and let

$m_k(y) = \mathbb{E}[f(T^k U) \mid Y = y]$ for a bounded vector readout f . The posterior orbit is

$$Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0}.$$

Theorem 2.2 proves a converse for every finite-state observer and a finite-state realization bound: an N -centre codebook, truncated at depth n , is executed by $Nn + 1$ states. If $e_M^2(Z)$ is its Hilbert-space quantization error, E_M^{av} is the optimal average prediction error above the full-history Bayes floor, and $W(n) = \sum_{k \geq n} w_k$, then

$$e_M^2(Z) \leq E_M^{\text{av}}, \quad E_{Nn+1}^{\text{av}} \leq e_N^2(Z) + \|f\|_\infty^2 W(n). \quad (1.1)$$

Consequently every finite positive quantization exponent is preserved by causal realization under a stretched-exponential renewal tail (Corollary 2.3). This conclusion requires neither smooth dynamics nor an invertible or noiseless observation map. Proposition 2.4 gives the exact finite-codebook condition for autonomous realization: closure under the left shift. Example 2.5 shows why such a condition cannot be discarded.

For expanding torus systems the state overhead can be removed at the level of sharp rates. In dimension d , let $T_b x = bx$ modulo one, with $b \geq 2$, Haar resets, a wrapped Gaussian acquisition of noise level σ , and the circle-coordinate readout Φ . The posterior Fourier attenuation is explicit. With $v_k = w_k \exp(-4\pi^2 \sigma^2 b^{2k})$ and

$$\Psi_v(n) = \sum_{k < n} v_k b^{-2(n-k)} + \sum_{k \geq n} v_k,$$

Theorem 3.1 identifies the optimal excess risk, up to constants, as $\Psi_v(n_M)$, where $1 + b^d + \dots + b^{dn_M} \leq M < 1 + b^d + \dots + b^{d(n_M+1)}$. The renewal law need not be geometric: a uniform bound on mean residual lifetime suffices. The lower bound permits arbitrary randomized nonstationary encoders and off-image codebook centres. The upper bound is an explicitly counted suffix machine.

In the geometric case $w_k = (1 - q)q^k$, the profile is

$$(1 - q)b^{-2n} \sum_{k < n} (b^2 q)^k + q^n.$$

This gives the three rates in (3.16) and a uniform critical window $q = b^{-2} e^{x/n}$ with crossover $H(x) = (e^x - 1)/x$. The exact suffix risk has a leading-coefficient limit; the optimal risk is uniformly comparable to that profile. The distinction is stated in Corollary 3.4. The same profile governs the noise floor, so noise level and memory budget can vary jointly. Smooth conjugates furnish nonconstant-slope expanding examples, and bi-Lipschitz readouts preserve the raw-risk law (Proposition 3.5).

A second application concerns observation geometry rather than transport. Suppose the first d power sums of d ordered real roots are measured through a correlated Gaussian pair with an unrestricted additive calibration. Theorem 4.1 proves

$$\mathcal{R}_M \asymp \min(1, \sigma^{2/d}) + M^{-2/d}.$$

On domains with separated clusters and at most r_* roots in a cluster, the statistical exponent becomes $2/r_*$, while the representation exponent remains $2/d$. Both exponents are proved from the actual moment map: Newton identities and a multiplicity-aware root argument give the upper modulus, and moment-matched colliding configurations give the lower bound. No global contact equivalence is inserted as a hypothesis.

1.2. Relation to earlier theories. Quantization of dynamical trajectories and the cost of preserving unstable directions are established themes. Nair and Evans [26], Tatikonda and Mitter [27], and the monograph of Matveev and Savkin [28] study estimation and control through communication channels. In particular, Propositions 3.1 and 5.3 of [27] compare the channel rate with expansion of unstable linear modes, using a geometric converse and a recursive quantizer

for a specified information pattern. Our geometric converse is related in method. Its resource is different: no fresh channel symbol is available between renewals, and the complete persistent state has only M values. A recursive channel code with real-valued centres or scale variables is not automatically an M -state machine. We prove the finite-state realization and its overhead instead of identifying these constraints.

Theorem 1.5 and Corollary 1.10 of Lindenstrauss and Tsukamoto [29] relate metric mean dimension to small-distortion information rates over invariant measures, under a tame covering-growth assumption. Here the law of an acquired orbit is fixed, its future has renewal weights, and the constraint is a single hard persistent cardinality rather than asymptotic mutual information per unit time. The two invariants need not agree. Our exponent theorem does not assert a new variational principle for metric mean dimension; it identifies when a weighted-orbit quantization exponent survives finite-state causal realization.

Predictive states and predictive compression are discussed in [5, 6, 24, 25]. Finite-state filter and control approximations, including their stability assumptions, are compared in the retained quantitative development with [8, 9, 10, 19, 21, 22]. The present posterior-orbit theorem uses regeneration in place of a global contracting filter metric. It does not remove the stability assumptions from those different, continuously observed models.

The nuisance argument in Theorem 7.1 belongs to invariant minimax theory. The role of asymptotically invariant probabilities is classical; see Bondar and Milnes [30]. In our Gaussian proof diffuse nuisance priors make the residual observation asymptotically parameter free. Its specific conclusion preserves each hard label budget M and the cross-covariance of the observed pair. This is not a new general Hunt–Stein principle, nor full experiment equivalence. Finally, singular moduli and testing bounds are classical tools [12]; the moment theorem derives their hypotheses in a coupled, changing-rank example. The mixture-estimation theorem of Heinrich and Kahn [31] is not invoked as a substitute for this derivation, since its sample experiment and unknown-weight parameter space differ.

Comparison	Resource and objective	Relation to the present statements
Linear data-rate bounds [27], Props. 3.1, 5.3	Symbols per time; asymptotic state observation under a specified control-information pattern	Expansion-based converse and recursive upper code; our Theorems 2.2, 3.1 instead charge every persistent state during no-data intervals.
Dynamical rate distortion [29], Thm. 1.5	Mutual information per time; orbit distortion; invariant measures	A variational principle for metric mean dimension, not an autonomous hard-state realization theorem. Our invariant is the fixed-law weighted posterior-orbit quantization exponent.
Invariant minimax reduction [30]	Group-invariant statistical decisions and asymptotically invariant averaging	Theorem 7.1 implements the Gaussian reduction without enlarging M ; Theorem 4.1 uses it in a specified singular model.
Finite-mixture modulus [31], Thm. 3.2	Distribution observations; unknown weights; Wasserstein loss	Our moment observations have equal known weights and Euclidean ordered-root loss. Lemmas 4.2 and 4.3 establish the required modulus and tests directly.

1.3. Organization and dependencies. Sections 2–4 contain the principal theorem chain and its complete proofs. The noisy expansion law uses the posterior-orbit converse, not the earlier scalar expansion theorem. The root-collision theorem uses only the exact Gaussian nuisance

quotient from the retained results, together with the two lemmas proved in its own section. None of these new results uses a historical billiard spectral theorem, a particle large-deviation principle, a nonlinear semigroup limit, or an unbounded-operator closure assertion.

The subsequent quantitative development retains the scalar checkpoint and causal phase separation, correlated calibration, nonuniform causal resolution, positive-refresh hidden Markov laws, acquisition examples, and unrestricted-control comparison, with their proofs. The appendices retain the measurable experiment calculus, predictive quotients, information inequalities, and model-interface boundaries. Thus the principal arguments can be read independently, while the full mathematical development remains available in this edition. The title denotes the common programme of causal experiments and resource-aware reductions; it does not identify an abstract reduction with the model-specific theorem needed to instantiate it.

2. POSTERIOR ORBITS AND FINITE-STATE REALIZATION

Let E and $\mathcal{Y}$ be standard Borel spaces, let μ be a probability on E , and let K be a probability kernel from E to $\mathcal{Y}$. Let $T : E \rightarrow E$ and $f : E \rightarrow \mathbb{R}^r$ be Borel, with $\|f\| \leq F < \infty$. At each renewal a fresh pair (U, Y) has law $\mu(du)K(u, dy)$. The observer acquires Y , not necessarily U . Between renewals the hidden state follows T . The successive cycle lengths $L_0, L_1, \dots$ are independent copies of a positive integer-valued random variable L , independent of all acquired pairs, and

$$\ell = \mathbb{E}L < \infty, \quad s_k = \mathbb{P}(L > k), \quad w_k = \frac{s_k}{\ell}, \quad W(n) = \sum_{k \geq n} w_k. \quad (2.1)$$

Thus $\sum_{k \geq 0} w_k = 1$. There is an acquisition at time 0. Subsequent acquisition times are $\tau_j = L_0 + \dots + L_{j-1}$. At time $t = \tau_j + k < \tau_{j+1}$, the target is $f(T^k U_j)$. The renewal indicator is observed, but the length of the current cycle is not announced at its beginning.

An M -state observer retains $I_t \in \{1, \dots, M\}$. Its update is an arbitrary Borel stochastic kernel of the preceding state, absolute time, renewal indicator, and, at a renewal only, the new observation. After the update, the decoder has the state, absolute time, and independent public randomness. It does not separately retain the last observation, the renewal indicator, or the renewal age. Private randomness is allowed, but any data-dependent persistent register, including an age or a random-seed position, is part of I_t . Program descriptions, real constants, and transient computation are not charged to M . This is a persistent-cardinality constraint, not a bound on total computational storage.

Choose simultaneous Borel versions

$$m_k(y) = \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0} \in \mathcal{H} := \ell^2(\mathbb{N}_0; \mathbb{R}^r). \quad (2.2)$$

A countable union of null sets makes these versions simultaneous; values there may be set to zero. Write

$$e_M^2(Z) = \inf_{c_1, \dots, c_M \in \mathcal{H}} \mathbb{E} \min_{i \leq M} \|Z(Y) - c_i\|_{\mathcal{H}}^2, \quad B = \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \quad (2.3)$$

The number B is the unavoidable error of the complete acquisition history. Let $A_t = t - \max\{\tau_j : \tau_j \leq t\}$, and let Y_t^* be the observation at that last renewal. Define

$$E_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2, \quad (2.4)$$

$$R_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|f(X_t) - \hat{f}_t\|^2. \quad (2.5)$$

Here and below an infimum is over the specified machine class; it need not be attained. All statements about e_M^2 use squared distortion.

Lemma 2.1 (Kernel realization). *The preceding machine class can be realized using a countable family of independent uniform random variables, independent of the experiment. After fixing this family and the history before a renewal, the first n outputs along a continuation with no further renewal belong to a set of at most M vectors in $(\mathbb{R}^r)^n$.*

Proof. For a finite-state transition kernel the inverse cumulative probabilities give a Borel function of the input and one uniform variable. For a general standard Borel output, embed its range as a Borel subset of $[0, 1]$, use the measurable inverse distribution function, and map back; a fixed default value handles the null exceptional set. Applying this construction to each time-indexed kernel uses only countably many uniforms. The product probability space supplies them jointly. Public uniforms are included in the same construction, with their prescribed sharing between update and decoder. Once they are fixed, the update and decoder are deterministic. Given the state immediately after a renewal, the no-renewal input sequence and absolute times determine the subsequent states and outputs. There are only M possible starting states. The argument uses a fixed time-indexed family, not an uncharged data-dependent position in a random tape. $\square$

Theorem 2.2 (Posterior-orbit converse and realization). *For the experiment just defined,*

$$R_M^{\text{av}} = B + E_M^{\text{av}}, \quad e_M^2(Z) \leq E_M^{\text{av}}. \quad (2.6)$$

In fact every randomized, time-dependent M -state machine satisfies the latter lower bound with its $\liminf$ time-average excess loss. Conversely, for all integers $N, n \geq 1$,

$$E_{Nn+1}^{\text{av}} \leq e_N^2(Z^{<n}) + F^2W(n) \leq e_N^2(Z) + F^2W(n), \quad (2.7)$$

where $Z^{<n}$ is the first n coordinates of Z . The upper bound is realized, up to an arbitrary positive error, by a stationary deterministic machine with states $\{1, \dots, N\} \times \{0, \dots, n-1\} \cup \{\partial\}$. In particular, for $M \geq 2$,

$$e_M^2(Z) \leq E_M^{\text{av}} \leq \inf_{1 \leq n \leq M-1} \left\{ e_{\lfloor (M-1)/n \rfloor}^2(Z) + F^2W(n) \right\}. \quad (2.8)$$

Proof. Conditional on the full acquired history, the renewal indicators, and independent coding randomness, the conditional mean of $f(X_t)$ is $m_{A_t}(Y_t^*)$. Indeed, the current acquired pair is independent of previous cycles, and survival of its cycle is independent of that pair. Conditional orthogonality gives, for every machine,

$$\mathbb{E} \|f(X_t) - \hat{f}_t\|^2 = \mathbb{E} \|f(X_t) - m_{A_t}(Y_t^*)\|^2 + \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2.$$

The Cesàro renewal-reward identity makes the average of the first term tend to B . Bounded rewards and $\mathbb{E}L < \infty$ suffice, including arithmetic or deterministic cycle lengths. This proves the equality in (2.6).

Fix n . At a renewal at time s , condition on the past and then use Lemma 2.1. Along the first n no-renewal steps the outputs are chosen from at most M decoder strings. The new Y has its original marginal law under this conditioning. Since survival to age k has probability s_k , the expected excess reward in those steps is at least

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_i \sum_{k < n} s_k \|m_k(Y) - c_{i,k}\|^2 = \ell e_M^2(Z^{<n}).$$

Randomizing the selection of a string cannot reduce its pointwise minimum. Integrating the conditioned variables preserves the bound. Let $u_s = \mathbb{P}(s \text{ is a renewal})$. Rewards arising from distinct last renewals are disjoint. Summing over $0 \leq s \leq N - n$ therefore gives a lower bound $\ell e_M^2(Z^{<n}) \sum_{s=0}^{N-n} u_s$ for the total excess loss up to time $N - 1$. The elementary renewal theorem, in its Cesàro form, gives $N^{-1} \sum_{s < N} u_s \rightarrow 1/\ell$. Consequently the lower limit of average loss is at least $e_M^2(Z^{<n})$.

Truncation cannot increase quantization error. Conversely, extending a codebook for the truncation by zero gives

$$e_M^2(Z^{<n}) \leq e_M^2(Z) \leq e_M^2(Z^{<n}) + F^2 W(n).$$

Letting $n \rightarrow \infty$ proves the converse, without a compactness assumption on the orbit image or a restriction to on-image centres.

For the upper bound choose an ε -optimal N -centre codebook for $Z^{<n}$. Coordinatewise projection permits its k th coordinate to be written $\sqrt{w_k} c_{i,k}$ with $\|c_{i,k}\| \leq F$; a zero-weight coordinate is immaterial. At a renewal, choose the first nearest centre, a Borel choice, and set the state to $(i, 0)$. At a nonrenewal change (i, k) to $(i, k+1)$ until $k = n-1$, after which the state is ∂ . The latter state is absorbing between renewals. Decode (i, k) as $c_{i,k}$ and ∂ as zero. The average excess reward is at most $e_N^2(Z^{<n}) + \varepsilon + F^2 W(n)$ by the same renewal-reward identity. This is an actual finite-state update; its age information is precisely the charged second coordinate. Let $\varepsilon \downarrow 0$. Unused states are permitted, which proves the last assertion. $\square$

Corollary 2.3 (Invariance of the quantization exponent). *Suppose $W(n) \leq A \exp(-cn^\beta)$ for some $A, c, \beta > 0$, and*

$$\alpha = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z)}{\log M} \quad \text{exists in } (0, \infty). \quad (2.9)$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log E_M^{\text{av}}}{\log M} = \alpha. \quad (2.10)$$

If $e_M^2(Z) \asymp M^{-\alpha}$, more explicitly

$$c_1 M^{-\alpha} \leq E_M^{\text{av}} \leq C_1 M^{-\alpha} (\log M)^{\alpha/\beta} \quad (M \geq 2),$$

with constants depending on the displayed hypotheses. For geometric renewals with parameter $p > 0$, the exponent conclusion also holds for the infimum of the supremum-in-time excess risk.

Proof. Choose any $s > \alpha$ and $n = \lceil ((s/c) \log M)^{1/\beta} \rceil$. Then $\log[(M-1)/n] / \log M \rightarrow 1$, whereas $W(n) \leq AM^{-s}$. Apply (2.8); the two logarithmic bounds converge to α . Under the two-sided power bound the same substitution gives the explicit inequalities, enlarging constants for finitely many small M .

For geometric renewals put $q = 1 - p$, so $w_k = pq^k$. A stationary reset-based machine has a nonnegative age-loss sequence d_k . With an initial acquisition at zero its loss at time t is

$$p \sum_{k < t} q^k d_k + q^t d_t \leq p^{-1} \sum_{k \geq 0} w_k d_k.$$

Use the upper machines in Theorem 2.2; the converse follows because a supremum is at least the lower time-average. $\square$

The corollary is an exponent theorem, not a uniform equivalence of errors for arbitrary orbits. It applies to arbitrary Borel dynamics and arbitrary noisy or partial reset observations. Its geometric invariant is the quantization exponent of the *posterior* future orbit; the tail assumption controls the cost of making a finite codebook causal. Computing that invariant in a particular dynamical system remains a separate mathematical task.

Proposition 2.4 (Exact autonomous realization). *Let a ray mean a sequence $c = (c_k)_{k \geq 0}$ in the ball $\{z \in \mathbb{R}^r : \|z\| \leq F\}$. Among stationary deterministic machines, the optimal average excess risk is*

$$\inf_{\substack{|C| \leq M \\ SC \subseteq \mathcal{C}}} \mathbb{E} \min_{c \in C} \sum_{k \geq 0} w_k \|m_k(Y) - c_k\|^2, \quad (Sc)_k = c_{k+1}. \quad (2.11)$$

The reset update may depend on the previous state; allowing that dependence does not alter this infimum.

Proof. For a fixed autonomous no-renewal update a and decoder d , a state i determines the ray $(d(a^k i))_{k \geq 0}$. These at most M rays are closed under S . Any reset rule selects one of them, possibly using the previous state; its reward is no smaller than the pointwise minimum over all the rays. Conversely, for a finite shift-closed set, retain the chosen ray as the state, apply S on a nonrenewal, and decode its first coordinate. The first nearest ray is a Borel reset rule independent of the previous state. Renewal reward proves the formula. This is a characterization of the stated autonomous class, not an assertion that all optimal randomized nonstationary machines are autonomous. $\square$

Example 2.5 (Orbit cardinality does not include an age register). Let $E = \{0, 1\}$, $U = 1$ deterministically, $T(0) = T(1) = 0$, and $f(x) = x$. Use geometric renewals with $0 < p < 1$ and a constant reset observation. The posterior orbit is the single ray $(1, 0, 0, \dots)$, so $e_1^2(Z) = 0$ and $B = 0$. Nevertheless a one-state decoder, having only absolute time and independent public coins, does not know whether a renewal occurred at that time. For every $t \geq 1$ the minimum squared error is $p(1 - p)$, since the target is an independent Bernoulli(p) renewal indicator. Two states suffice for zero error: write 1 at each renewal and 0 otherwise. Thus the orbit converse cannot be reversed with the same cardinality without a realizability condition. The finite-chain overhead in (2.7) and the closure in (2.11) address a genuine obstruction.

3. NOISY EXPANDING EXPERIMENTS AND A UNIFORM CRITICAL WINDOW

The realization overhead in Theorem 2.2 is not always necessary. We now identify a class in which symbolic closure yields matching errors, including the acquisition noise and the renewal tail. Let $b \geq 2$ and $d \geq 1$ be integers, let $\mathbb{T}^d = (\mathbb{R}/\mathbb{Z})^d$, and put

$$T_b x = bx \pmod{1}, \quad \Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d. \quad (3.1)$$

At a renewal U is Haar uniform and the observation is

$$Y = U + \sigma G \pmod{1}, \quad G \sim N(0, I_d), \quad \sigma \geq 0. \quad (3.2)$$

The noise, the fresh states, and the renewal lengths are mutually independent. The target is $\Phi(X_t) \in \mathbb{R}^{2d}$, not a discontinuous choice of coordinates on the torus. Both average raw risk and average excess risk have the meanings of Section 2.

Haar invariance implies that Y and G are independent: conditional on G , the translate of U is Haar. Consequently the posterior law of U given $Y = y$ is that of $y - \sigma G$ modulo one, and

$$m_k(y) = a_k \Phi(b^k y), \quad a_k = \exp(-2\pi^2 \sigma^2 b^{2k}), \quad B = d \sum_{k \geq 0} w_k (1 - a_k^2). \quad (3.3)$$

This calculation uses the Gaussian characteristic function, so it also specifies exactly which noise information is used by the decoder. Write

$$v_k = w_k a_k^2, \quad \Psi_v(n) = \sum_{k < n} v_k b^{-2(n-k)} + \sum_{k \geq n} v_k, \quad n \geq 0. \quad (3.4)$$

For $v = w$, use the notation Ψ_w . We impose the quantitative renewal condition

$$W(n) \leq A w_n \quad (n \geq 0) \quad (3.5)$$

for some finite $A \geq 1$. In terms of the cycle length this is a uniform bound on the mean residual lifetime, conditional on survival. It includes geometric renewals, mixtures with geometric parameters bounded away from zero, and bounded renewal lengths. It does not include arbitrary polynomial renewal tails.

Put $a = b^d$ and

$$S_n = 1 + a + \cdots + a^n, \quad n_M = \max\{n \geq 0 : S_n \leq M\}. \quad (3.6)$$

The distinction between a^n and S_n counts, rather than suppresses, the persistent information about remaining precision.

Theorem 3.1 (Sharp posterior-orbit law). *Under (3.5), for every $M \geq 1$ and $\sigma \geq 0$,*

$$\frac{\Psi_v(n_M)}{8Ab^6} \leq E_M^{\text{av}} \leq \frac{d\pi^2}{3} \Psi_v(n_M), \quad (3.7)$$

$$B + \frac{\Psi_v(n_M)}{8Ab^6} \leq R_M^{\text{av}} \leq B + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (3.8)$$

The lower bounds hold for all randomized, nonstationary machines. The upper bounds are attained by the finite suffix machine described below. Its exact average raw risk with S_n states is

$$U_{n,\sigma} = B + d \sum_{k < n} v_k \left[1 - \text{sinc}^2(\pi b^{-(n-k)}) \right] + d \sum_{k \geq n} v_k, \quad (3.9)$$

where $\text{sinc}(z) = \sin(z)/z$ and $\text{sinc}(0) = 1$. For geometric renewals, (3.8) also holds for $R_M^{\text{sup}} := \inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E} \|\Phi(X_t) - \hat{f}_t\|^2$, with an initial acquisition at zero. The supremum raw risk of the displayed suffix machine is exactly $U_{n,\sigma}$.

Lemma 3.2 (Orbit separation and small balls). *Suppose $v_k = w_k a_k^2$ with a_k nonincreasing and (3.5) holds. For Haar Y the quantization error of $(\sqrt{v_k} \Phi(b^k Y))_{k \geq 0}$ satisfies*

$$e_M^2 \geq \frac{\Psi_v(n_M)}{8Ab^6}. \quad (3.10)$$

Proof. First, $\sum_{k \geq n} v_k \leq A v_n$. Also Ψ_v is nonincreasing, and

$$\Psi_v(n+1) \geq b^{-2} \Psi_v(n). \quad (3.11)$$

These follow directly from (3.4) and monotonicity of a_k . For $x, y \in \mathbb{T}^d$, write $\delta = \max_j \text{dist}_{\mathbb{T}}(x_j, y_j) \in [0, 1/2]$. If $\delta > 0$, set $K = \lfloor \log_b(1/(2\delta)) \rfloor$. In a coordinate attaining δ , for $0 \leq k \leq K$ the unreduced difference has modulus $b^k \delta \leq 1/2$. Hence $|\sin(\pi b^k \delta)| \geq 2b^k \delta$, and the squared orbit distance $D^2(x, y)$ obeys

$$\begin{aligned} D^2(x, y) &= 4 \sum_{k \geq 0} v_k \sum_{j=1}^d \sin^2(\pi b^k (x_j - y_j)) \\ &\geq 16\delta^2 \sum_{k=0}^K v_k b^{2k} \geq \frac{4}{b^2} \sum_{k=0}^K v_k b^{-2(K-k)} \geq \frac{4}{Ab^2} \Psi_v(K). \end{aligned} \quad (3.12)$$

The middle inequality uses $\delta > b^{-(K+1)}/2$; the last uses the residual-tail bound and $A \geq 1$.

Let $h = [2(2M)^{1/d}]^{-1}$ and $K_h = \lfloor \log_b(1/(2h)) \rfloor = \lfloor \log_a(2M) \rfloor$. Since $M < S_{n_M+1}$ and $a \geq 2$, one has $K_h \leq n_M + 2$. Whenever $\delta \geq h$, the associated $K \leq K_h$; thus (3.11) and (3.12) give

$$D^2(x, y) \geq \gamma \Psi_v(n_M), \quad \gamma = \frac{4}{Ab^6}.$$

If $\Psi_v(n_M) = 0$, the conclusion is immediate. Otherwise take a Hilbert ball of radius r , with $r^2 = \gamma \Psi_v(n_M)/16$. Two preimages in the same ball have orbit distance at most $2r$, so their torus sup-distance is less than h . Fixing one preimage, all preimages lie in a torus cube of side $2h$, whose Haar measure is $1/(2M)$. This argument applies even when the centre of the Hilbert ball is not on the orbit image. The union of M such balls has preimage measure at most $1/2$. Every M -centre quantizer therefore has mean squared error at least $r^2/2 = \Psi_v(n_M)/(8Ab^6)$. $\square$

Proof of Theorem 3.1. The lower bound follows from Theorem 2.2, (3.3), and Lemma 3.2. The orthogonality identity then adds B to the excess risk.

For the upper bound use the alphabet $\{0, \dots, b-1\}^d$. The states are all words of lengths $0, \dots, n$, exactly S_n states. At a renewal retain the first n base- b digits of each coordinate of Y , grouped into one word. Use the terminating expansion at ambiguous endpoints; those endpoints have Haar measure zero. At a nonrenewal delete the first letter, leaving the empty word fixed. For a nonempty word w of length l , let $c(w)$ be the midpoint of its b -adic cube, and decode it as

$$a_{n-l} \operatorname{sinc}(\pi b^{-l}) \Phi(c(w)).$$

Decode the empty word as zero. The age before exhaustion is $n-l$, so there is no separate age register. The update does not use the renewal law.

Conditional on this suffix at age $k < n$, the vector $b^k Y$ is uniform on its cube. Averaging each trigonometric coordinate over an interval gives its midpoint value times $\operatorname{sinc}(\pi b^{-(n-k)})$. The displayed output is therefore the conditional mean of $\Phi(b^k U)$ under the suffix. Its raw squared error is

$$d \left[1 - a_k^2 \operatorname{sinc}^2(\pi b^{-(n-k)}) \right].$$

After exhaustion the raw error is d . Renewal reward proves (3.9). For $0 \leq t \leq 1/2$,

$$\frac{4}{3} t^2 \leq 1 - \operatorname{sinc}^2(\pi t) \leq \frac{\pi^2}{3} t^2. \quad (3.13)$$

Indeed, if V, V' are independent uniform variables on an interval of length t , the middle expression is $2\mathbb{E} \sin^2(\pi(V - V'))$. Use $2|u| \leq |\sin(\pi u)| \leq \pi|u|$ for $|u| \leq 1/2$ and $\mathbb{E}(V - V')^2 = t^2/6$. This proves the average upper bounds, including $n = 0$.

For geometric renewals put $q = 1 - p$. The raw age-loss sequence just computed is nondecreasing: a_k decreases and, as k increases before exhaustion, the sinc factor decreases as well. The age at time t has the law of $\min(G_0, t)$, where $\mathbb{P}(G_0 = k) = pq^k$. Therefore its expected raw loss increases to $\sum_k pq^k d_k = U_{n,\sigma}$ and never exceeds this number. The lower bound for supremum loss follows from the average converse. $\square$

Corollary 3.3 (Joint statistical and persistent-state law). *Under the hypotheses of Theorem 3.1,*

$$R_M^{\text{av}} \asymp_{b,d,A} B + \Psi_w(n_M). \quad (3.14)$$

The same holds for raw supremum risk under geometric renewals. Moreover a suffix decoder which does not know σ or the renewal law attains this order of raw risk.

Proof. Every coefficient in Ψ_v is between zero and one. Thus

$$0 \leq \Psi_w(n) - \Psi_v(n) \leq \sum_k w_k (1 - a_k^2) = B/d.$$

This and (3.8) prove the equivalence. For the noise-independent decoder use $\operatorname{sinc}(\pi b^{-l}) \Phi(c(w))$, omitting a_{n-l} . Compare with the hypothetical target $\Phi(b^k Y)$. At age k ,

$$\mathbb{E} \|\Phi(b^k U) - \Phi(b^k Y)\|^2 = 2d(1 - a_k) \leq 2d(1 - a_k^2).$$

The squared triangle inequality and the noiseless suffix bound give average raw risk at most $4B + (2d\pi^2/3)\Psi_w(n)$. For geometric renewals the same bound holds for supremum loss: both the noise-discrepancy sequence $1 - a_k^2$ and the noiseless suffix raw-loss sequence are nondecreasing, so each time expectation is bounded by its stationary-age average. The machine uses only b, d, n and the observed digits. $\square$

3.1. Geometric renewals and the critical window. For $w_k = pq^k$, $p = 1 - q$, the noiseless profile is

$$\Psi_q(n) = pb^{-2n} \sum_{k=0}^{n-1} (b^2 q)^k + q^n. \quad (3.15)$$

Theorem 3.1 is uniform for q in any fixed compact subset of $(0, 1)$, since $A = 1/p$ is then bounded. In particular it remains valid when q and σ depend on M .

Corollary 3.4 (Phase regimes and crossover). *For each fixed $0 < q < 1$, at $\sigma = 0$ the optimal average and supremum raw risks have orders*

$$\begin{cases} M^{-2/d}, & q < b^{-2}, \\ (1 + \log M)M^{-2/d}, & q = b^{-2}, \\ M^{-\log(1/q)/(d \log b)}, & q > b^{-2}. \end{cases} \quad (3.16)$$

Let $q_n = b^{-2} \exp(x/n)$ and $H(x) = (e^x - 1)/x$, with $H(0) = 1$. Uniformly for x in bounded intervals,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) \longrightarrow (1 - b^{-2})H(x). \quad (3.17)$$

For the explicit S_n -state suffix machine, its exact noiseless raw risk satisfies the stronger coefficient limit

$$\frac{b^{2n}}{n} U_{n,0} \longrightarrow \frac{d\pi^2}{3} (1 - b^{-2})H(x). \quad (3.18)$$

The optimal risks are bounded above and below by constant multiples of the profile in (3.17), uniformly in this window; no equality of their leading constants is asserted.

Proof. Since $S_n \asymp_b b^{dn}$, evaluating the geometric sum in (3.15) proves (3.16). In the critical window,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) = (1 - q_n) \frac{1}{n} \sum_{k=0}^{n-1} e^{xk/n} + \frac{e^x}{n}.$$

The Riemann sum tends uniformly on bounded x -intervals to $\int_0^1 e^{xt} dt = H(x)$. For the exact suffix risk use $1 - \text{sinc}^2(\pi t) = (\pi^2/3)t^2 + O(t^4)$, uniformly for $0 \leq t \leq 1/2$. After multiplication by b^{2n} , the error sum is bounded by a constant times

$$\sum_{k < n} e^{xk/n} b^{-2(n-k)},$$

which is uniformly bounded on bounded x -intervals. The exhaustion term $db^{2n}q_n^n = de^x$ is bounded as well. Dividing by n proves (3.18). The final assertion is exactly the uniform two-sided bound of Theorem 3.1. $\square$

For completeness, the noise floor has the same transition. For $0 < \sigma \leq 1$, set $n_\sigma = \lfloor \log_b(1/\sigma) \rfloor$; for $\sigma > 1$ set $n_\sigma = 0$. From $1 - e^{-4\pi^2 z^2} \asymp \min(1, z^2)$ and $b^{-(n_\sigma+1)} < \sigma \leq b^{-n_\sigma}$ one obtains

$$B \asymp_{b,d} \Psi_q(n_\sigma), \quad R_M \asymp_{b,d,q} \Psi_q(n_\sigma) + \Psi_q(n_M). \quad (3.19)$$

At $\sigma = 0$ the first term is interpreted as zero. The comparison in the second formula is uniform on compact q -intervals. Thus the small-noise orders of B are respectively σ^2 , $\sigma^2(1 + \log(1/\sigma))$, and $\sigma^{\log(1/q)/\log b}$ in the three regimes. This describes filtering uncertainty and expanding transport in the same experiment, including joint limits, rather than adding an unrelated noisy model.

3.2. Nonconstant slopes and other readouts. The preceding law is not confined to a constant-slope coordinate presentation. The next statement records both a genuine robustness class and its precise boundary.

Proposition 3.5 (Transport under conjugacy and metric readouts). *Let $h : \mathbb{T}^d \rightarrow E'$ be a Borel isomorphism. Transport the reset law, observation (3.2), dynamics, and readout by h ; thus the new observation is $h(Y)$, the dynamics are $hT_b h^{-1}$, and the readout is Φh^{-1} . All the preceding finite-state risks are unchanged. Suppose instead that E' is compact and the new continuous readout g satisfies, for some $0 < c \leq C < \infty$,*

$$c\|\Phi(x) - \Phi(y)\| \leq \|g(hx) - g(hy)\| \leq C\|\Phi(x) - \Phi(y)\|. \quad (3.20)$$

Then its optimal raw squared risks, for the same observations and state budgets, lie between $c^2/4$ and $4C^2$ times those for Φ . This assertion concerns raw risk, not equality of posterior excess risk for nonlinear readouts.

Proof. Apply h^{-1} to each acquired observation within the transient update computation. This and its inverse transform machines without changing a persistent state. For the readout assertion, replace any output z for Φ by a nearest point $\Phi(x_z)$ on its compact image, choosing measurably. For a true state x , $\|\Phi(x_z) - \Phi(x)\| \leq 2\|z - \Phi(x)\|$. Output $g(hx_z)$ and use the upper inequality in (3.20). The inverse construction projects onto the compact image of g and uses the lower inequality. Both transformations act only on the decoder, use no extra states, and are valid conditionally on any coding randomness. $\square$

For example, in one dimension take

$$h_\varepsilon(x) = x + \frac{\varepsilon}{2\pi} \sin(2\pi x) \pmod{1}, \quad |\varepsilon| < \frac{b-1}{b+1}.$$

This is a smooth circle diffeomorphism and the conjugate map has nonconstant derivative

$$(h_\varepsilon T_b h_\varepsilon^{-1})'(h_\varepsilon x) = b \frac{1 + \varepsilon \cos(2\pi b x)}{1 + \varepsilon \cos(2\pi x)} > 1.$$

Coordinatewise products give higher-dimensional examples, with nonuniform physical reset density and transported nonadditive observation noise. The ordinary circle embedding of $h_\varepsilon x$ satisfies (3.20). These are nonconstant-slope expanding experiments, but their Lyapunov exponent remains $\log b$ by conjugacy. We do not identify their law with a pressure theorem for arbitrary nonconjugate expanding maps. The general posterior-orbit theorem still applies to such maps; its quantization invariant would have to be evaluated separately.

4. CALIBRATED MOMENT OBSERVATIONS AT ROOT COLLISIONS

We next derive, rather than postulate, a singular observation modulus. Let

$$\Theta_d = \{\theta \in [0, 1]^d : \theta_1 \leq \cdots \leq \theta_d\}, \quad P_j(\theta) = \sum_{i=1}^d \theta_i^j, \quad 1 \leq j \leq d,$$

and write $P = (P_1, \dots, P_d)$. The data are the actually observed pair

$$Y = P(\theta) + \eta + \varepsilon_Y, \quad W = \eta + \varepsilon_W, \quad \begin{pmatrix} \varepsilon_Y \\ \varepsilon_W \end{pmatrix} \sim N(0, \sigma^2 \Sigma_0), \quad (4.1)$$

where $\eta \in \mathbb{R}^d$ is unrestricted, $\sigma \geq 0$ is known, and Σ_0 is a fixed positive-definite $2d$ by $2d$ matrix. Let

$$V_0 = (\Sigma_0)_{YY} + (\Sigma_0)_{WW} - (\Sigma_0)_{YW} - (\Sigma_0)_{WY} > 0.$$

An encoder of the pair retains one of M labels. Its decoder estimates θ , with squared Euclidean loss. Public and private randomness are permitted under the same hard label constraint as above.

Denote the minimax risk, taking a supremum over both θ and η , by $\mathcal{R}_M(\Theta, \sigma)$ when the parameter set is Θ . Projection permits decoder values to lie in Θ_d , so the loss is bounded by d .

Theorem 7.1, proved in the retained development, reduces this problem *exactly for every* M to

$$D = Y - W \sim N(P(\theta), \sigma^2 V_0). \quad (4.2)$$

For $\sigma > 0$ this is its correlated Gaussian nuisance quotient; for $\sigma = 0$ the difference is noiseless and the reverse comparison follows by fixing the nuisance to any constant. The reduction is an equality of minimax risks for decisions about θ . It is not a Blackwell equivalence for the full parameter (θ, η) .

Theorem 4.1 (Collision strata and label resolution). *For every fixed $d \geq 1$ and positive-definite Σ_0 , there are $0 < c \leq C < \infty$, independent of M and σ , such that*

$$c \left\{ \min(1, \sigma^{2/d}) + M^{-2/d} \right\} \leq \mathcal{R}_M(\Theta_d, \sigma) \leq C \left\{ \min(1, \sigma^{2/d}) + M^{-2/d} \right\}. \quad (4.3)$$

More generally, let $I_1, \dots, I_s \subset [0, 1]$ be ordered closed intervals of positive lengths, with positive gaps between consecutive intervals. Prescribe $r_j \geq 1$ roots in I_j , with $\sum_j r_j = d$, and let $\Theta(I, r)$ be the resulting ordered parameter set. If $r_ = \max_j r_j$, then*

$$\mathcal{R}_M(\Theta(I, r), \sigma) \asymp_{d, V_0, I, r} \min(1, \sigma^{2/r_*}) + M^{-2/d}. \quad (4.4)$$

The constants in the stratified assertion may deteriorate when an interval width or a gap tends to zero. Neither assertion assumes a two-sided global contact equivalence.

The model is genuinely coupled. The determinant of the derivative of P at distinct ordered roots is

$$\det DP(\theta) = d! \prod_{i < j} (\theta_j - \theta_i). \quad (4.5)$$

Thus the rank changes on the collision strata. There is no invertible smooth global coordinate change that makes this a uniformly regular θ -experiment. The next lemma supplies the inverse modulus needed for the statistical theorem. Its root-perturbation argument is classical in method; the point here is to carry it through the actual calibrated observation and hard-label decision problem.

Lemma 4.2 (Derived inverse moment moduli). *There is a finite C_d such that for all $\theta, \theta' \in \Theta_d$,*

$$\|\theta - \theta'\| \leq C_d \|P(\theta) - P(\theta')\|^{1/d}. \quad (4.6)$$

For $\theta, \theta' \in \Theta(I, r)$ and $\delta = \|P(\theta) - P(\theta')\|$ sufficiently small, the coordinates in cluster j satisfy

$$\|\theta^{(j)} - \theta'^{(j)}\| \leq C_{d, I, r} \delta^{1/r_j}. \quad (4.7)$$

Both statements include multiple roots and preserve their multiplicities.

Proof. Let $e_j(\theta)$ be the elementary symmetric polynomials, with $e_0 = 1$. Newton's identities give

$$j e_j = \sum_{i=1}^j (-1)^{i-1} e_{j-i} P_i.$$

On $[0, 1]^d$, $|P_i| \leq d$ and $|e_j| \leq \binom{d}{j}$. For an explicit finite Lipschitz bound define $L_0 = 0$ and

$$L_j = \frac{1}{j} \sum_{i=1}^j \left(d L_{j-i} + \binom{d}{j-i} \right), \quad 1 \leq j \leq d.$$

Induction gives $|e_j(\theta) - e_j(\theta')| \leq L_j \delta$. For the monic polynomials $Q_\theta(z) = \prod_i (z - \theta_i)$ and $Q_{\theta'}$, it follows that

$$|Q_\theta(z) - Q_{\theta'}(z)| \leq A_d \delta \quad (|z| \leq 2), \quad A_d = 2^d \max_{1 \leq j \leq d} L_j. \quad (4.8)$$

If $\delta = 0$, Newton's identities identify the two multisets of roots, and their ordered vectors coincide. Suppose $\delta > 0$ is small and put $\rho = (2A_d\delta)^{1/d} < 1$. Take the union of the complex disks of radius ρ about the roots of Q_θ . At its boundary every factor has modulus at least ρ , so $|Q_\theta| \geq \rho^d = 2A_d\delta > |Q_\theta - Q_{\theta'}|$. Rouché's theorem on each connected component gives the same number of zeros, counted with multiplicity, for the two polynomials. One may increase ρ arbitrarily slightly to avoid tangent boundaries and then take a limit; the piecewise smooth boundary version follows equivalently from the argument principle. A component containing at most d disks has diameter at most $2d\rho$. Match the roots within each component. For real ordered vectors, sorting a matching cannot increase its maximum displacement: an inverted pair can be uncrossed with no larger maximum distance. Thus $\|\theta - \theta'\| \leq 2d\sqrt{d}\rho$. For larger δ , compactness and the bound $\|\theta - \theta'\| \leq \sqrt{d}$ extend the inequality with a larger constant.

For the stratified assertion let $\Delta > 0$ be the smallest gap; when $s = 1$ use the preceding argument with $r_1 = d$. Around roots in I_j use disks of radius

$$\rho_j = \left[2A_d(2/\Delta)^{d-r_j}\delta \right]^{1/r_j}.$$

For small δ these radii are less than $\min(1, \Delta/3)$. At the boundary of their union the r_j factors from this cluster have modulus at least ρ_j , and all other factors have modulus at least $\Delta/2$. Hence the product again exceeds $A_d\delta$. Rouché gives equal root counts in each component, and its diameter is at most $2r_j\rho_j$. Matching and sorting within the cluster proves (4.7). $\square$

Lemma 4.3 (Indistinguishable colliding configurations). *For every $r \geq 1$ there exist two distinct real vectors $a_1 < \dots < a_r$ and $b_1 < \dots < b_r$ in $(-1/4, 1/4)$ such that*

$$\sum_i a_i^j = \sum_i b_i^j \quad (0 \leq j < r), \quad \sum_i a_i^r \neq \sum_i b_i^r. \quad (4.9)$$

If these vectors are scaled by h and translated to an interior point x_0 of a prescribed cluster, their root separation is a positive constant times h , whereas the difference of all moments of orders $1, \dots, d$ is $O(h^r)$, for fixed $d \geq r$.

Proof. Choose any distinct a_i in the indicated interval and let $Q(z) = \prod_i (z - a_i)$. Around each a_i choose a small closed interval, with disjoint interiors and endpoints still in $(-1/4, 1/4)$, on which Q has opposite nonzero signs at the two endpoints. For any sufficiently small nonzero constant c , the polynomial $Q + c$ retains these sign changes. It therefore has one real root in each interval, and these are all its r roots. Call them b_i . The monic coefficients through degree one of Q and $Q + c$ agree; only the constant coefficient changes. Newton's identities imply (4.9), with the difference at order r equal up to sign to $rc \neq 0$. For every $j \leq d$, expand $\sum_i (x_0 + ha_i)^j - \sum_i (x_0 + hb_i)^j$ by the binomial theorem. All terms of degree less than r in h vanish. The remaining terms are $O(h^r)$. The Euclidean separation of the two root vectors is $h\|a - b\| > 0$. $\square$

Proof of Theorem 4.1. Work first in the difference experiment (4.2). On either compact parameter set take a measurable least-squares minimizer

$$\tilde{\theta} \in \operatorname{argmin}_{\vartheta \in \Theta} \|D - P(\vartheta)\|.$$

Existence follows from compactness and continuity; the measurable minimum theorem gives a Borel selection. With $D = P(\theta) + \sigma V_0^{1/2}G$, $G \sim N(0, I_d)$, minimality gives

$$\|P(\tilde{\theta}) - P(\theta)\| \leq 2\sigma\|V_0^{1/2}G\|. \quad (4.10)$$

The global modulus and Gaussian moments yield mean squared error at most $C\sigma^{2/d}$, also bounded by d . On $\Theta(I, r)$, (4.7), extended by a bounded constant outside its small- δ range, yields an

upper bound $C \sum_j \sigma^{2/r_j} \leq C' \sigma^{2/r_*}$ for $0 < \sigma \leq 1$. For clarity, that extension is valid since $\|\theta^{(j)} - \theta'^{(j)}\|^2 \leq r_j$ and $\delta \geq \delta_0$ implies $r_j \leq r_j \delta_0^{-2/r_j} \delta^{2/r_j}$. The cap by the diameter handles $\sigma > 1$.

Let $N = \lfloor M^{1/d} \rfloor \geq M^{1/d}/2$. The N^d midpoint product grid in $[0, 1]^d$ has squared covering radius at most $d/(4N^2)$. Project its centres onto Θ , which in both cases is a closed convex subset. This does not increase their distances to points of Θ . Quantize $\tilde{\theta}$ to the first nearest projected centre and retain its index. The squared triangle inequality adds at most $C_d M^{-2/d}$ and multiplies the statistical bound by at most two. This proves the upper bounds with at most M labels, including the noiseless case.

For a statistical lower bound take a cluster of size r_* , or the single cluster of size d for the global assertion. Place the configurations of Lemma 4.3 at an interior point of that cluster; fix all other roots in their intervals. For sufficiently small $\sigma > 0$ take $h = c_0 \sigma^{1/r_*}$. Their squared parameter distance is at least $c_1 h^2$ and the Kullback–Leibler divergence of the two difference laws is

$$\frac{1}{2\sigma^2} \|V_0^{-1/2}(P(\theta) - P(\theta'))\|^2 \leq Ch^{2r_*}/\sigma^2.$$

Choose $c_0 > 0$ so that this is at most $1/8$. Pinsker's inequality and the two-point testing bound give risk at least $c_2 \sigma^{2/r_*}$ even without a label constraint. For example, classifying an estimator by the closer of the two parameters shows that its maximum squared risk is at least one eighth of their squared distance times one minus their total variation distance. For larger σ , the Gaussian experiment is a garbling of the one at any smaller positive fixed noise level: add an independent normal with covariance $(\sigma^2 - \sigma_0^2)V_0$ before encoding. Minimax risk is therefore nondecreasing in σ , giving the cap $c \min(1, \sigma^{2/r_*})$.

It remains to prove a representation lower bound independent of noise and calibration. The parameter set contains a d -dimensional box of positive volume, obtained by placing all roots in disjoint small subintervals inside their prescribed intervals. Put the uniform prior on that box. Conditional on independent public randomness, a decoder has at most M possible deterministic centres; private output randomness can be replaced by its mean for squared loss. A union of M Euclidean balls of radius $c_3 M^{-1/d}$ covers at most half the box, with $c_3 > 0$ depending only on its volume and d . Pointwise, every distribution of an encoded label has loss at least the squared distance to the nearest centre. Integrate first over the box and then over the public randomness. The Bayes, hence minimax, risk is at least $c_4 M^{-2/d}$. Combining the two lower bounds by $\max(u, v) \geq (u + v)/2$ proves the result. Finally apply the exact nuisance quotient to return to the observed pair. $\square$

Remark 4.4 (Statistical meaning of the collision exponent). The exponent $1/r_*$ is derived from moment cancellation and multiplicity-sensitive root stability; it is not an assumed contact order. The representation exponent remains $2/d$ because each allowed stratum-domain contains an open set of d distinct roots. The paired experiment has equal known weights and noisy observations of its first d power sums. It is not the experiment of independent samples from a finite mixture with unknown weights. Rates in the latter experiment, such as those of Heinrich and Kahn [31], concern a different likelihood and different local parameter sets. Nor is a polynomial root argument a substitute for an unsmoothed billiard local limit theorem. The present theorem verifies an actual changing-rank observation geometry within the calibrated reduction, without importing either of those assertions.

Remark 4.5 (Uniformity in the retained contact theorem). The constants of Theorem 7.2 depend on its displayed metric-equivalence constants λ, Λ . Their independence of contact orders, widths, or noise scales is conditional on uniform control of λ, Λ in those regimes. It is not a claim of uniform physical conditioning. In the moment model the dependence on dimension, covariance conditioning, cluster gaps, and interval widths is made explicit at the level of the hypotheses in

Theorem 4.1; no uniform limit through a collapsing cluster gap is inferred from the stratified statement.

COMPLEMENTARY QUANTITATIVE DEVELOPMENT

The following sections retain the preceding quantitative theory and its proofs. The scalar doubling result provides an exact checkpoint comparison; the correlated nuisance quotient is also used in Section 4. The later filter and control results have separate assumptions and are not premises of the posterior-orbit realization theorem.

5. REGENERATION AND THE GEOMETRY OF FUTURE PREDICTIONS

The obstruction in this section concerns information already acquired and subsequently discarded. It does not arise from estimating an unknown transition probability. We first give a converse for a general regenerative experiment and then compute its consequence for an expanding map.

5.1. A retained-label experiment. Let E be a standard Borel space, $T : E \rightarrow E$ a Borel map, μ a probability measure, and $f : E \rightarrow [0, 1]$ a Borel readout. Fix $0 < p < 1$ and write $q = 1 - p$. At time zero an observer acquires $U_0 \sim \mu$ and $X_0 = U_0$. At each integer time $t \geq 1$, an independent mode R_t is observed, with $\Pr(R_t = 1) = p$. If $R_t = 1$, a fresh independent $U_t \sim \mu$ is acquired and $X_t = U_t$. Otherwise $X_t = T(X_{t-1})$; there is no further observation of X_t . All acquisitions and modes are independent of the coding randomness. The full-history prediction is $f(X_t)$.

A causal M -label machine has a state $L_t \in \{1, \dots, M\}$. At time zero it encodes U_0 . Subsequently its update can use the previous label, the clock, the current mode, and, at a reset, the current U_t . After the update only its label, the clock, and independent public coding randomness are available for decoding. In particular the current acquisition is no longer available at the checkpoint. Private coding randomness is allowed. Past modes, reset times, previous acquisitions, and previous decoder outputs are not additional persistent registers. Time-dependent programs are allowed in the lower bound. Any random-generator state or data-dependent tape position that persists also counts as part of the label. Private coins may equivalently be supplied as independent, time-indexed uniforms.

One may realize the loss physically by selecting, after this compression, a Bernoulli terminal probe of mean $f(X_t)$. The probe is not an additional observation in the continuing experiment. Conditional expectation then identifies squared prediction excess with

$$L_t^{\text{pred}} = |f(X_t) - \hat{f}_t|^2.$$

Projection of the decoder onto $[0, 1]$ cannot increase this loss. Define

$$\mathcal{R}_M^{\text{ca}}(p; T, \mu, f) = \inf_{\text{causal } M\text{-label machines}} \sup_{t \geq 0} \mathbb{E} L_t^{\text{pred}}.$$

The clock is free; the elapsed time since the last reset is not free.

5.2. The future-orbit converse. For $u \in E$ set

$$Z_q(u) = (\sqrt{p} q^{k/2} f(T^k u))_{k \geq 0} \in \ell^2, \quad e_M^2(Z_q) = \inf_{c_1, \dots, c_M \in \ell^2} \int \min_i \|Z_q(u) - c_i\|_{\ell^2}^2 \mu(du). \quad (5.1)$$

The map is Borel and has norm at most one. Its geometry records every future readout that must be recovered from one label if no new acquisition occurs. In contrast, the checkpoint geometry concerns only $f(X_t)$.

Theorem 5.1 (Regenerative future-orbit lower bound). *Every causal M -label machine in the preceding experiment satisfies*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbb{E} L_t^{\text{pred}} \geq e_M^2(Z_q). \quad (5.2)$$

Consequently $\mathcal{R}_M^{\text{ca}}(p; T, \mu, f) \geq e_M^2(Z_q)$. The assertion holds for arbitrary independent public and private coding randomization and does not require a stationary machine or an invariant law for T .

Proof. Let $e_{M,K}^2$ be the quantization error of the first $K+1$ coordinates of Z_q . Appending the coordinates $\sqrt{p}q^{k/2}/2$, $k > K$, to any truncated codebook shows that

$$e_M^2(Z_q) - \frac{1}{4}q^{K+1} \leq e_{M,K}^2 \leq e_M^2(Z_q). \quad (5.3)$$

Approximate minimizers suffice for this argument; existence is not needed.

Fix a possible reset time $j \geq 1$. Condition on its entire past, on the occurrence of that reset, and on the independent coding random tapes. The new U_j still has law μ . The update assigns it one of at most M labels. Along the subsequent word of K nonreset modes, every such label determines a sequence of $K+1$ decoder values. Future random choices are part of the fixed coding tapes; the nonreset word does not contain new data. The weighted squared error of these values, integrated over U_j , is therefore at least $e_{M,K}^2$. This remains true when the assignment itself is randomized, since fixing its independent random tape produces an ordinary assignment to the same number of centres. Integrating back over the past and the tapes gives

$$p \sum_{k=0}^K q^k \mathbb{E}[L_{j+k}^{\text{pred}} \mid R_j = 1, R_{j+1} = \cdots = R_{j+k} = 0] \geq e_{M,K}^2.$$

The displayed conditional expectations also integrate over the unconditioned past at j ; future modes beyond $j+k$ play no role.

For each checkpoint t , the events that its last reset occurred at $j = t - k$, $0 \leq k \leq K$, are disjoint. Sum their nonnegative contributions over $1 \leq t \leq N$ and retain only $1 \leq j \leq N - K$. Independence of the modes gives

$$\sum_{t=1}^N \mathbb{E} L_t^{\text{pred}} \geq (N - K) e_{M,K}^2.$$

First let $N \rightarrow \infty$ at fixed K , and then let $K \rightarrow \infty$ in (5.3). The supremum of expected checkpoint losses dominates their lower limiting average, proving the consequence. $\square$

The theorem is a converse for arbitrary causal machines, not only for machines obtained by quantizing E . The centres in (5.1) may lie off the orbit image. In particular, replacing their quantization problem by a covering problem for the current state would lose information needed for this lower bound.

6. A SHARP CAUSAL-MEMORY TRANSITION FOR AN EXPANDING EXPERIMENT

Take

$$E = [0, 1), \quad \mu = \text{Leb}|_{[0,1)}, \quad T(x) = 2x - \lfloor 2x \rfloor, \quad f(x) = x. \quad (6.1)$$

The fresh acquisition is an analog uniform variable. A machine in the upper bound below reads only a finite binary prefix of it. The map is piecewise expanding; it is not asserted to be globally Lipschitz for the Euclidean metric on $[0, 1)$. The proof will account explicitly for the jumps at dyadic boundaries.

Let $\mathcal{R}_M^{\text{cp}}(t)$ be the optimal expected squared excess when an encoder can use the entire acquisition history at checkpoint t and then send one of M labels. Put

$$\Phi_q(M) = \begin{cases} M^{-2}, & 0 < q < 1/4, \\ (1 + \log M)M^{-2}, & q = 1/4, \\ M^{-\kappa(q)}, & 1/4 < q < 1, \end{cases} \quad \kappa(q) = \frac{\log(1/q)}{\log 2}. \quad (6.2)$$

All logarithms except those explicitly marked otherwise are natural.

Theorem 6.1 (Identical checkpoint geometry, three causal rates). *For the regenerative experiment (6.1), every integer $M \geq 1$, and every $t \geq 0$,*

$$\mathcal{R}_M^{\text{cp}}(t) = \frac{1}{12M^2}. \quad (6.3)$$

For each $0 < q < 1$, constants $0 < c_q \leq C_q < \infty$ give

$$c_q \Phi_q(M) \leq \mathcal{R}_M^{\text{ca}}(1 - q; T, \mu, f) \leq C_q \Phi_q(M) \quad (M \geq 1). \quad (6.4)$$

The lower bound already holds for the lower limiting temporal mean in (5.2). The upper bound is attained in order by a single deterministic finite program which does not use p or q . The program has at most M persistent labels, including every state used to record the number of retained binary digits.

The threshold is $p = 3/4$. Below that reset frequency the causal label exponent is smaller than the checkpoint exponent; at the threshold a logarithmic factor is necessary. The current acquired prediction is uniform in all these regimes. Thus its dimension and even its entire probability law do not determine the optimal causal memory law.

6.1. An explicit finite machine. For an integer $B \geq 0$ use the label set

$$\mathcal{W}_B = \bigcup_{b=0}^B \{0, 1\}^b, \quad |\mathcal{W}_B| = 2^{B+1} - 1,$$

including the empty word. At each reset retain the first B binary digits of U_t , using the terminating expansion at a dyadic rational. At a nonreset delete the leading digit, or leave the empty word unchanged. Decode a word of length b as the midpoint of its binary cylinder; the empty word decodes as $1/2$.

Lemma 6.2 (Exact risk of the suffix machine). *For this machine, $\sup_{t \geq 0} \mathbb{E} L_t^{\text{pred}}$ equals*

$$U_B(q) = \frac{1}{12} \left[p 4^{-B} \sum_{k=0}^{B-1} (4q)^k + q^B \right], \quad (6.5)$$

with the sum empty when $B = 0$. Equality holds at every $t \geq B$.

Proof. A uniform variable has independent fair binary digits outside the null set of dyadic rationals. If the most recent reset was k steps ago, then the label retains its next $(B - k)_+$ digits. Conditional on this label and the age, the current state is uniform on the corresponding cylinder, so its midpoint has squared error $4^{-(B-k)_+}/12$. At time t the age probabilities are pq^k for $0 \leq k < t$, and q^t for $k = t$, because time zero was an acquisition. Equivalently this age has the law of $\min(G, t)$, where $\Pr(G = k) = pq^k$. The function $k \mapsto 4^{-(B-k)_+}/12$ is nondecreasing and constant for $k \geq B$. Its expectation increases to (6.5) and is already equal to it for $t \geq B$. $\square$

Choose the largest B such that $2^{B+1} - 1 \leq M$. Then $(M + 1)/4 < 2^B \leq (M + 1)/2$. If $4q < 1$, the sum in (6.5) is bounded, and $q^B \leq 4^{-B}$. If $4q = 1$, the expression is $4^{-B}(pB + 1)/12$. If $4q > 1$, it is at most $[1 + p/(4q - 1)]q^B/12$. These estimates prove the upper bound in (6.4) for every integer budget, not just powers of two.

The label can be stored as an integer whose leading sentinel records its word length. Only $\lceil \log_2(2^{B+1} - 1) \rceil$ persistent bits are needed. Binary prefix acquisition and output formatting use $O(B+1)$ transient bits; deletion can be performed in $O(B+1)$ bit operations. The only read-only integer parameter is B . Neither a real-valued codebook nor the last reset time is hidden in the program. The cost of observing the reset mode and reading the new prefix is an acquisition cost, distinct from these memory bounds.

6.2. The lower bound without a restriction on encoders. Define, for $0 < \delta \leq 1$,

$$F_q(\delta) = \begin{cases} \delta^2, & q < 1/4, \\ \delta^2(1 + \log(1/\delta)), & q = 1/4, \\ \delta^{\kappa(q)}, & q > 1/4. \end{cases}$$

Each function is increasing on $(0, 1]$.

Lemma 6.3 (Separation of future orbits). *For $u, v \in [0, 1)$,*

$$\|Z_q(u) - Z_q(v)\|_{\ell^2}^2 \geq \frac{p}{32} F_q(|u - v|). \quad (6.6)$$

Proof. The case $u = v$ is immediate. Write $\delta = |u - v|$. If $\delta \leq 1/2$, let $n = \lfloor \log_2(1/(2\delta)) \rfloor$. For $0 \leq k \leq n$, the difference of the fractional parts of $2^k u$ and $2^k v$ has absolute value either $2^k \delta$ or $1 - 2^k \delta$. Both are at least $2^k \delta$, since $2^k \delta \leq 1/2$. Consequently

$$\|Z_q(u) - Z_q(v)\|^2 \geq p\delta^2 \sum_{k=0}^n (4q)^k. \quad (6.7)$$

For $q < 1/4$ the $k = 0$ term alone suffices, for every δ . For $q = 1/4$ and $\delta \leq 1/4$, the integer $n + 1$ is at least $\log(1/\delta)/(2 \log 2)$ and hence at least $(1 + \log(1/\delta))/4$. For $\delta \geq 1/4$, the initial term $p\delta^2$ is at least $p\delta^2(1 + \log(1/\delta))/(1 + \log 4)$. This proves a stronger constant than the stated one in the critical case.

For $q > 1/4$ and $\delta \leq 1/2$, one has $2^n \delta > 1/4$ and $q^n \geq (2\delta)^{\kappa(q)}$. The last term in (6.7) is therefore at least $p\delta^{\kappa(q)}/16$. For $\delta > 1/2$, the initial term gives $p\delta^2 \geq p\delta^{\kappa(q)}/4$, since $0 < \kappa(q) < 2$. All cases imply (6.6). $\square$

Proof of Theorem 6.1. Lebesgue measure is preserved by the doubling map, and a reset also has Lebesgue law. Thus X_t is uniform for every t . Given the complete history it is known exactly. Scalar quantization of the uniform law by M midpoints has error $1/(12M^2)$. Conversely, after ordering arbitrary centres their Voronoi cells are intervals of lengths a_i with $\sum_i a_i = 1$; the integral error on an interval is at least $a_i^3/12$. Convexity gives $\sum_i a_i^3 \geq M^{-2}$. Random assignment cannot improve on the nearest centre, and conditioning on an independent public seed proves the same bound for randomized encoders and decoders. This establishes (6.3).

For the causal lower bound put $h = 1/(2M)$ and $r = \frac{1}{4}\sqrt{pF_q(h)/32}$. Consider any radius- r ball in ℓ^2 , centred anywhere. Two points of the orbit image lying in this ball have distance at most $2r$. By Lemma 6.3, their preimages cannot be separated by h or more. The preimage of the ball therefore has Lebesgue measure at most h . A union of M such balls covers at most half the acquired mass, and hence

$$e_M^2(Z_q) \geq r^2/2 = \frac{p}{1024} F_q(1/(2M)).$$

For $q \leq 1/4$ this last profile is at least $\Phi_q(M)/4$; for $q > 1/4$ it is $2^{-\kappa(q)}\Phi_q(M) \geq \Phi_q(M)/4$. Thus $c_q = p/4096$ is one possible lower constant. Theorem 5.1 applies to every causal machine and proves the lower assertion including the temporal-mean version. The suffix machine and Lemma 6.2 give the upper bound as already computed. $\square$

The converse uses Euclidean differences only up to the first relevant dyadic separation; it does not replace the doubling map by an invalid global Lipschitz estimate. Nor does it assume that a general machine stores an interval or a binary word. These restrictions are used only for the explicit upper construction. The same algorithm works for all $p \in (0, 1)$, so an unknown reset frequency does not change its order of performance at any fixed true p .

At $p = 1$ every step is a fresh checkpoint and the exact risk is $1/(12M^2)$. The assertion above concerns $p > 0$; its constants need not remain uniform as $p \downarrow 0$. This dependence is displayed in the finite geometric sum (6.5), rather than suppressed in an unspecified stability constant.

7. AN EXACT CALIBRATION QUOTIENT AND SINGULAR RESOLUTION

The regenerative experiment concerns excess prediction loss relative to a full-history observer. We now study squared error relative to an unknown parameter. The distinction is essential: calibration uncertainty persists even before any label constraint is imposed.

7.1. Correlated paired measurements. Let Θ be a compact metric space, let $f : \Theta \rightarrow \mathbb{R}^m$ be continuous, and observe

$$Y = f(\theta) + \eta + \varepsilon_Y, \quad W = \eta + \varepsilon_W, \quad \theta \in \Theta, \quad \eta \in \mathbb{R}^m. \quad (7.1)$$

The joint centred Gaussian noise has a known positive definite covariance

$$\Sigma = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YW} \\ \Sigma_{WY} & \Sigma_{WW} \end{pmatrix}.$$

No independence between the two channels or their coordinates is imposed. The nuisance η is unrestricted and is not available to the observer. Both measurements are part of the acquisition. The procedure retains one of M labels and decodes an action in a standard Borel action space; independent public and private randomization is allowed.

Put

$$D = Y - W, \quad V = \Sigma_{YY} + \Sigma_{WW} - \Sigma_{YW} - \Sigma_{WY}, \quad (7.2)$$

$$C = \Sigma_{WY} - \Sigma_{WW}, \quad A = CV^{-1}, \quad R = \Sigma_{WW} - CV^{-1}C^\top. \quad (7.3)$$

The matrices V and R are positive definite. A direct Gaussian regression calculation gives the invertible transformation

$$(Y, W) \mapsto (D, Z), \quad Z = W - AD, \quad D \sim N(f(\theta), V), \quad Z \sim N(\eta - Af(\theta), R), \quad (7.4)$$

where D and Z are independent for fixed (θ, η) .

For a bounded nonnegative Borel loss $\ell(\theta, a) \leq L$, write $\mathcal{V}_M^{\text{pair}}$ for the minimax risk in (7.1), with supremum over (θ, η) , and $\mathcal{V}_M^D$ for the minimax risk from the sole observation $D \sim N(f(\theta), V)$, with the same action space, loss, label budget, and randomization convention.

Theorem 7.1 (Exact minimax elimination of additive calibration). *For every integer $M \geq 1$,*

$$\mathcal{V}_M^{\text{pair}} = \mathcal{V}_M^D. \quad (7.5)$$

Thus correlated measurement noise enters the decision problem through the covariance of the observed difference, not through the two marginal variances considered separately.

Proof. Taking differences implements every D -procedure on the paired experiment without changing its label count, so $\mathcal{V}_M^{\text{pair}} \leq \mathcal{V}_M^D$. For the reverse inequality fix any paired procedure and any $T > 0$. Average its risk over an independent nuisance $\eta \sim N(0, T^2 I_m)$. For each θ , this average is bounded above by its worst-nuisance risk. Under this averaging, (7.4) has product law

$$N(f(\theta), V) \otimes N(-Af(\theta), T^2 I_m + R).$$

Replace the second factor by the parameter-independent law $N(0, T^2 I_m + R)$. The relative entropy between the two second factors is at most $\|Af(\theta)\|_2^2 / (2T^2)$, so their total variation distance is at most $\|Af(\theta)\|_2 / (2T)$. This follows also when A is singular. Since $f(\Theta)$ is compact, the bound is uniform in θ .

A D -encoder can sample this parameter-independent replacement Z using private randomness, reconstruct $W = Z + AD$ and $Y = D + W$, and apply the original paired encoder. Its decoder is unchanged and still sees at most M labels. Data processing and boundedness of the loss imply that its maximum risk is at most the maximum paired risk plus

$$\frac{L}{2T} \sup_{\theta \in \Theta} \|Af(\theta)\|_2.$$

Let $T \rightarrow \infty$ and then take the infimum over paired procedures. This proves the opposite inequality in (7.5). $\square$

This is an equality of minimax decision problems for θ under an unrestricted additive nuisance. It is not a sufficiency assertion for (θ, η) , nor an equivalence of the complete experiments for estimating η . The distinction avoids identifying a nuisance-dependent conditional law with a parameter-independent postprocessing kernel.

7.2. Contact geometry in the noise metric. Let now

$$\Theta = \prod_{j=1}^d [0, a_j], \quad a_j > 0, \quad k_j \in \mathbb{N}, \quad k_j \geq 1, \quad v_j > 0.$$

Assume that, for fixed constants $0 < \lambda \leq \Lambda < \infty$,

$$\lambda \|b(\theta) - b(\theta')\|_2 \leq \|V^{-1/2}(f(\theta) - f(\theta'))\|_2 \leq \Lambda \|b(\theta) - b(\theta')\|_2, \quad b_j(\theta) = \theta_j^{k_j} / v_j. \quad (7.6)$$

This is a two-sided condition on the actual observation map in the noise metric. It is stronger than a rank statement at one point, but it allows nonlinear, nonseparable embeddings of the contact coordinates. No coordinate independence of the statistical experiment is required.

With $a_{(1)} \geq \dots \geq a_{(d)}$ the ordered widths, put

$$S = \sum_{j=1}^d \min\{a_j^2, v_j^{2/k_j}\}, \quad Q_M = \max_{1 \leq l \leq d} \left(\frac{\prod_{i=1}^l a_{(i)}}{M} \right)^{2/l}. \quad (7.7)$$

Write $\mathcal{R}_M$ for the minimax squared Euclidean parameter risk in (7.1), with estimates projected onto Θ . Let ω_l denote the volume of the Euclidean unit ball in $\mathbb{R}^l$.

Theorem 7.2 (Correlated, nonseparable contact and label law). *Under (7.6), for every integer $M \geq 1$,*

$$c_{d,\Lambda}(S + Q_M) \leq \mathcal{R}_M \leq 2K_{m,\lambda}S + 8dQ_M, \quad K_{m,\lambda} = \max\{1, 4m/\lambda^2\}. \quad (7.8)$$

One may take

$$c_{d,\Lambda} = \frac{1}{2} \min \left\{ \frac{3c^2}{32}, \min_{1 \leq l \leq d} \frac{1}{2} (2\omega_l)^{-2/l} \right\}, \quad c = \min\{1/2, 1/(2\Lambda)\}.$$

The comparison constants are independent of the contact orders, widths, noise scales, label budget, and unknown calibration. A deterministic minimum-distance estimate followed by one fixed product codebook attains the upper bound.

Proof. The squared loss is bounded on Θ , so Theorem 7.1 reduces the minimax problem exactly to $D \sim N(f(\theta), V)$.

Estimation. Choose a measurable minimizer

$$\tilde{\theta} \in \operatorname{argmin}_{u \in \Theta} \|V^{-1/2}(D - f(u))\|_2.$$

The minimum exists by compactness. For an explicit Borel selection, subdivide the parameter box dyadically. At each level choose the first closed child of the previously chosen box on which the minimum equals the global minimum. Such a child exists. The minimum over any fixed compact box is continuous in D , so all these finitely many equality tests are Borel. The selected nested boxes have diameters tending to zero; their unique limiting point is a Borel minimizer. For $G = V^{-1/2}(D - f(\theta)) \sim N(0, I_m)$, minimality and the triangle inequality imply

$$\|V^{-1/2}(f(\tilde{\theta}) - f(\theta))\|_2 \leq 2 \|G\|_2.$$

It follows from (7.6) and $|x - y|^{k_j} \leq |x^{k_j} - y^{k_j}|$ for nonnegative x, y that

$$|\tilde{\theta}_j - \theta_j|^2 \leq \min\{a_j^2, (2v_j \|G\|_2 / \lambda)^{2/k_j}\}.$$

Since $\mathbb{E} \|G\|_2^{2/k_j} \leq m^{1/k_j}$, summing gives $\sup_{\theta} \mathbb{E} \|\tilde{\theta} - \theta\|_2^2 \leq K_{m,\lambda} S$.

Compression. Set $\delta = 2\sqrt{Q_M}$ and divide coordinate j into $n_j = \max\{1, \lfloor a_j/\delta \rfloor\}$ equal intervals. Their product has at most M cells: if l widths exceed δ , then $\prod_j n_j \leq \prod_{i \leq l} a_{(i)}/\delta^l \leq M/2^l$, and the case $l = 0$ uses one cell. Each side length is at most 2δ . The midpoint codebook therefore has squared radius at most $4dQ_M$. Quantize θ to it. The squared triangle inequality proves the upper bound in (7.8).

Statistical lower bound. Set $r_j = \min\{a_j, v_j^{1/k_j}\}$ and use the 2^d parameter vertices $\theta_j(\xi) = cr_j \xi_j$, $\xi \in \{0, 1\}^d$. For adjacent vertices differing only in coordinate j , (7.6) bounds their Gaussian relative entropy by

$$\frac{1}{2} \Lambda^2 (cr_j)^{2k_j} / v_j^2 \leq \frac{1}{2} \Lambda^2 c^{2k_j} \leq \frac{1}{8}.$$

Their total variation distance is at most $1/4$. Thresholding an estimate at $cr_j/2$ gives a test of bit j ; an incorrect bit entails squared error at least $c^2 r_j^2 / 4$. Pairing adjacent vertices gives average error probability at least $3/8$, even after compression. Summing over j yields a minimax lower bound $3c^2 S / 32$.

Representation lower bound. Put uniform probability on Θ . For each independent public seed an averaged decoder has at most M centres, whatever noisy observation produced its label. Projection onto the l widest coordinates shows that M balls of radius r cover at most $M\omega_l r^l / \prod_{i \leq l} a_{(i)}$ of the parameter mass. With $r = (\prod_{i \leq l} a_{(i)} / (2M\omega_l))^{1/l}$, at least half the mass has squared error at least r^2 . Maximizing over l and integrating over the seed gives $\mathcal{R}_M \geq [\min_l \frac{1}{2} (2\omega_l)^{-2/l}] Q_M$. Combining this bound with the statistical bound proves the stated lower constant. $\square$

Example 7.3 (Nonseparable contact observations). In two dimensions let

$$b(\theta) = (\theta_1^2/v_1, \theta_2^3/v_2), \quad H(x_1, x_2) = (x_1, x_2 + \alpha \sin x_1), \quad f(\theta) = V^{1/2} H(b(\theta)).$$

Both H and its inverse have Lipschitz constant at most $1 + |\alpha|$. Thus (7.6) holds with $\lambda = (1 + |\alpha|)^{-1}$ and $\Lambda = 1 + |\alpha|$. Theorem 7.2 gives a matched law with statistical term $\min(a_1^2, v_1) + \min(a_2^2, v_2^{2/3})$ and the two-dimensional anisotropic label floor, despite the coupling of the two mean coordinates. For an actual correlated pair with any chosen positive definite V , take independent Gaussian variables $\xi_D \sim N(0, V)$ and $\xi_R \sim N(0, R)$, with $R > 0$, and any matrix A . Set $\varepsilon_W = A\xi_D + \xi_R$ and $\varepsilon_Y = (I + A)\xi_D + \xi_R$. The joint noise covariance is positive definite, and its observed difference has exactly covariance V .

For a scalar channel, correlation coefficient ρ gives $V = \sigma^2 + \tau^2 - 2\rho\sigma\tau$. Ignoring the cross covariance can therefore change the resolution scale. The independent-coordinate experiment,

including its zero-noise endpoints, remains Theorem 10.1. The positive definite hypothesis here is used for the exact Gaussian regression and does not replace those previously proved endpoints.

The observation map and covariance in this section are specified parts of the experiment. The unknown quantity is the additive calibration, not an arbitrary unknown map f or an unknown covariance. The theorem derives the statistical and label rates from the two-sided contact geometry; it does not infer that geometry from a finite derivative rank.

8. ACQUIRED GEOMETRY AND NONUNIFORM CAUSAL RESOLUTION

8.1. The experiment, the metric, and the resource. Let D_t be separable metric spaces with metrics d_t and their Borel sigma fields. A prepared experiment has a known initial predictive state s_0 and jointly Borel updates

$$s_{t+1} = F_t(s_t, a_t, y_{t+1}), \quad F_t(D_t, a, y) \subset D_{t+1}.$$

All actions admitted at a time have a common continuation interface on D_t . The maps must be defined at every representative used below, not only almost surely under the exact experiment. The current input may include a publicly observed mode, but its past realizations are not available to a decoder unless charged to the retained label.

The physical prediction is $z_t = \Pi_t(s_t) \in \mathbb{R}^{q_t}$, with

$$\|\Pi_t(s) - \Pi_t(s')\|_2 \leq K_t d_t(s, s').$$

For a finite query distribution, coordinates are multiplied by the square roots of the query probabilities. The excess squared prediction loss is then exactly $\|z_t - \hat{z}_t\|_2^2$. This follows by conditioning on the full acquisition history; the cross term with the future outcome has conditional mean zero. The query is drawn only after compression.

Fix an exploration law independent of coding randomness and denote the resulting acquired law of z_t by μ_t . A lower bound always refers to this law. An upper bound may hold for a larger family of feedback policies; in that case its stability hypotheses must hold for that family. A parameter-uniform statement requires the same implemented update for all parameters, as in Section F. Alternatively, a Bayesian unknown parameter can be included in the predictive state.

For each t , choose a fixed Borel quantizer $q_t : D_t \rightarrow C_t \subset D_t$, where $|C_t| \leq M_t$ and

$$d_t(s, q_t(s)) \leq r_t \quad (s \in D_t).$$

Finite covers with a first-minimum tie rule give such maps. Let the implemented update $\tilde{F}_t$ take values in D_{t+1} and have uniform error at most η_{t+1} at representatives. The machine is

$$\hat{s}_0 = s_0, \quad \hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (8.1)$$

Only the index of $\hat{s}_t$ persists. A prescribed sequence of arbitrary codebooks is not automatically a finite uniform program. An infinite-time finite-program assertion below uses a fixed codebook, a periodic finite list, or a separately specified finite algorithm generating and decoding the codebooks.

For a fixed visible input word, let $\Phi_{j,t} : D_j \rightarrow D_t$ be the exact composition of updates from time j to time t , and $\Phi_{t,t} = \text{id}$. Assume specified measurable bounds $\Lambda_{j,t} \geq 0$ satisfy

$$d_t(\Phi_{j,t}(x), \Phi_{j,t}(x')) \leq \Lambda_{j,t} d_j(x, x'), \quad \Lambda_{t,t} = 1. \quad (8.2)$$

The bounds can depend on that word. Taking an uncountable supremum and silently declaring it measurable is unnecessary. The hypothesis is on specified measurable bounds valid for every pair of states in the domain.

8.2. A resolution theorem. Write $[x]_+ = \max(x, 0)$. If $\nu \leq \mu_t$ is a subprobability measure of mass b , put

$$A_\nu(r) = \sup_{c \in \mathbb{R}^q} \nu(B(c, r)), \quad \mathcal{L}_t(M) = \sup_{\nu \leq \mu_t, r > 0} r^2 [b - MA_\nu(r)]_+.$$

One may use any collection of specified submeasures in place of the supremum over all of them; no existence of an optimizing submeasure is asserted or required.

Theorem 8.1 (Acquired-law resolution and exact-suffix stability). *Under the preceding hypotheses the following statements hold.*

(i) *Every checkpoint encoder with at most M labels, including independent public and private coding randomization, has expected excess at least $\mathcal{L}_t(M)$. Consequently every one-machine procedure with stage budgets M_t has maximum expected excess at least $\sup_t \mathcal{L}_t(M_t)$.*

(ii) *Along every admitted common input word the machine (8.1) satisfies*

$$d_t(s_t, \hat{s}_t) \leq \sum_{j=1}^t \Lambda_{j,t}(r_j + \eta_j). \quad (8.3)$$

In particular its expected checkpoint excess is at most

$$K_t^2 \left\| \sum_{j=1}^t \Lambda_{j,t}(r_j + \eta_j) \right\|_{L^2}^2 \leq K_t^2 \left(\sum_{j=1}^t \|\Lambda_{j,t}\|_{L^2} (r_j + \eta_j) \right)^2. \quad (8.4)$$

The probability and all the norms in this assertion are those of the actual run of the implemented controller.

(iii) *Suppose the same integer budget M is used at every time. Let $\psi_t(M) > 0$. If fixed constants $a, A, b > 0$ give covers with $r_t + \eta_t \leq A\psi_t(M)$ and acquired submeasures ν_t of mass at least b such that*

$$MA_{\nu_t}(a\psi_t(M)) \leq b/2, \quad \sup_{t,M} \sum_{j=1}^t K_t \|\Lambda_{j,t}\|_{L^2} \frac{\psi_j(M)}{\psi_t(M)} \leq G < \infty,$$

then checkpoint expected risks are bounded below by $ba^2\psi_t(M)^2/2$, and one causal machine has expected excess at most $A^2G^2\psi_t(M)^2$ at each time. Thus the optimal maximum expected checkpoint risk is comparable to $\sup_t \psi_t(M)^2$. Replacing the L^2 bounds by bounds valid for all histories gives the analogous worst-history upper assertion.

Proof. Condition on an independent public seed. Averaging a randomized decoder conditional on its label and that seed cannot increase squared loss. There are at most M prediction centres $c_1, \dots, c_M$. Randomized assignment cannot improve on the distance to the nearest centre. The ν -mass of their r -balls is at most $MA_\nu(r)$; all remaining mass has squared error at least r^2 . This proves (i), first for a fixed seed and then after integration. The seed is independent of the acquired law. Every causal label is a checkpoint encoding of the entire prefix, so the same lower bounds hold before taking the infimum over causal machines.

For (ii), freeze the actual input word and set $w_j = \Phi_{j,t}(\hat{s}_j)$ for $0 \leq j \leq t$. Then $w_0 = s_t$, $w_t = \hat{s}_t$, and

$$w_{j-1} = \Phi_{j,t}(F_{j-1}(\hat{s}_{j-1}, a_{j-1}, y_j)).$$

The two arguments compared here belong to D_j . Their distance is at most $r_j + \eta_j$. Applying (8.2) and summing the t triangle inequalities proves (8.3). No hypothetical controller is reevaluated at a different state: all suffixes use the same actual word. The Lipschitz readout and the triangle inequality in L^2 give (8.4). The finite-sum argument also applies at each time of an infinite run. Substituting the hypotheses of (iii) into (i) and (ii) proves (iii). In particular the conclusion is $\sup_t \mathbb{E} L_t$, not $\mathbb{E} \sup_t L_t$. $\square$

The theorem deliberately separates the upper and lower geometries. A small-dimensional chart can supply a converse without covering the whole domain; conversely, a small cover does not imply that the experiment acquires mass at that scale. Certified perturbations of the model enter η_j only when measured on the stated domain in its stated metric.

Corollary 8.2 (Finite-horizon anisotropic transfer). *Assume the global geometry, acquired rectangular marginals, and causal compatibility hypotheses (G), (A), (C) of the attainable-geometry theorem in [15]. Its all-integer-budget checkpoint profile and its finite-horizon causal conclusion follow from Theorem 8.1. Explicitly, with its notation,*

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left(M^{-1} \prod_{i=1}^{\ell} s_{t,i} \right)^{2/\ell}, \quad d_t(s_t, \hat{s}_t) \leq C \sum_{j=1}^t \left(\prod_{i=j}^{t-1} L_i \right) Q_j(M_j)^{1/2}.$$

The resulting comparison constants are those of the fixed horizon.

Proof. The bounded-format thin-rectangle covering lemma of [15], whose statement and proof are reproduced in the source supplement, gives $r_t(M) \leq C\sqrt{Q_t(M)}$ with representatives in the reachable set. To recall the integer-budget step, its entropy bound is $C \sum_{j=0}^{p_t} r^{-j} \prod_{i \leq j} s_{t,i}$. At $r = L\sqrt{Q_t(M)}$, all terms of positive degree sum to at most $M/2$ for a sufficiently large fixed L . For $M \geq 2C$ the constant term fits the other half; the remaining finitely many budgets are covered by one point and the bound $\sqrt{Q_t(M)} \geq s_{t,1}/M$.

For each acquired rectangular marginal in (A), the mass in a union of M balls of radius r is at most $C_\ell M r^\ell / \prod_{i \leq \ell} s_{t,i}$ of its normalized mass. Choose r a sufficiently small multiple of $(\prod_{i \leq \ell} s_{t,i}/M)^{1/\ell}$. A fixed fraction of the actual subprobability remains. The bounded left inverse of the physical query map and the bounded coordinate transformation transfer this lower bound to prediction error. Maximizing over ℓ gives cQ_t . Zero products need no converse; if every scale vanishes the domain is a singleton. Causal compatibility gives $\Lambda_{j,t} \leq \prod_{i=j}^{t-1} L_i$. Apply (ii), and use the finite number of times for the maximum-risk bound. This reproduces the transfer conclusion; the experiment-specific proofs of (G) and (A) remain those of [15]. $\square$

8.3. Intermittent contraction. Let $\mathcal{H}_t$ contain the visible past through time t . The next criterion is useful when some steps are not contracting at all.

Theorem 8.3 (A conditional block criterion). *Assume each same-input map is nonexpansive. Fix an integer $b \geq 1$ and $0 < p \leq 1$, $0 \leq \rho < 1$. Starting at any deterministic time j , partition the subsequent inputs into consecutive blocks of length b . For each block starting at k , suppose there is an event $E_k \in \mathcal{H}_{k+b}$ with*

$$\Pr(E_k \mid \mathcal{H}_k) \geq p, \quad \text{Lip}(\Phi_{k,k+b}) \leq \rho \text{ on } E_k,$$

and the block is nonexpansive otherwise. These hypotheses refer to the actual policy or hold uniformly over the specified policy family. Set $q = 1 - p(1 - \rho^2)$. Then valid suffix bounds satisfy

$$\|\Lambda_{j,t}\|_{L^2} \leq q^{\lfloor (t-j)/b \rfloor / 2}, \quad \sum_{j=1}^t \|\Lambda_{j,t}\|_{L^2} \leq \frac{b}{1 - \sqrt{q}}.$$

Thus a fixed radius- r codebook, with numerical defect at most η , gives

$$\sup_{t \geq 1} \mathbb{E} d_t(s_t, \hat{s}_t)^2 \leq \frac{b^2(r + \eta)^2}{(1 - \sqrt{q})^2}.$$

No independence between blocks is required. If every block contracts by ρ on every history, the corresponding pathwise bound is $b(r + \eta)/(1 - \rho)$.

Proof. For a complete block use the factor $Z_k = \rho \mathbf{1}_{E_k} + \mathbf{1}_{E_k^c}$. Its conditional second moment is at most q . For the first m complete blocks after j ,

$$\mathbb{E} \prod_{\ell=0}^{m-1} Z_{j+\ell b}^2 \leq q^m.$$

Indeed the product through the penultimate block is measurable at the start of the last, so conditional expectation removes the last factor with multiplier at most q ; induction completes the argument. The unfinished final block is nonexpansive. Products of these factors therefore give the required measurable suffix bounds. Each possible value of the integer part occurs at most b times in the sum over j . Sum the geometric series and apply (8.4) with the identity readout. The deterministic assertion is the same calculation without expectations. $\square$

Arbitrarily long strings of unsuccessful blocks have positive probability in many examples satisfying this theorem. A uniform expected checkpoint bound therefore cannot be reinterpreted as a pathwise bound by suppressing the norm in the hypothesis. Conversely, expansion at individual steps is allowed by Theorem 8.1 whenever the exact suffix bounds are summable; the nonexpansive block criterion is a sufficient condition, not the definition of the general class.

9. POSITIVE AND INTERMITTENT HIDDEN MARKOV EXPERIMENTS

9.1. Projective stability. For positive probability vectors define

$$h(p, q) = \max_i \log(p_i/q_i) - \min_i \log(p_i/q_i).$$

It is a metric on the interior of the simplex. For a positive likelihood vector r , normalization of $r_i p_i$ does not change these log-ratios apart from a common additive constant. Bayes multiplication is consequently an isometry in h .

Lemma 9.1 (Positive-matrix contraction). *For a positive matrix A , put*

$$R(A) = \max_{i,j,k,l} \frac{A_{ik}A_{jl}}{A_{il}A_{jk}}, \quad \vartheta(A) = \frac{\sqrt{R(A)} - 1}{\sqrt{R(A)} + 1}.$$

The normalized map $p \mapsto Ap/(\mathbf{1}^\top Ap)$ is $\vartheta(A)$ -Lipschitz in h . A nonnegative matrix mapping positive vectors to positive vectors is nonexpansive. Also $\|p - q\|_1 \leq h(p, q)$.

Proof. We include the finite-dimensional argument for the classical projective estimate [18]. Write $p_i = e^{x_i} / \sum e^{x_i}$. The derivative of $\log \sum_k A_{ik} e^{x_k}$ is the probability row $w_{ik} = A_{ik} e^{x_k} / \sum_l A_{il} e^{x_l}$. For two rows i, j , the ratio w_{ik}/w_{jk} has maximum divided by minimum at most $R(A)$. If probability vectors u, v have ratios in $[m, M]$, with $m \leq 1 \leq M$, convexity of $|x - 1|$ gives

$$\|u - v\|_{\text{TV}} \leq \frac{(M - 1)(1 - m)}{M - m}.$$

Subject to $M/m \leq R$, the right side is at most $(\sqrt{R} - 1)/(\sqrt{R} + 1)$; increasing the endpoint ratio and then putting $M = Rm$ reduces the maximization to $m = R^{-1/2}$. Consequently the difference of two derivative rows applied to a vector v is at most $\vartheta(A) \text{osc}(v)$ in absolute value. Integrate along the segment joining two log-coordinate vectors and take the maximum difference between rows. Normalization subtracts a common constant and does not change oscillation. This proves the contraction. For nonnegative matrices the same derivative argument uses only $\|w_i - w_j\|_{\text{TV}} \leq 1$.

For the last assertion join q to p by the normalized exponential path with logarithmic increment $v_i = \log(p_i/q_i)$. Its derivative has coordinates $p_i(s)(v_i - \sum_l p_l(s)v_l)$, whose ℓ^1 norm is at most $\text{osc}(v)$. Integrating gives the assertion. $\square$

Use the experiment and physical terminal probe in Section G, now with

$$d \geq 1, \quad 0 < \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (9.1)$$

All known constants and both binary report signs retain their original meaning. Write $n = d + 1$, $\tau = 1/4$, and define

$$\rho_{d,\gamma} = \frac{1 - \gamma}{1 - \gamma + 2\gamma/n}, \quad B_\epsilon = \log \frac{1 + \epsilon}{1 - \epsilon}, \quad R_\epsilon = \frac{B_\epsilon}{1 - \rho_{d,\gamma}}, \quad (9.2)$$

$$D_{d,\gamma} = 2^{-d}(5/4)^{d+2}(n/\gamma)^{d+1} \cosh^{2d}(1). \quad (9.3)$$

Let v_d denote the volume of the Euclidean unit ball in $\mathbb{R}^d$.

Theorem 9.2 (The memory law for every positive refresh rate). *For (9.1), every integer $M \geq 1$, and every $t \geq 1$,*

$$\begin{aligned} c_{d,\gamma}\epsilon^2 M^{-2/d} &\leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_{d,\gamma}\epsilon^2 M^{-2/d}, \\ c_{d,\gamma}\epsilon^2 M^{-2/d} &\leq \mathfrak{R}_{M,\infty}^{\text{av}} \leq \mathfrak{R}_{M,\infty}^{\text{max}} \leq C_{d,\gamma}\epsilon^2 M^{-2/d}. \end{aligned}$$

One possible choice is

$$c_{d,\gamma} = \frac{\tau^2}{2d}(2v_d D_{d,\gamma})^{-2/d}, \quad C_{d,\gamma} = \frac{\tau^2}{d} \left(\frac{64}{3}\right)^2 (1 - \rho_{d,\gamma})^{-4}. \quad (9.4)$$

One deterministic fixed-codebook machine gives the upper bounds for all histories and all times. The lower bounds use independent uniform commands and allow independent coding randomization. At $\epsilon = 0$, one label is exact. If the terminal amplitude is multiplied by ϵ , all bounds acquire the additional factor ϵ^2 .

Proof. Put $a = 1 - \gamma$, $b = \gamma/n$. The transition matrix has diagonal entries $a + b$ and off-diagonal entries b . Its maximum cross-ratio is $((a + b)/b)^2$, attained by two distinct diagonal positions. Lemma 9.1 gives coefficient $a/(a + 2b) = \rho_{d,\gamma}$. Multiplication by the likelihood does not alter h . Since $P_\gamma \nu = \nu$,

$$h(F(p, u, y), \nu) \leq \rho_{d,\gamma} h(p, \nu) + B_\epsilon.$$

Thus $D = \{p : h(p, \nu) \leq R_\epsilon\}$ is an invariant compact subset of the positive simplex. The initialization ν belongs to D .

In log-odds coordinates $x_i = \log(p_i/p_0)$, the domain is exactly $\text{osc}(0, x_1, \dots, x_d) \leq R_\epsilon$ and lies in $[-R_\epsilon, R_\epsilon]^d$. Divide this cube into k^d equal cells, $k = \lfloor M^{1/d} \rfloor$. Select one point of D in each occupied cell. Two points in a cell have projective distance at most $4R_\epsilon/k$, so the fixed codebook has radius $r_M \leq 8R_\epsilon M^{-1/d}$. Its points are admissible normalized beliefs, although they need not be reachable at a particular time. Quantize after each exact update. The suffix bound is $\rho_{d,\gamma}^{t-j}$, whence

$$\sup_t h(p_t, \hat{p}_t) \leq \frac{8B_\epsilon}{(1 - \rho_{d,\gamma})^2} M^{-1/d}.$$

For $0 \leq \epsilon \leq 1/4$, integration of $2/(1 - \epsilon^2)$ gives $B_\epsilon \leq (8/3)\epsilon$. The physical loss is $(\tau^2/d) \|p_{t,+} - \hat{p}_{t,+}\|_2^2$ and is at most $(\tau^2/d) h(p_t, \hat{p}_t)^2$. This proves the upper constant.

For the converse the pre-report coordinates satisfy $q_i \geq \gamma/n$. The command-to-posterior Jacobian in Lemma G.4, with this floor in place of m_* , gives a density bounded by $D_{d,\gamma}\epsilon^{-d}$ for $p_{t,+}$. For clarity, the actual joint density of command and sign is $2^{-(d+1)} D_y$ and

$$|\det D_u F(p, u, y)_+| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \text{sech}^2 u_i.$$

The odds $p'_i/p'_0 = q_i(1 + y\epsilon \tanh u_i)/q_0$ make each sign's map injective. Changing variables, summing the two signs, and integrating over every previous history gives exactly (9.3); no conditioning

cost is removed. The mass of M Euclidean balls of radius r is at most $Mv_d D_{d,\gamma} \epsilon^{-d} r^d$. Take $r = \epsilon(2Mv_d D_{d,\gamma})^{-1/d}$. At least half the mass remains, and the physical factor is τ^2/d . This proves the lower constant and, by checkpoint relaxation, its causal version. At zero signal the posterior stays ν . Scaling the terminal probe scales its squared loss, proving the last assertion. $\square$

The dependence on mixing and dimension in (9.4) is part of the statement. Since $1 - \rho_{d,\gamma} = 2\gamma/[n(1 - \gamma) + 2\gamma]$, the upper constant is explicit even for weak refresh. The theorem is uniform in M, t and ϵ for each positive γ ; it does not assert matching powers of γ in a simultaneous $\gamma \downarrow 0$ limit. This is a strict enlargement of the refresh interval, not a relabeling of the old ℓ^1 estimate. At zero refresh the finite-horizon theorem remains available whenever its separate geometry and stability hypotheses hold.

9.2. An experiment with unbounded noncontracting intervals. At each time first draw an observed mode I_t , independently of the hidden state and of all past variables, with $\Pr(I_t = 1) = p > 0$. For $I_t = 1$ execute the preceding refresh and commanded observation. For $I_t = 0$ apply any prescribed permutation of the hidden states and return an uninformative fair sign. Commands are independent uniform for the converse. The upper construction accepts every admitted command word. The mode is read in the current update, then discarded except for its effect on the retained label. The terminal probe is unchanged.

Corollary 9.3 (Intermittent-refresh expected resolution). *In this experiment, for every integer $M \geq 1$,*

$$pc_{d,\gamma} \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,\infty}^{\text{av}} \leq \frac{\tau^2}{d} \frac{(64/3)^2 \epsilon^2 M^{-2/d}}{(1 - \rho_{d,\gamma})^2 (1 - \sqrt{1 - p(1 - \rho_{d,\gamma}^2)})^2}.$$

The same lower bound holds at every positive checkpoint. The upper bound uses one fixed codebook. For $0 < p < 1$, arbitrarily long noncontracting intervals are admitted.

Proof. Permutations fix ν and preserve h ; hence the invariant domain D and its cover from the preceding proof work in both modes. A good step contracts by $\rho_{d,\gamma}$, a bad step is an isometry. Theorem 8.3 with $b = 1$ gives the upper bound. Conditional on the current good mode, the density calculation in the preceding proof is uniform over the prior history. The good-mode submeasure has mass p and density at most $pD_{d,\gamma} \epsilon^{-d}$. The same balls leave at least $p/2$ of its mass, proving the lower bound. Independent Bernoulli modes permit every finite string of bad steps. $\square$

9.3. Discrete and singular acquisition. For a finite command alphabet of size K , a fixed time t has at most $(2K)^t$ command/sign histories in the ordinary refreshing experiment. Its checkpoint prediction is therefore exactly encodable at that budget. This elementary fact precludes a positive $M^{-2/d}$ lower bound for all M at fixed t in that different experiment.

A singular command law has a different outcome. Let $d = 1$, $\gamma = 1$, and let $u = 2V - 1$, where V is the standard middle-thirds Cantor random variable. Set $s = \log 2 / \log 3$. The hidden state is freshly prepared at each step, but the binary command observation and the terminal probe are still physical observations as above.

Proposition 9.4 (A singular acquired dimension). *For this experiment there are constants $0 < c < C < \infty$ such that for every $t \geq 1$, $M \geq 1$, and $0 \leq \epsilon \leq 1/4$,*

$$c\epsilon^2 M^{-2/s} \leq R_{M,t}^{\text{av}} \leq \mathfrak{R}_{M,\infty}^{\text{max}} \leq C\epsilon^2 M^{-2/s}.$$

Here worst histories use commands in the support of the prescribed Cantor law. One fixed codebook attains the upper bound.

Proof. A level- m Cantor cylinder has length 3^{-m} and mass 2^{-m} . At that scale an interval of length comparable to 3^{-m} meets only a bounded number of cylinders. Thus the mass of an interval

of radius r is at most Cr^s , while the support has a cover by 2^m intervals of length 3^{-m} . Affine rescaling to $[-1, 1]$ changes only constants.

For either sign the posterior coordinate is $(1 + y\epsilon \tanh u)/(2 + y\epsilon \tanh u)$. Its derivative in absolute value is $\epsilon \operatorname{sech}^2 u/(2 + y\epsilon \tanh u)^2$, between two fixed positive multiples of ϵ when $\epsilon > 0$. Each sign therefore gives a bi-Lipschitz image at scale ϵ . The actual sign weight is between $3/8$ and $5/8$. The sum of the two image measures has concentration at most $C(r/\epsilon)^s$; each image has the corresponding covers. Allocate half the available centres to each image, adjusting the fixed constant for the finitely many budgets below four. The acquired-law lower bound and this cover give the asserted rate. Since $\gamma = 1$, the new posterior does not depend on the old label. The same quantizer is an exact one-step causal construction at every time. At $\epsilon = 0$ it is constant. $\square$

Neither the domain dimension of the command nor a Jacobian rank alone determines these three rates. For a mixture with finitely many additional atoms, assign one centre to each atom and use the remaining centres for the singular component; the submeasure converse retains its actual mixture weight. More generally Theorem 8.1 uses the full concentration profile without assigning a single dimension.

10. SINGULAR STATISTICAL RESOLUTION WITH MEASURED CALIBRATION

We now change the target and the loss explicitly. Let $\theta \in \Theta = \prod_{j=1}^d [0, a_j]$, $a_j > 0$, and let $\eta \in \mathbb{R}^d$ be an unrestricted unknown additive calibration. Fix integers $k_j \geq 1$. The experiment consists of the $2d$ observations

$$Y_j = \theta_j^{k_j} + \eta_j + \sigma_j Z_j, \quad W_j = \eta_j + \tau_j Z'_j, \quad 1 \leq j \leq d, \quad (10.1)$$

with independent standard normal variables and known $\sigma_j, \tau_j \geq 0$. A procedure observes the pair, retains one of M labels, and decodes an estimate of θ . Only the label and independent public coding randomness are available to the decoder. Both Y and W count as acquisitions. In particular W is not a free read-only value of η .

Put $v_j = (\sigma_j^2 + \tau_j^2)^{1/2}$, and let $a_{(1)} \geq \dots \geq a_{(d)}$ be the ordered widths. Define

$$S = \sum_{j=1}^d \min\{a_j^2, v_j^{2/k_j}\}, \quad Q_M = \max_{1 \leq \ell \leq d} \left(M^{-1} \prod_{i=1}^{\ell} a_{(i)} \right)^{2/\ell}. \quad (10.2)$$

The risk here is squared parameter error, not the prediction excess used in the preceding sections.

Theorem 10.1 (A matched contact, calibration, and label law). *For the actual experiment (10.1), write*

$$\mathcal{R}_M = \inf_{M\text{-label procedures}} \sup_{\theta \in \Theta, \eta \in \mathbb{R}^d} \mathbb{E}_{\theta, \eta} \|\hat{\theta} - \theta\|_2^2.$$

For every integer $M \geq 1$,

$$c_d(S + Q_M) \leq \mathcal{R}_M \leq 2S + 8dQ_M, \quad (10.3)$$

where one may take

$$c_d = \frac{1}{2} \min \left\{ \frac{3}{128}, \min_{1 \leq \ell \leq d} \frac{1}{2} (2\omega_\ell)^{-2/\ell} \right\},$$

where ω_ℓ denotes the volume of the Euclidean unit ball in $\mathbb{R}^\ell$. The constants are independent of $a_j, k_j, \sigma_j, \tau_j, M$ and the unknown calibration. The upper construction is deterministic. The lower bound allows independent public and private randomization.

Proof. Estimation before compression. The actually observed differences $D_j = Y_j - W_j$ have laws $N(\theta_j^{k_j}, v_j^2)$ independent of η . Define

$$\tilde{\theta}_j = (\operatorname{proj}_{[0, a_j^{k_j}]} D_j)^{1/k_j}.$$

For nonnegative x, y , $|x^{1/k} - y^{1/k}| \leq |x - y|^{1/k}$. Projection cannot increase the distance to $\theta_j^{k_j}$. Concavity of x^{1/k_j} and $\mathbb{E}Z^2 = 1$ consequently give

$$\mathbb{E}|\tilde{\theta}_j - \theta_j|^2 \leq \min\{a_j^2, v_j^{2/k_j} \mathbb{E}|Z|^{2/k_j}\} \leq \min\{a_j^2, v_j^{2/k_j}\}.$$

This is a finite-sample bound, including degenerate zero-noise coordinates.

An integer-budget cover. Set $\delta = 2\sqrt{Q_M}$ and partition coordinate j into $n_j = \max\{1, \lfloor a_j/\delta \rfloor\}$ equal intervals. Let ℓ be the number of widths greater than δ . If $\ell = 0$ the product of the n_j is one. Otherwise

$$\prod_j n_j \leq \prod_{i \leq \ell} a_{(i)}/\delta^\ell \leq M/2^\ell \leq M.$$

Every interval has length at most 2δ , so its midpoint product codebook has squared covering radius at most $d\delta^2 = 4dQ_M$. Quantize $\tilde{\theta}$ to this codebook. The squared triangle inequality proves the upper bound in (10.3).

The statistical obstruction with unknown calibration. Write $r_j = \min\{a_j, v_j^{1/k_j}\}$. Coordinates with $r_j = 0$ contribute zero and can be omitted from this step. For $\omega \in \{0, 1\}^d$, set

$$\theta_j(\omega) = \frac{1}{2}r_j\omega_j, \quad m_j = (r_j/2)^{k_j}, \quad \eta_j(\omega) = -\frac{\tau_j^2}{v_j^2}m_j\omega_j.$$

These are admissible alternatives in the same unknown-calibration experiment. Adjacent vertices differ in the mean pair at coordinate j by $(m_j\sigma_j^2/v_j^2, -m_j\tau_j^2/v_j^2)$. Their Gaussian divergence is

$$\frac{1}{2} \left(\frac{m_j^2\sigma_j^2}{v_j^4} + \frac{m_j^2\tau_j^2}{v_j^4} \right) = \frac{m_j^2}{2v_j^2} \leq \frac{1}{8}.$$

A zero-variance channel in this expression has identical means and is simply omitted. Pinsker's inequality bounds the variation distance by $1/4$. Any estimate determines a test of bit j by thresholding at $r_j/4$. An incorrect test has squared coordinate error at least $r_j^2/16$. Pair adjacent vertices and average: their average error probability is at least $(1 - 1/4)/2 = 3/8$. Summing over coordinates gives average squared risk at least $3 \sum_j r_j^2/128$. Compression and independent randomization cannot improve the testing distance. Since maximum risk dominates this finite average, $\mathcal{R}_M \geq 3S/128$.

The representation obstruction. Fix $\eta = 0$ and put uniform probability on the parameter box. At each fixed public seed an averaged decoder has at most M centres. Regardless of what noisy observations led to the label, its error at θ is at least the distance to the nearest centre. Project onto the ℓ widest coordinates. The fraction of this rectangular uniform law covered by M balls of radius r is at most $M\omega_\ell r^\ell / \prod_{i \leq \ell} a_{(i)}$. Choose $r = (\prod_{i \leq \ell} a_{(i)} / (2M\omega_\ell))^{1/\ell}$. At least half the mass has squared error at least r^2 . Maximize over ℓ and integrate over the seed. This proves $\mathcal{R}_M \geq [\min_\ell \frac{1}{2}(2\omega_\ell)^{-2/\ell}]Q_M$. Combining the two lower bounds by $\max(x, y) \geq (x + y)/2$ proves the result. $\square$

The testing alternatives explain why the calibration variance adds to the signal variance. Holding the nuisance artificially fixed in that step would prove only the weaker obstruction from σ_j , and would miss the cost of measuring calibration. The encoder never reads the unknown nuisance. Its read-only constants are the specified noise levels, contact orders, and parameter box.

For $d = 1$, $a_1 = 1$, and $v_1 \asymp N^{-1/2}$, the theorem gives $\mathcal{R}_M \asymp N^{-1/k_1} + M^{-2}$. Hence the statistical amplitude scale is $N^{-1/(2k_1)}$, and order $N^{1/(2k_1)}$ labels suffice to reach it. When $k_1 = 2$, this recovers the quadratic-contact amplitude scale in the physical-ray result of [16], now with an explicit finite-label constraint and an actually measured unrestricted additive calibration in (10.1). It is a comparison of rates in these specified experiments, not an assertion of their Le Cam

equivalence. The nonnegative branch is part of the parameter space: an even-power observation on a signed branch would have an additional identifiability obstruction.

11. OPTIMIZATION UNDER A RETAINED-LABEL CONSTRAINT

The following result supplies a policy comparison in addition to the common-controller statements of Section F. Let D be a compact metric predictive-state domain and $\mathcal{A}$ a finite action set. The report kernel is $g(dy | s, a)$ and the state update is $F(s, a, y) \in D$. Assume these are Borel, and, uniformly in a, y ,

$$0 \leq c(s, a) \leq C_0, \quad |c(s, a) - c(s', a)| \leq L_c d(s, s'), \quad d(F(s, a, y), F(s', a, y)) \leq l d(s, s'),$$

$$\|g(\cdot | s, a) - g(\cdot | s', a)\|_{\text{TV}} \leq \kappa d(s, s').$$

The current conditional expected physical cost is $c(s, a)$. The predictive state is sufficient for future costs and experiments under all admitted policies. Thus the belief-state dynamic program represents the optimum among all history-dependent policies. Fix discount $0 < \beta < 1$ with $\beta l < 1$, and define

$$L_* = \frac{L_c + \beta \kappa C_0 / (1 - \beta)}{1 - \beta l}.$$

Let $\mathcal{C} \subset D$ have at most M points and a radius- r Borel quantizer q . Assume the initial state is in $\mathcal{C}$; it can be reserved as a centre. The finite reference decision process uses cost $c(c, a)$ and next state $qF(c, a, y)$ with report distribution $g(dy | c, a)$. It has an optimal stationary policy $\hat{a}(c)$ because its state and action spaces are finite. Its transition probabilities are the specified integrals of these kernels; numerical approximation is addressed below.

Theorem 11.1 (A finite reference policy versus unrestricted control). *Implement $a_t = \hat{a}(\hat{s}_t)$ in the actual experiment and update $\hat{s}_{t+1} = qF(\hat{s}_t, a_t, y_{t+1})$. Let V^* be the unrestricted optimal discounted value and $J(\hat{a})$ the actual cost of this retained-label controller. Then*

$$0 \leq J(\hat{a}) - V^*(s_0) \leq 2L_* \sum_{t \geq 0} \beta^t \mathbb{E} d(s_t, \hat{s}_t) + \frac{2\beta L_* r}{(1 - \beta)^2}. \quad (11.1)$$

In particular, if the fixed-codebook construction has $\sup_t \mathbb{E} d(s_t, \hat{s}_t) \leq Gr$, the upper bound is $2L_ Gr / (1 - \beta) + 2\beta L_* r / (1 - \beta)^2$. Only one of M labels persists; no exact value function or full history is an input to the implemented policy.*

Proof. The exact Bellman operator on bounded Borel functions is

$$TV(s) = \min_a \{c(s, a) + \beta \int V(F(s, a, y)) g(dy | s, a)\}.$$

It is a contraction in the uniform norm. Starting at zero, its iterates have values in $[0, C_0 / (1 - \beta)]$. If an iterate has Lipschitz constant L , the next has constant at most $L_c + \beta l L + \beta \kappa C_0 / (1 - \beta)$: compare the same report argument first and then the two report laws, using the range bound on V . The minimum over the same finite action set preserves the bound. It follows that V^* is L_* -Lipschitz, and each optimal action-value function $Q^*(\cdot, a)$ is also L_* -Lipschitz.

Let $\hat{V}$ be the finite reference value. At a centre, replacing $F(c, a, y)$ by $qF(c, a, y)$ in V^* changes a Bellman expression by at most $\beta L_* r$. The finite Bellman contraction therefore implies

$$\sup_{c \in \mathcal{C}} |\hat{V}(c) - V^*(c)| \leq \frac{\beta L_* r}{1 - \beta} =: D_0.$$

For every centre and action, the exact action value and the finite reference action value differ by at most $\beta(L_* r + D_0) = \beta L_* r / (1 - \beta) = D_0$. Minimizing the latter consequently yields

$$Q^*(c, \hat{a}(c)) - V^*(c) \leq 2D_0.$$

For an actual state s and retained centre c , Lipschitz continuity of both Q^* and V^* gives $Q^*(s, \hat{a}(c)) - V^*(s) \leq 2L_*d(s, c) + 2D_0$.

Along the actual trajectory, the conditional definition of Q^* makes the discounted sum of these residuals equal to $J(\hat{a}) - V^*(s_0)$. More explicitly the finite sum telescopes, and its final term has absolute value at most $\beta^{T+1}C_0/(1-\beta)$, tending to zero. Sum the bound and let T increase. This proves (11.1). The policy is a finite table indexed by the retained representative. The exact value entered only this comparison proof. $\square$

A stationary centre may be reserved at a cost of one label: for $M \geq 2$ use the preceding covers with $M-1$ centres and add s_0 . A power-law covering rate is unchanged up to its stated fixed factor; for $M=1$ the bounded-cost estimate is available. If an implemented finite table is within ξ in each action value of the exact reference table, choosing its minimizing action adds at most $2\xi/(1-\beta)$ to the bound. This numerical certificate is separate from computing that table. Theorems 8.1 and 8.3 provide G when their assumptions hold under the lifted policy. A prediction lower bound is not a matching lower bound for every control cost; no such identification is used here.

12. FINITE DESCRIPTIONS AND NUMERICAL IMPLEMENTATION

We distinguish five resource coordinates: persistent labels, program description, transient workspace, numerical error, and physical acquisitions. The first coordinate is determined by the resolution laws; the others are not silently identified with it. We give a finite-description construction for the positive-refresh upper bound, so its codebook does not require an independently stored list of M arbitrary real vectors.

Suppose γ, ϵ are rational known inputs and $\epsilon > 0$. Choose a rational R' with $R_\epsilon \leq R' \leq 2R_\epsilon$. Such a number is computable from the displayed elementary formula. For an integer $k \geq 1$, set $\delta = R'/k$ and use the log-odds grid

$$\mathcal{J}_k = \{j \in \{-k, \dots, k\}^d : \text{osc}(0, j_1, \dots, j_d) \leq k\}, \quad p_i(j) = \frac{e^{\delta j_i}}{1 + \sum_{l=1}^d e^{\delta j_l}}, \quad (12.1)$$

with p_0 the reciprocal denominator. Its domain is the invariant ball of radius R' .

Proposition 12.1 (An implicit codebook). *There is a finite-description codebook in this ball with at most M representatives and projective radius at most $12R'M^{-1/d}$. Its persistent state uses at most $\lceil \log_2 M \rceil$ bits. Its read-only description consists of d , the rational calibration, R' , an integer k , and fixed algorithms for enumeration and elementary functions; it contains no list of M arbitrary real numbers. Certified update error η and quantization error at most $r + \eta$ give projective filtering error at most $(r + 2\eta)/(1 - \rho_{d,\gamma})$.*

Proof. For $M \geq 3^d$ choose $k = \lfloor (M^{1/d} - 1)/2 \rfloor$. Then $(2k+1)^d \leq M$ and $k \geq M^{1/d}/6$. Round each log-odds coordinate toward zero in multiples of δ . The result remains in the oscillation ball: when both signs occur, both extreme absolute values decrease; when all coordinates have one sign, their extreme absolute value decreases. Each coordinate changes by at most δ , so projective distance changes by at most $2\delta \leq 12R'M^{-1/d}$. For $M < 3^d$ use only ν ; its radius R' is at most $3R'M^{-1/d}$. Lexicographic enumeration of the finite integer set labels the grid. The model and this enumeration are finite descriptions of all its representatives. The stated error is the geometric-series estimate, with total one-step defect $r + 2\eta$. $\square$

All log-odds and softmax functions in this construction are computable on their compact domains. With digitally represented commands, interval arithmetic can evaluate the update and finitely many distances until a representative within $r + \eta$ is certified, for any prescribed positive rational η . The fixed positive slack ensures termination even at a tie; exact comparison of two computable reals is not required. This is an existence and accuracy construction, not a uniform

polynomial-time claim. The bound charges the counter needed to enumerate the grid and the precision of the computed numbers as transient workspace. The required precision can depend on $d, \gamma, \epsilon, M, \eta$.

A command rounded with coordinate error at most h changes each log likelihood by at most $\epsilon h/(1-\epsilon)$ and hence changes the posterior projective distance by at most $2\epsilon h/(1-\epsilon)$. This supplies an explicit additional one-step defect. It does not replace the analog acquisition law in the lower bound by a finite alphabet while keeping the same converse. The sampling law and the numerical interface must be declared separately. For a noncomputable real calibration, an exact read-only real convention would be a different resource model. The finite-description assertion above uses rational calibration and a numerical certificate instead.

The paired singular experiment uses $2d$ scalar measurements and one terminal label. Its box codebook is generated by d integer subdivision counts rather than an arbitrary centre list. Computing the differences and roots uses transient numerical workspace. A controller in Theorem 11.1 additionally has a read-only policy table; its construction and precision are not included in the label count.

13. COMPARISON OF RESULTS AND MATHEMATICAL DEPENDENCIES

The relationship to earlier work is most precise at the level of the conclusion and the retained information.

Antecedent	Existing conclusion used for comparison	Conclusion proved here
[15], attainable-geometry transfer	Anisotropic acquired-law profile and a compatible finite-horizon label machine	Exact-suffix bounds and intermittent infinite-horizon resolution; the old transfer is Corollary 8.2
[8]	Quantization methods for nonlinear filtering	A matching converse for acquired prediction labels, not merely a finite signal grid
[19, Theorems 12, 16, 17]	Finite-memory policy approximation from filter stability	A policy computed on a finite representative model with an explicit retained-label budget and actual-state error term
[20, Theorems 9, 27]	Approximate information states and approximate dynamic programs with performance bounds	Theorem 11.1 is a specialized quantitative policy-transfer result, used together with the acquired-law converse
[16], physical-ray and score experiments	Quadratic-contact scale, Gaussian comparison, information retained by contact scores	Finite-sample contact orders, unrestricted measured additive calibration, and a matched label floor in one specified experiment

Recent finite-window developments include continuous-space approximations [21] and refined window-policy bounds [22]. The fixed-policy belief-cost comparison in [23] is another nearby, but distinct, target. None is replaced by a claim that the stability recurrence, quantization principle, or Bellman comparison was invented here. The specific mathematical assertions to compare are the joint hypotheses and conclusions of Theorems 8.1, 8.3, 9.2, and 10.1.

There is no backward dependence on a later theta manuscript. The only model-specific imported geometric estimate used in a corollary is the bounded-format covering result of the fixed A1 edition; the new principal theorems have proofs here. The A2 comparisons keep its physical cone and noise experiment distinct from (10.1). The earlier Sinai/vector-roof, particle large-deviation, kinetic-semigroup, unbounded operator, and labelled-phase problems keep their original dependency

order. A proof of finite-label filtering does not assert any of those model-specific conclusions. The contribution to those chains is a proved resolution and decision interface, usable after their own experiment and domain hypotheses have been established.

FOUNDATIONAL DEVELOPMENT AND BOUNDARY EXAMPLES

The following appendices give the measurable, statistical, and asymptotic foundations used above. Their statements and proofs are retained from the first edition. The strong-refresh result in Appendix G is extended by Theorem 9.2; the additional statistical results are Theorems 7.1 and 7.2. The final interface appendix distinguishes these results from separate model-specific spectral, large-deviation, and operator problems.

APPENDIX A. POSITIVE CAUSAL EXPERIMENTS

A.1. Types and primitive data. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. All primitive spaces below are standard Borel. Time is initially $t = 0, \dots, N$, with the countable-time extension stated separately. The unknown parameter is $\lambda \in \Lambda$; it is not a random variable until a prior is supplied. The hidden state, applied action, and next report have spaces X_t, A_t, Y_{t+1} . A visible history is

$$h_t = (a_0, y_1, \dots, a_{t-1}, y_t) \in H_t.$$

A public initial report, clock increments, and observed cost records can be included in this notation. An action-availability rule, when present, is part of the interface. We usually write a common action space to avoid encoding that rule repeatedly.

Definition A.1 (Raw experiment). A raw experiment consists of a parameter space Λ , preparations $\mu_\lambda \in \mathcal{P}(X_0)$, and jointly measurable probability kernels

$$K_t^\lambda(dx', dy, dc \mid x, a) \tag{A.1}$$

on $X_{t+1} \times Y_{t+1} \times \mathcal{C}$, together with a class of executable future tests and a resource interface. The nonnegative vector c records charged increments, such as raw preparations or physical time. When it is observed, it is included in y ; when it is not observed, it is integrated in the visible law but is still counted in the resource account. Every kernel has total mass one. A failed measurement and a terminating measurement are reports, not missing mass.

The following six requirements will be referred to as the foundational axioms. They specify the admissible language rather than analytic conclusions.

- A0 (measurable interfaces).** Primitive spaces are standard Borel; histories and continuation availability are specified.
- A1 (positive experiment).** Preparation and transition–observation kernels are nonnegative and normalized as in (A.1).
- A2 (causality).** The next action uses only the visible past and permitted independent randomness. Stopping is nonanticipative.
- A3 (executable tests).** The designated future tests have actual protocols and are closed under the one-step extensions needed for prediction.
- A4 (preparation and accounting).** Average risks and acquisition laws refer to a declared preparation and exploration strategy. Failures, pilots, and accepted-event probabilities are not removed from their costs.
- A5 (charged implementation).** Persistent labels, numerical precision, workspace, read-only calibration, and simulator state have separate accounts.

In particular these axioms do not assert strict positivity of each report, finite-dimensionality, stationarity, differentiability, a spectral gap, or a large-deviation principle. If the physical state is not Markov in the coordinates initially proposed, its entire necessary past can be included in X_t . This does not prove closure for a smaller observable state.

A.2. Strategies, paths, and stopping. A strategy is a sequence of measurable kernels $\alpha_t(da | h_t)$ supported on available actions. It cannot depend on the unknown true parameter. A finite-state strategy may have an additional explicit internal state; an unrestricted history strategy need not satisfy a finite-memory budget.

Proposition A.2 (Path construction). *For every parameter and causal strategy there is a unique consistent finite-dimensional path law with iterated-kernel expression*

$$\mu_\lambda(dx_0) \prod_{t=0}^{N-1} \alpha_t(da_t | h_t) K_t^\lambda(dx_{t+1}, dy_{t+1}, dc_{t+1} | x_t, a_t). \quad (\text{A.2})$$

For countably many discrete times these laws extend uniquely to the product path sigma field. Deterministic evolution and deterministic observations are included by taking Dirac kernels.

Proof. For finite N , integrate bounded measurable cylinder functions from the last variable backwards. At each step measurability follows from the kernel property, positivity is preserved, and the integral of one is one. The resulting probability measures have consistent marginals because the last kernel integrates to one. The standard kernel-extension theorem gives the countable product law; this is the Ionescu–Tulcea construction, as treated in [4]. Uniqueness follows first on cylinder events and then on their generated sigma field. A Dirac kernel makes the corresponding integral substitution along the deterministic trajectory, without changing the remaining preparation or measurement randomness. $\square$

A bounded stopping time is implemented by an absorbing stopped symbol and zero subsequent increments. Thus the same construction handles optional termination. For an unbounded stopping time, an almost-surely finite stopped path can be considered by truncation; quantitative infinite-horizon error bounds still require a summable bound or a separate limiting argument. A random number of collisions is not silently substituted for a deterministic large-deviation speed.

A.3. Prepared and parameter-uniform questions. For statistical decision theory we retain the whole family $(P_\lambda^\alpha)_\lambda$. A decision rule is one rule, independent of the true parameter. For Bayesian prediction we instead specify a prior $\eta(d\lambda)$ and use the joint hidden state (λ, x_t) with preparation $\eta(d\lambda)\mu_\lambda(dx_0)$. Fixed known calibration is permitted, but it is not a substitute for an unknown parameter.

This distinction also applies to infima over encoders. The infimum for a fixed preparation need not be achieved by the same encoder for all priors. Likewise, a family of codebooks indexed by a known experimental calibration is not an estimator of an unknown calibration. Every quantitative theorem below states which situation is intended.

A.4. Instruments and evidence. After a preparation has been fixed, write the hidden state simply as x . For a report event $B \subset Y_{t+1}$, define the positive instrument

$$(\mathbb{T}_{t,a,B}f)(x) = \int \mathbf{1}_B(y)f(x')K_t(dx', dy | x, a).$$

It is countably additive in B for nonnegative f , positive in f , and $\mathbb{T}_{t,a,Y}1 = 1$. With finite reports write $\mathbb{T}_{t,a,y} = \mathbb{T}_{t,a,\{y\}}$. Starting with $b_0 = \mu$, define the unnormalized history functional by

$$b_{t+1}(f) = b_t(\mathbb{T}_{t,a_t,y_{t+1}}f), \quad Z_t = b_t(1). \quad (\text{A.3})$$

These functionals do not include the probability of the chosen command. They are likelihood/evidence functionals conditional on the realized command sequence; the actual joint history law also includes the strategy kernels in (A.2).

Proposition A.3 (Normalized update). *For finite reports and $Z_t > 0$, let $\pi_t = b_t/Z_t$. If $q_t(y \mid \pi_t, a) = \pi_t(\mathbb{T}_{t,a,y}1) > 0$, then*

$$\pi_{t+1}(f) = \frac{\pi_t(\mathbb{T}_{t,a,y}f)}{\pi_t(\mathbb{T}_{t,a,y}1)}, \quad Z_{t+1} = Z_t q_t(y \mid \pi_t, a). \quad (\text{A.4})$$

The formula remains valid for adaptive actions. When the evidence is zero, the unnormalized functional is zero and no normalized posterior is specified by this formula.

Proof. Divide (A.3) by its value at one. Positivity shows that a positive functional of total mass zero vanishes on every bounded test. For adaptive actions, condition on the visible history: the strategy's probability of the current action is fixed by that history and is independent of the hidden state conditional on it. This factor cancels in the posterior. An induction in (A.2) verifies the conditional expectation identity for every bounded hidden-state test. $\square$

For a non-atomic report space, conditioning is by disintegration of the joint next-state/report law, not by dividing the mass of a singleton. Under a fixed preparation a standard Borel conditional distribution exists almost everywhere. A single Borel update jointly in all proposed states, inputs, and reports is additional implementation data, or must be justified by an applicable parametrized disintegration theorem. We will not use a simultaneous version across a nondominated parameter family without that justification. All explicit quantitative models below have finite reports and Borel formulas.

A.5. Preparing a conditional experiment. If an event has acceptance probability $p > 0$, independently repeating a preparation until acceptance implements the conditional law. Its preparation count T satisfies $\mathbb{P}(T > n) = (1-p)^n$ and $\mathbb{E}T = 1/p$. A cap B instead creates an abort symbol with probability $(1-p)^B$. This is a different experiment from exact conditioning. The cap is necessary for a uniform pathwise preparation budget. No lower bound or implementation claim below replaces an actual report word by a probability-one conditioning event.

APPENDIX B. FUTURE TESTS AND PREDICTIVE QUOTIENTS

B.1. Operator words and adaptive protocols. A finite future protocol starts from the present time, uses available actions causally, and returns a bounded terminal readout. Its conditional expected readout, as a function of the current hidden state, is a future test. A terminal hidden-state probe is allowed only if an actual measurement kernel producing it is part of the protocol. For finite report alphabets, a deterministic depth- m protocol is a tree of actions and terminal readouts. Its test has the recursion

$$f = \sum_y \mathbb{T}_{t,a,y} f_y. \quad (\text{B.1})$$

Randomized protocols integrate such recursions. Let $\mathcal{V}_t$ be the real linear space generated by the designated future tests, including constants. We require the one-step closure

$$\mathbb{T}_{t,a,y} \mathcal{V}_{t+1} \subset \mathcal{V}_t \quad (\text{B.2})$$

for the continuations being retained. Where full future laws are intended, the tests include indicator readouts from a determining event class and its needed bounded-monotone closure.

A fixed finite menu used to assess prediction loss can be much smaller than $\mathcal{V}_t$. Such a menu does not automatically define a recursively sufficient state. One must either prove its closure/completeness or keep a larger predictive state for the online construction.

Definition B.1 (Predictive equivalence). Two prepared beliefs at time t are predictively equivalent when they have the same continuation interface and

$$\pi \equiv_t \tilde{\pi} \iff \pi(f) = \tilde{\pi}(f) \quad (f \in \mathcal{V}_t).$$

The continuation interface includes the legal future action tree, remaining budget, and any public clock that affects future tests. The predictive quotient is the set of these equivalence classes, or equivalently the image of $\pi \mapsto (\pi(f))_{f \in \mathcal{V}_t}$ together with that interface.

Theorem B.2 (Minimal future prediction). *Under (B.2), predictively equivalent beliefs give the same laws for all designated future protocols. After an action and a report of positive probability, their updated beliefs remain predictively equivalent. Conversely, equality of those protocol laws implies equality of their future tests. If a deterministic history statistic r_t determines all these predictions and the continuation interface, the predictive state factors uniquely through r_t on its attained image.*

Proof. For a one-step report, its probability is the value of $\mathbf{T}_{t,a,y}1 \in \mathcal{V}_t$, and is therefore common. For a next-state test f , both the numerator and denominator of (A.4) have the same values under equivalent beliefs. The conditional next predictions are equal. Induct over the depth of a protocol tree using (B.1). Randomized action choices use the same conditional randomization at corresponding visible histories and can be integrated. Bounded monotone convergence extends equality from finite determining tests to the designated generated events. Conversely, expectations of each designated bounded readout are determined by its law, hence agree; equality extends linearly to $\mathcal{V}_t$.

If $r_t(h) = r_t(\tilde{h})$, sufficiency of the statistic for these tasks implies equality of all future predictions and interfaces. Thus each fibre of r_t is contained in a predictive-equivalence class. Define the factor on the image of r_t by the predictive class of any history in that fibre. It is well defined and is the unique such map. This last assertion is a set-theoretic universal property. Its measurable version requires an actual measurable quotient realization, as specified below. $\square$

The theorem does not give a normalized continuation at a zero-probability report. One can extend an update arbitrarily there for a chosen measurable implementation, but such an extension is not canonical information. For worst-history statements, admissible histories and these extensions must be explicitly specified.

B.2. Safe measurable realizations. The complete posterior on a standard Borel hidden space is a safe measurable state under a fixed experiment and appropriate conditional versions. Its predictive quotient may be simpler, but not every abstract measurable quotient is standard Borel. We use the following concrete sufficient condition.

Theorem B.3 (Compact continuous realization). *Let B_t be compact metric admissible belief domains and let $(f_{t,j})_{j \geq 1}$ be continuous bounded evaluation functions on B_t that determine predictive equivalence. Put*

$$e_t(b) = (f_{t,j}(b))_{j \geq 1}, \quad S_t = e_t(B_t),$$

after rescaling the functions into $[-1, 1]$. Suppose action spaces are compact metric, reports finite, and $F_t : B_t \times A_t \times Y_{t+1} \rightarrow B_{t+1}$ is continuous for each report and respects predictive equivalence. Then S_t is compact metrizable and the induced update $\bar{F}_t : S_t \times A_t \times Y_{t+1} \rightarrow S_{t+1}$ is continuous for each report. Every continuous evaluation constant on predictive fibres descends to a continuous function on S_t .

Proof. The countable product $[-1, 1]^{\mathbb{N}}$ is compact metrizable, and e_t is continuous. Its image is compact, hence closed and metrizable. A continuous surjection from a compact space to a Hausdorff space is a closed quotient map. The same holds for $e_t \times \text{id}_{A_t}$ because its domain is compact and its target Hausdorff. For each report, $e_{t+1} \circ F_t$ is continuous and constant on fibres

of this quotient map. The definition of quotient topology therefore gives a unique continuous descended map. The final assertion is the same argument without the action factor. $\square$

If the hidden state space is compact and the belief domain is compact in weak convergence, continuous hidden-state tests give the required continuous evaluations. With continuous positive likelihoods bounded away from zero, Bayes updating is continuous. In concrete finite models this is verified directly. A countable dense set of actions can determine all finite words when the corresponding evaluations are continuous; this must be checked, not inferred from standard Borel structure alone.

For finite histories the quotient is finite and the measurable issue is trivial. For noncompact or nondominated settings we state results on an already supplied Borel predictive realization rather than claiming a universal smooth quotient.

B.3. Finite remaining-test coordinates. The following elementary result is useful when a finite representation exists, including representations with redundant coordinates.

Proposition B.4 (Positive finite-coordinate update). *Let $\psi_t = (1, \psi_{t,1}, \dots, \psi_{t,q_t})$ be bounded tests. Suppose*

$$\mathbb{T}_{t,a,y}1 = b_{t,a,y}\psi_t, \quad \mathbb{T}_{t,a,y}\psi_{t+1,i} = c_{t,a,y,i}\psi_t,$$

with uniformly bounded coefficient rows and jointly Borel dependence on the input. On a domain of beliefs where every relevant report probability is at least $\kappa > 0$, the coordinate vector $v = (\pi\psi_{t,i})_{i \geq 1}$ updates by

$$v'_i = \frac{c_{t,a,y,i}(1, v)^\top}{b_{t,a,y}(1, v)^\top}. \quad (\text{B.3})$$

It is uniformly Lipschitz on segments joining such moment vectors when the same report floor holds on their belief mixtures. Linear independence of the tests is not required.

Proof. Equation (B.3) follows from (A.4). A segment between two belief moment vectors represents their convex mixtures. The denominator is a report probability under that mixture and is at least κ . The numerator, coefficient rows, and coordinate vectors are bounded. Differentiating the quotient bounds its derivative by a constant depending only on these bounds and κ . Integrate that derivative along the segment. Redundant coefficient representations have the same evaluations on actual moment vectors and their mixtures, so they give the same update there. $\square$

In the static latent-variable case the instruments are multiplication by report likelihoods. In a dynamic model they are positive transfer operators. In either case the update uses current coordinates, command, and report, not a recovered history tape. An observer who stores the entire vector v at arbitrary real precision has not yet complied with a finite-label budget; that is the problem of Sections E and F.

B.4. Three different fibres. It is useful to record the types before adding geometry. Predictive fibres are fibres of $h_t \mapsto s_t(h_t)$. Statistical fibres are equivalence classes of parameters with equal laws under the specified strategy class. Feature fibres are fibres of a chosen map $\mathbf{O} : \Lambda \rightarrow Z$. Equality of features can be strictly weaker than equality of laws, and equality of a fixed-prior posterior prediction need not be parameter-uniform sufficiency. Their derivatives, when defined, have different domains.

APPENDIX C. EXECUTABLE REDUCTIONS AND CHARGED RESOURCES

C.1. Finite-state implementations. A synchronized causal machine between experiments has an internal state r_t . At a target step it receives the target action a'_t , chooses a source action according to a parameter-independent kernel depending on (r_t, a'_t) , receives the source report, and outputs a target report while updating r_t . Random seeds are independent of the parameter and

physical preparation. A stored seed that encodes previous reports is part of r_t . Every raw source step and every retained state is charged.

This synchronized model suffices for the results below. A macro-step that uses a random number of source steps requires an explicit causal stopping protocol and its preparation certificate. Such a protocol can be included, but a bound for one synchronized source step is not silently reused for it.

A target controller with state z_t receives the target reports and updates z_t . Its joint implementation with the machine stores (r_t, z_t) . Thus $|r_t| \leq R_t$ and $|z_t| \leq M_t$ give the direct product bound $R_t M_t$. This is an upper bound for one realization, not a claim that no smaller implementation is possible.

Definition C.1 (Causal reduction certificate). For experiments over a common parameter space, a certificate $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ consists of a parameter-independent causal machine, a lift L_Γ of every admitted target controller to its source implementation, and a monotone resource map ρ_Γ . Under target controller α , write $\hat{P}_\lambda^{\Gamma, \alpha}$ for the simulated target-history law obtained from the joint machine/source execution. An ε -certificate satisfies

$$\sup_{\lambda, \alpha} \left\| \hat{P}_\lambda^{\Gamma, \alpha} - P_{\mathfrak{F}, \lambda}^\alpha \right\|_{\text{TV}} \leq \varepsilon, \quad (\text{C.1})$$

and implementing a target budget B costs at most $\rho_\Gamma(B)$. The default certificate is pathwise. Expected-cost and tail-budget certificates are different types and must be stated separately.

For randomized strategy lifts, the right object is the joint causal execution. It need not be representable by a terminal kernel on the raw source history alone that is independent of the controller. Once the lift and controller are fixed, its visible pushforward is unambiguous. This avoids a misleading notation that would hide feedback inside a supposedly static garbling.

An exact certificate has $\varepsilon = 0$. The identity machine gives the identity reduction. A reparametrization $r : \Lambda' \rightarrow \Lambda$ pulls back an experiment family and is kept separate from data reduction. If the reparametrization is not injective it introduces parameter redundancy; it is not loss of observed data.

C.2. The path-kernel estimate. We use the convention $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. For $0 \leq f \leq L$ this implies $|Pf - Qf| \leq L \|P - Q\|_{\text{TV}}$.

Lemma C.2 (Predictable telescoping). *Let P and Q be laws on the same finite visible history tree, with the same initial law and the same feedback action kernels. Suppose their conditional report kernels at history h_t and action a_t are $p_t(\cdot | h_t, a_t)$ and $q_t(\cdot | h_t, a_t)$. If*

$$\|p_t(\cdot | h_t, a_t) - q_t(\cdot | h_t, a_t)\|_{\text{TV}} \leq d_t(h_t, a_t),$$

then

$$\|P - Q\|_{\text{TV}} \leq \min \left\{ 1, \mathbb{E}_P \sum_{t=0}^{N-1} d_t(H_t, A_t) \right\}. \quad (\text{C.2})$$

If both systems become absorbing under the same visible stopping rule σ , the summand can be multiplied by $\mathbf{1}_{\{t < \sigma\}}$.

Proof. Fix a terminal measurable test $0 \leq f \leq 1$. Replace the report kernels one at a time, using a P -prefix and a Q -suffix. The suffix conditional expectation of f is a measurable function between zero and one. The change at step t , after integrating the common feedback action kernel, is therefore bounded in absolute value by $\mathbb{E}_P d_t(H_t, A_t)$. Summing the telescoping differences proves the bound for f . Take the supremum over event indicators. After the visible rule has stopped, the kernels are the same absorbing kernel and their local error is zero. Histories of zero probability under P contribute nothing; the kernels are nevertheless specified on all admitted histories so that all intermediate laws exist. $\square$

The expectation is under the actual reference experiment. In particular a rare report carrying a large conditional error is not automatically charged as if it occurred at every time. No maximal-coupling measurability theorem is needed for this proof. The action policy may be discontinuous: the same kernel is applied on the same history in each telescoping term.

Theorem C.3 (Composition and risk transfer). *Suppose $\Gamma_1 : \mathfrak{E} \rightarrow \mathfrak{F}$ and $\Gamma_2 : \mathfrak{F} \rightarrow \mathfrak{G}$ have certificates (ε_1, ρ_1) and (ε_2, ρ_2) uniformly over the controller classes needed by the lifts. Their serial implementation has certificate*

$$(\min\{1, \varepsilon_1 + \varepsilon_2\}, \rho_1 \circ \rho_2).$$

Exact reductions compose associatively as executions, up to relabeling of product states. For a common target $\tau(\lambda)$ and bounded loss $0 \leq \ell \leq L$,

$$\mathfrak{R}_{\mathfrak{E}}(\rho_{\Gamma}(B)) \leq \mathfrak{R}_{\mathfrak{F}}(B) + L\varepsilon \quad (\text{C.3})$$

for either minimax risk or risk under a common prior.

Proof. Execute the second machine as a controller of the first. Their stored states form a product and their source actions remain nonanticipative and parameter independent. The first certificate applies to this enlarged controller class by hypothesis. The subsequent machine/controller execution is a Markov postprocessing of its visible history together with its independent randomization, so it contracts total variation. Compare first with an exact $\mathfrak{F}$ run and then with the intended $\mathfrak{G}$ run. The triangle inequality gives the displayed defect. Budget maps compose in the same order because they map target requirements to source requirements. Regrouping three nested machine states changes only their names, proving associativity.

For a target procedure within budget B , run its source implementation and use the same final decision. Its risk at each parameter differs by at most $L\varepsilon$. The resource certificate makes it admissible under $\rho_{\Gamma}(B)$. Take the supremum over parameters, or integrate against the common prior, and then take the infimum over target procedures. An approximating sequence of procedures suffices if an optimum is not attained. $\square$

A risk certificate concerns the same decision target and loss on both sides. It does not compare excess errors relative to different full-history predictors. An expected preparation constraint cannot be transported by a pathwise certificate without checking its type. Conversely, total-variation closeness alone does not transfer an unbounded cost expectation.

C.3. Checking an exact state reduction locally.

Proposition C.4 (One-step realization). *Let $r_t : X_t \rightarrow X'_t$, $g_t : Y_t \rightarrow Y'_t$, and an action lift $\iota_t : A'_t \rightarrow A_t$ be measurable. Suppose the preparations push forward correctly and, for every admitted source state and target action,*

$$(r_{t+1}, g_{t+1})_{\#} K_t^{\lambda}(\cdot \mid x, \iota_t a') = K_t^{\lambda}(\cdot \mid r_t x, a'). \quad (\text{C.4})$$

Suppose also that continuation interfaces, recorded costs, and designated readouts match. Then applying the lifted actions and report maps gives an exact causal reduction for every target feedback strategy.

Proof. At time zero the reduced state laws agree. Conditional on a source state and on a lifted action, (C.4) gives the joint reduced next-state/report law. Integrate against the common current reduced law. Since the target controller reads the reduced history, its next action law is the same in both systems. Induction proves equality on all finite cylinders, and hence on the path sigma field. Readout and cost matching give the asserted experiment and resource identity. $\square$

The state map need not be observable to implement the reduction: the machine only needs the action lift and report map. The hidden-state map is a proof device for the law identity. If the proposed lift instead requires the unobserved state, it is not an executable reduction of this type.

C.4. Terminal and causal equivalence. For a fixed terminal family, a garbling G is reversible if there exists a parameter-independent kernel R such that $P_\lambda GR = P_\lambda$. This is statistical equivalence. Requiring R to operate causally and charging its memory gives a stronger notion. Requiring suitable bounds in both directions gives resource-aware causal equivalence. These notions are not identified by definition.

Independent parallel experiments may be implemented by product machines when their preparations and physical randomness are genuinely independent conditional on the parameter. Persistent cardinalities multiply and costs add. Shared nuisance variables or coupled preparations require the joint experiment; replacing them by an independent product changes the object.

APPENDIX D. POSTERIOR GEOMETRY AND INFORMATION LOSS

D.1. Posterior barycentres. The following statements concern a common preparation. Let Z be a hidden target, Y an original observation, and W a parameter-independent postprocessing of Y . Conditional independence is denoted by the Markov chain $Z \rightarrow Y \rightarrow W$.

Theorem D.1 (Posterior barycentre and convex order). *For bounded measurable $f_1, \dots, f_q$ of Z , put*

$$p_Y = (\mathbb{E}[f_i(Z) | Y])_i, \quad p_W = (\mathbb{E}[f_i(Z) | W])_i.$$

Then

$$p_W = \mathbb{E}[p_Y | W]. \quad (\text{D.1})$$

Consequently $\mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$ for every convex function for which the expectations are defined. Moreover,

$$\text{Cov}(p_Y) - \text{Cov}(p_W) = \mathbb{E} \text{Cov}(p_Y | W) \succeq 0. \quad (\text{D.2})$$

Proof. The Markov property gives $\mathbb{E}[f_i(Z) | Y, W] = \mathbb{E}[f_i(Z) | Y]$. Taking conditional expectation with respect to W proves (D.1). Conditional Jensen yields the convex-order inequality. Subtract the common mean and expand the orthogonal decomposition $p_Y - \mathbb{E}p_Y = (p_Y - p_W) + (p_W - \mathbb{E}p_W)$; its cross covariance is zero by conditioning on W . This gives (D.2). $\square$

For a measure-valued posterior, the same identity holds against every bounded test. It is a conditional barycentre, not a function sending each original posterior to a single coarser posterior. In particular the image of a finite set of original posteriors may generate a continuous family of conditional barycentres. This is why convex-order monotonicity is not a statement about the cardinality of a predictive representation.

D.2. Relative entropy and recovery. For $P \ll Q$, set $D(P||Q) = \int \log(dP/dQ) dP$, allowing $+\infty$; set it to infinity otherwise. Let $G : X \rightsquigarrow W$ be a Markov kernel and form the joint laws $\bar{P}(dx, dw) = P(dx)G(dw | x)$ and $\bar{Q}(dx, dw) = Q(dx)G(dw | x)$.

Theorem D.2 (Entropy loss and a common recovery kernel). *If $D(P||Q) < \infty$, then*

$$D(P||Q) = D(PG||QG) + \int D(\bar{P}(dx | w)||\bar{Q}(dx | w)) PG(dw). \quad (\text{D.3})$$

In particular, entropy decreases under G . Suppose a family P_λ is dominated by one P_0 , every $D(P_\lambda||P_0)$ is finite, and

$$D(P_\lambda G||P_0 G) = D(P_\lambda||P_0) \quad \text{for every } \lambda.$$

Then $R(w, dx) = \bar{P}_0(dx | w)$ is a parameter-independent recovery kernel satisfying $P_\lambda GR = P_\lambda$ for every λ .

Proof. The joint likelihood ratio is $d\bar{P}/d\bar{Q}(x, w) = dP/dQ(x)$. Disintegrating both joint laws gives the factorization into the marginal likelihood ratio on W and the conditional likelihood ratio on X . Take logarithms and integrate. Standard truncation of the densities justifies the formula when logarithms are not bounded; finite total entropy makes the nonnegative conditional entropy integrable. This proves (D.3). Alternatively the marginal ratio is the conditional expectation under $\bar{Q}$ of dP/dQ , and conditional Jensen proves the inequality directly, also for infinite entropy.

For the family, equality makes the conditional entropy vanish $P_\lambda G$ -almost everywhere. Hence the two conditional distributions coincide there. The conditional law under the one reference P_0 is the same measurable kernel for all parameters; $P_\lambda G \ll P_0 G$ handles its null-set convention. Integrating that conditional law against $P_\lambda G$ recovers P_λ . $\square$

This is a sufficient recovery criterion under a common reference. An isolated equality for one pair does not certify an unrelated parameter family. Nor does it make the recovery kernel causal or memory free.

D.3. Scores and Fisher information. To avoid hiding regularity in notation, first take a finite observation space with $p_\lambda(y) > 0$ and continuously differentiable probabilities on an open parameter domain in $\mathbb{R}^k$. Let

$$s_\lambda(y) = \nabla_\lambda \log p_\lambda(y), \quad \mathcal{I}_\lambda = \mathbb{E}_\lambda[s_\lambda s_\lambda^\top].$$

If $G(w | y)$ is a parameter-independent finite channel, restrict to output symbols having positive probability; identically impossible symbols can be removed.

Theorem D.3 (Score projection). *The processed score and information satisfy*

$$s_\lambda^G(W) = \mathbb{E}_\lambda[s_\lambda(Y) | W], \tag{D.4}$$

$$\mathcal{I}_\lambda - \mathcal{I}_\lambda^G = \mathbb{E}_\lambda \text{Cov}_\lambda(s_\lambda(Y) | W) \succeq 0. \tag{D.5}$$

For a direction v , equality of information in that direction is equivalent to $v^\top s_\lambda(Y)$ being determined by W almost surely.

Proof. Differentiate $q_\lambda(w) = \sum_y p_\lambda(y)G(w | y)$ and divide by $q_\lambda(w)$. The coefficients are precisely the conditional probabilities of Y given W , proving (D.4). Differentiating $\sum_y p_\lambda(y) = 1$ gives $\mathbb{E}_\lambda s_\lambda = 0$. The conditional variance identity now gives (D.5). Its quadratic form in v is the mean conditional variance of $v^\top s_\lambda$, which vanishes exactly under the stated measurability condition. $\square$

The same proof works in a dominated differentiable model if differentiation under the integral is justified and the score lies in L^2 . More generally, the QMD version is the classical score-projection theorem described by Pollard [11]; when it is invoked, quadratic-mean differentiability is an additional statistical hypothesis. The raw experiment axioms by themselves impose no differentiability.

Information equality in (D.5) is a statement at a parameter point. It is not the recovery conclusion of Theorem D.2. Even information preservation throughout a model need not make a statistic sufficient without stronger assumptions; see [11].

Under a smooth parameter map $\lambda = r(\zeta)$, the information pulls back as $Dr^\top \mathcal{I} Dr$. Under a deterministic linear map of a random vector, its covariance pushes forward as $A \Sigma A^\top$. These are different operations on different objects. Neither formula says that an arbitrary covariance equals inverse Fisher information.

D.4. Sequential information in finite positive models. A useful controlled specialization follows without differentiating the controller. Suppose the conditional report probabilities are finite, positive, and differentiable, and the feedback policy has no explicit unknown-parameter dependence. Along a history define the local score

$$\xi_{t+1} = \nabla_\lambda \log p_\lambda(Y_{t+1} \mid H_t, A_t).$$

The policy factors contribute no score. Conditional normalization gives $\mathbb{E}_\lambda[\xi_{t+1} \mid H_t, A_t] = 0$.

Proposition D.4 (Sequential score decomposition). *If the local scores are square integrable, the path score is $S_N = \sum_{t < N} \xi_{t+1}$, and*

$$\mathbb{E}_\lambda[S_N S_N^\top] = \sum_{t < N} \mathbb{E}_\lambda[\xi_{t+1} \xi_{t+1}^\top]. \quad (\text{D.6})$$

For bounded stopping, each term is restricted to active steps. No independence of successive reports is required.

Proof. The likelihood factors into conditional report probabilities and parameter-independent policy factors. Its logarithmic derivative is the displayed sum. For $s < t$, ξ_{s+1} is measurable before ξ_{t+1} , so their cross moment is zero by the conditional mean-zero property. Expanding the square proves (D.6). A bounded stopping indicator is predictable for the next increment; the same argument applies. $\square$

This is an information identity for the actual controlled law. It does not make an information-optimal strategy automatically prediction-memory optimal, and it does not supply an asymptotic normal approximation.

APPENDIX E. ATTAINED PREDICTION AND CHECKPOINT QUANTIZATION

E.1. The physical loss. Fix a preparation and an exploration law, independently of coding randomness. At time t choose a future query J independently after the label is formed, with probabilities $w_j > 0$, $\sum_j w_j = 1$. Conditional on history and query, a bounded response Z_j has mean $m_j(h)$. Squared loss is used, and only the selected query is executed. Set

$$p(h) = (\sqrt{w_j} m_j(h))_{j=1}^q, \quad \nu = \text{Law}(p(H_t)).$$

The weighted decoder vector is denoted by c . Subtracting the complete history predictor's risk leaves $\|p(h) - c\|^2$.

An M -label checkpoint encoder reads the complete prefix once and retains one of M symbols. Its decoder receives the label and the subsequent query, not the discarded history or the acquisition commands. A public seed may specify a codebook but is independent of the preparation. Private coins are allowed and are either discarded or included in the retained label. We assume a bounded closed convex prediction domain containing all $p(h)$; projection into it can only improve squared error.

Theorem E.1 (Exact checkpoint identity). *The optimal mean checkpoint excess risk equals*

$$R_M^{\text{chk}} = e_M^2(\nu) := \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp). \quad (\text{E.1})$$

Independent public or private coding randomization does not decrease this infimum. An arbitrarily near-optimal deterministic nearest-centre encoder is measurable.

Proof. Condition on history, query, and the code randomness. The elementary identity

$$\mathbb{E}[(Z_j - a)^2 \mid h, J = j] = \text{Var}(Z_j \mid h, J = j) + (m_j(h) - a)^2$$

reduces the loss to the displayed Euclidean distortion. For each label, replace a randomized decoder by its conditional mean over decoder coins; convexity of squared norm does not increase

the loss. Given fixed decoder centres, random assignment by the encoder is no better than the nearest centre. Hence every fixed public-seed codebook has risk at least the right-hand side of (E.1); averaging over the independent seed preserves this inequality.

Conversely choose centres within any prescribed positive tolerance of the infimum. Encode by the first index attaining the least distance. Finite comparisons of continuous distance functions make this rule Borel. The resulting distortion is the integrand in (E.1). Let the tolerance decrease to zero. A codebook indexed by an unknown true parameter is not produced by this argument: ν is the one declared preparation law. $\square$

For worst-history loss the analogous deterministic checkpoint quantity is a covering radius, not the integral in (E.1). The mean and worst-history laws have the same order only under additional geometric and mass conditions.

E.2. A small-ball lower bound.

Lemma E.2 (Mass obstruction). *Suppose a probability law ν in $\mathbb{R}^d$ satisfies $\nu(B(c, r)) \leq Ar^d$ for every centre c and radius $r > 0$. Then for every integer $M \geq 1$,*

$$e_M^2(\nu) \geq \frac{1}{2}(2AM)^{-2/d}. \quad (\text{E.2})$$

In particular, a Lebesgue density bounded by $B\epsilon^{-d}$ gives $e_M^2(\nu) \geq c_{d,B}\epsilon^2 M^{-2/d}$ for $\epsilon > 0$.

Proof. The union of M balls of radius $r = (2AM)^{-1/d}$ has probability at most one half. On its complement every centre is at least distance r away. Integrating the nearest-centre squared distance gives at least $r^2/2$. For the density assertion use $A = Bv_d\epsilon^{-d}$, where v_d is the volume of the Euclidean unit ball. $\square$

The lemma permits centres outside the support and therefore already covers arbitrary decoders. It does not condition away a small probability of reaching the difficult region. If the useful density lower geometry instead occurs only on an event of probability β , the eventual risk lower bound must carry that mass.

Lemma E.3 (A positive attained chart). *Suppose a coordinate projection of a prediction law dominates β times uniform probability on a d -dimensional cube of side a , with $\beta, a > 0$. If that projection has operator norm at most L , then*

$$e_M^2(\nu) \geq c_d \beta a^2 L^{-2} M^{-2/d}.$$

Proof. For any centres in the original prediction space, project the centres onto the cube's coordinate space. Distances there are at most L times the original distances. The uniform cube law has density a^{-d} and satisfies Lemma E.2. Domination by β and rescaling prove the assertion. $\square$

E.3. Whole-image covers and all integer budgets. Let $D \subset \mathbb{R}^d$ be nonempty, compact, and contained in a cube of side $2a$. Divide each coordinate interval into $k = \lfloor M^{1/d} \rfloor$ pieces. There are at most $k^d \leq M$ cells. Choose one point of D in each nonempty cell. Any point is within Euclidean distance $2a\sqrt{d}/k$ of its representative. Since $k \geq M^{1/d}/2$, we obtain a radius $4a\sqrt{d}M^{-1/d}$ for every integer $M \geq 1$. The corresponding ℓ^1 bound is $4adM^{-1/d}$. Neither a large-budget asymptotic nor an integer-budget relaxation is needed.

More generally, finitely many Lipschitz images of compact cubes have such covers with exponent equal to the largest domain dimension. Allocate centres between the finitely many charts, use the preceding grid, and enlarge the constant for budgets smaller than the number of charts. This is a sufficient global description, not a consequence of local rank alone. A nonlinear or disconnected image cannot be replaced by a tangent plane without checking its entire cover.

E.4. Causal lower bounds use one exploration law. Let $\mathcal{C}_M$ be causal encoders with at most M_t labels at time t . Under the fixed exploration strategy, their label is also a measurable, possibly randomized, encoding of the complete time- t prefix. Therefore

$$\inf_{C \in \mathcal{C}_M} \max_{1 \leq t \leq N} \mathbb{E} \ell_t^C \geq \max_{1 \leq t \leq N} e_{M_t}^2(\nu_t). \quad (\text{E.3})$$

Indeed the checkpoint lower inequality holds for every single C and all t before taking the infimum. The same law of exploration supplies all marginals ν_t . Changing the exploration strategy with the encoder, or with the checkpoint after the infimum, would define another problem.

E.5. Calibrated anisotropy. A useful model interface is provided by [15]. There, under whole-image covering, positive attained mass for each initial scale flag, and uniformly stable raw updates, the scale profile is

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left(\frac{s_{t,1} \cdots s_{t,\ell}}{M} \right)^{2/\ell}.$$

The physical prediction metric is fixed; small singular values are not removed by an unbounded coordinate whitening. The finite-horizon causal bound follows by propagating the covering radii, as in the next section. We cite this refined profile as a conditional model interface, not as an unproved universal consequence of a list of singular values. The dynamic model of Section G will establish its own isotropic scale, density bound, and invariant cover directly from the experiment.

APPENDIX F. CAUSAL QUANTIZATION WITH FEEDBACK AND STOPPING

F.1. Admissible state domains. Consider a predictive realization $s_t(h_t)$ with a known initial state s_0 . Let (D_t, d_t) be admissible metric state domains containing every attainable state. They may also contain normalized beliefs that are not attained at that precise time. This causes no problem if all updates and readouts are genuinely defined on D_t . What is forbidden is evaluating an update at an artificial point where it has no experimental meaning or no stated extension.

Assume common action availability on each domain, or work within a fixed continuation-interface component. For each input (a, y) , let $F_t(\cdot, a, y) : D_t \rightarrow D_{t+1}$ be Borel and satisfy

$$d_{t+1}(F_t(s, a, y), F_t(\tilde{s}, a, y)) \leq L_t d_t(s, \tilde{s}). \quad (\text{F.1})$$

The known-calibration formulas implementing F_t are independent of the unknown parameter. If there is a parameterized law, write the next-report kernel as $g_t^\lambda(dy \mid s, a)$ and assume

$$\left\| g_t^\lambda(\cdot \mid s, a) - g_t^\lambda(\cdot \mid \tilde{s}, a) \right\|_{\text{TV}} \leq \kappa_t d_t(s, \tilde{s}) \quad (\text{F.2})$$

uniformly over the relevant parameter domain. The same Borel state update must represent every law for a parameter-uniform statement. For a single prepared Bayesian experiment there is no additional parameter supremum.

Choose finite representative sets $C_t \subset D_t$ with $|C_t| \leq M_t$ and Borel quantizers $q_t : D_t \rightarrow C_t$ satisfying $d_t(s, q_t s) \leq r_t$. A numerical update $\tilde{F}_t(c, a, y) \in D_{t+1}$ may replace the exact update at a representative, with

$$d_{t+1}(\tilde{F}_t(c, a, y), F_t(c, a, y)) \leq \eta_{t+1}. \quad (\text{F.3})$$

Set $\hat{s}_0 = s_0$ and implement

$$\hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (\text{F.4})$$

The persistent state is only the index of $\hat{s}_t$ in C_t .

Theorem F.1 (Finite-state emulation). *Under (F.1)–(F.3), define*

$$b_0 = 0, \quad b_{t+1} = L_t b_t + r_{t+1} + \eta_{t+1}, \quad b_t = \sum_{j=1}^t (r_j + \eta_j) \prod_{i=j}^{t-1} L_i. \quad (\text{F.5})$$

Then the following assertions hold.

(i) On every admitted common command/report sequence, $d_t(s_t, \hat{s}_t) \leq b_t$. If the physical prediction readout is K_t -Lipschitz, its excess squared error is at most $K_t^2 b_t^2$.

(ii) Define the finite-state reference experiment to use the same representative update (F.4), but to draw its next report from $g_t^\lambda(\cdot \mid \hat{s}_t, a_t)$. For any common causal feedback controller, the visible path laws of the actual experiment with the representative machine and of this finite-state reference experiment satisfy

$$\left\| P_{N,\lambda}^\alpha - \hat{P}_{N,\lambda}^\alpha \right\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=0}^{N-1} \kappa_t b_t \right\}. \quad (\text{F.6})$$

The histories here include the applied actions, reports, and retained labels. A shared visible stopping rule replaces the deterministic sum by $\mathbb{E}_{P_\lambda^\alpha} \sum_{t < \sigma \wedge N} \kappa_t b_t$.

(iii) Any additional causal simulator is charged as in Theorem C.3. A feedback controller with D_t own states is implemented with at most $D_t M_t$ persistent pairs. For a common bounded path-dependent loss, the risk-transfer error is at most its range times the right side of (F.6).

Proof. For (i), insert successively the exact update from the retained representative and its numerical approximation. The triangle inequality, (F.1), the numerical error, and the covering radius give

$$d_{t+1}(s_{t+1}, \hat{s}_{t+1}) \leq L_t d_t(s_t, \hat{s}_t) + \eta_{t+1} + r_{t+1}.$$

Induction proves (F.5). The physical readout bound follows by its Lipschitz property and the conditional squared-loss identity. This argument remains valid when the actions depend on the retained label: it compares the exact posterior and the representative along the actual same sequence of applied commands and reports.

For (ii), at any common visible prefix the applied controller has the same action kernel. The representative $\hat{s}_t$ is the same measurable function of that prefix in both experiments. In the actual law, the exact predictive state is s_t ; in the finite-state reference law, the next-report kernel is evaluated at $\hat{s}_t$. Their conditional variation distance is at most $\kappa_t b_t$ by (i) and (F.2). The next label is a common deterministic function of the action, report, and current label, so appending it does not increase the conditional variation distance. Apply Lemma C.2. Stopping makes the conditional kernels identical after the stopping time, which gives the predictable active-step bound.

For (iii), pair the retained representative with the controller's internal state. Each next action and update then reads only this pair and current inputs. The state cardinality is bounded by the product. The bounded-loss and extra-simulator statements are precisely the conclusions of Theorem C.3. $\square$

The theorem is a directional emulation statement. It does not construct a reverse kernel from the finite-state reference experiment to the raw physical data. It therefore does not claim two-sided Le Cam equivalence. Nor does it estimate an unknown predictive update. The implemented update and representatives are known; uniformity over a statistical family is available only under the explicitly common update conditions above.

F.2. What feedback does and does not require. There are two different uses of a finite filter. One can predict on an arbitrarily selected actual input sequence; this uses only Theorem F.1(i). One can also use a compressed controller to choose inputs. That still gives accurate filtering along its own actual run, but comparison with the original full-information optimal controller is another problem. Part (ii) compares a common controller across two experiments and so supplies a valid route through experiment simulation. It does not require the controller to be Lipschitz, because identical prefixes produce identical action laws. A discontinuous policy evaluated at two different approximate states is not being assumed to choose the same action.

For example, minimizing a terminal common decision loss over all D_t -state controllers in the finite-state reference experiment gives a source controller with $D_t M_t$ states and the corresponding (F.6) risk allowance. This need not approximate the unrestricted optimal source policy, whose comparison would require a separate richness theorem for the controller class.

F.3. Uniform prediction and finite-horizon path comparison.

Corollary F.2 (Contractive implementation). *If $L_t \leq \rho < 1$, $r_t \leq r$, and $\eta_t \leq \eta$ uniformly, then*

$$\sup_t d_t(s_t, \hat{s}_t) \leq \frac{r + \eta}{1 - \rho}.$$

For a bounded physical readout with uniform Lipschitz constant K , the prediction excess is at most $K^2(r + \eta)^2/(1 - \rho)^2$ at every time. If also $\kappa_t \leq \kappa$, the path-law error on N steps is at most $\min\{1, N\kappa(r + \eta)/(1 - \rho)\}$.

Proof. Sum the geometric series in (F.5), then apply the two parts of Theorem F.1. $\square$

A uniform per-checkpoint prediction bound is not uniform total-variation closeness of two infinite path laws. The last display may grow with N . Two processes that are arbitrarily close at each step can remain statistically distinguishable from a sufficiently long record. If a general construction uses a different arbitrary codebook at each time, it is a time-indexed machine description, not automatically a finite uniform program. The dynamic construction in the next section uses one fixed invariant domain and one fixed codebook, so that distinction does not obstruct its infinite-time statement.

F.4. Model or calibration perturbations. Suppose an implementable reference update F_t^0 has $d(F_t(s, a, y), F_t^0(s, a, y)) \leq \delta_{t+1}$ on the relevant domain, and its report law differs by at most ζ_t at the same state. Repeating the proof replaces $r_j + \eta_j$ by $r_j + \eta_j + \delta_j$ in (F.5) and $\kappa_t b_t$ by $\kappa_t b_t + \zeta_t$ in (F.6). This is a deterministic certificate, not an unpaid pilot. If a preliminary estimate yields such bounds only on an event of probability $1 - p$, bounded risks receive an additional allowance of at most their range times p , and the pilot's data and persistent state must be charged.

The perturbation must remain in the same physical metric and admissible domain. A reparametrization with an unbounded inverse near a degeneracy cannot be inserted while keeping the old constants.

APPENDIX G. A DYNAMIC MODEL WITH A UNIFORM MEMORY LAW

G.1. The experiment and its resource convention. Fix $d \geq 1$, let $E_d = \{0, 1, \dots, d\}$, and write $\nu_i = 1/(d + 1)$. The known parameters satisfy

$$4/5 \leq \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (\text{G.1})$$

Prepare $X_0 \sim \nu$. At each acquisition step the hidden state is updated by the transition matrix

$$P_\gamma(i, j) = (1 - \gamma)\mathbf{1}_{\{i=j\}} + \gamma\nu_j. \quad (\text{G.2})$$

A command $u_t \in [-1, 1]^d$, chosen before the report, gives a binary report $Y_t \in \{-1, 1\}$ with likelihoods

$$\ell_y(u, i) = \begin{cases} 1/2, & i = 0, \\ (1 + y\epsilon \tanh u_i)/2, & 1 \leq i \leq d. \end{cases} \quad (\text{G.3})$$

Conditional on the hidden path and commands, the report randomizations are independent. The resulting joint transition/report kernel is $P_\gamma(i, j)\ell_y(u, j)$. It is an instance of Definition A.1; all report probabilities lie between $3/8$ and $5/8$.

For the acquisition lower bound the commands are independent uniform on $[-1, 1]^d$, independent of the hidden process and coding randomness. The upper-bound filter will work on every admitted

command/report sequence, including adaptively chosen commands. It does not change the fixed exploration law used for the lower bound.

At any selected time $t \geq 1$ the acquisition can stop and one terminal probe is executed without an intervening refresh. After the retained label is formed, draw J uniformly from $\{1, \dots, d\}$. Probe j produces $Z \in \{0, 1\}$ with

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau \mathbf{1}_{\{i=j\}}, \quad \tau = 1/4. \quad (\text{G.4})$$

This is one noisy binary readout, not observation of X_t or of the full posterior. Its choice is independent of the acquisition. Each acquisition and the terminal probe costs one recorded experimental step. Evaluating maximum-checkpoint risk means that a run executes only its selected probe; it does not execute counterfactual probes at all times.

Let $p_t = \text{Law}(X_t \mid H_t)$, and write $p_{t,+} = (p_{t,1}, \dots, p_{t,d})$. The probe's full-history success probability is $1/2 + \tau p_{t,j}$. If a decoder uses candidate coordinates c_j , its excess loss is

$$\frac{\tau^2}{d} \|p_{t,+} - c\|_2^2. \quad (\text{G.5})$$

Arbitrary probe predictions are allowed; projecting them onto the interval $[1/2, 1/2 + \tau]$ cannot worsen loss. Thus the decoder coordinate formulation loses no candidates for the infimum.

Let $R_{M,t}^{\text{av}}$ and $R_{M,t}^{\text{max}}$ be optimal checkpoint risks under the stated exploration and over all admitted histories, respectively. Let $\mathfrak{R}_{M,N}^{\text{av}}$ and $\mathfrak{R}_{M,N}^{\text{max}}$ be the corresponding best maximum risks for one causal M -label encoder at times $1, \dots, N$. The clock, d, γ, ϵ , and the finite read-only program are known; no history or unknown calibration is stored in those constants. At time zero one fixed initial state is allowed. For $N = \infty$ replace maxima by suprema and use one machine for all times.

Theorem G.1 (Refreshing experiment memory law). *For every fixed $d \geq 1$, constants $0 < c_d \leq C_d < \infty$ exist such that, for every (G.1), every integer $M \geq 1$, every $N \in \mathbb{N} \cup \{\infty\}$, and every $1 \leq t \leq N$,*

$$c_d \epsilon^2 M^{-2/d} \leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}, \quad (\text{G.6})$$

$$c_d \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,N}^{\text{av}} \leq \mathfrak{R}_{M,N}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}. \quad (\text{G.7})$$

The upper bounds are realized by one deterministic finite-label filter using a fixed codebook for all positive times. The lower bounds allow independent public and private coding randomization. At $\epsilon = 0$ all four risks are zero with one label. Constants are uniform in time, horizon, calibration within (G.1), and the integer budget.

The proof will derive the cover, the actual attained density, and the stable update separately. This also supplies a concrete dynamic realization of the general emulation theorem.

G.2. Explicit posterior recursion. Set $p_0 = \nu$. Given the previous posterior p , the distribution just before the next report is

$$q = (1 - \gamma)p + \gamma\nu, \quad q_i \geq m_* := \frac{4}{5(d+1)}. \quad (\text{G.8})$$

For a command and report put

$$r_0 = 1, \quad r_i = 1 + \gamma \epsilon \tanh u_i \quad (i \geq 1), \quad D_y = q_0 + \sum_{i=1}^d q_i r_i.$$

Then the probability of that report is $D_y/2$, and

$$F(p, u, y)_0 = \frac{q_0}{D_y}, \quad F(p, u, y)_i = \frac{q_i r_i}{D_y} \quad (i \geq 1). \quad (\text{G.9})$$

Every denominator belongs to $[1 - \epsilon, 1 + \epsilon]$. These formulas hold for every belief in the full simplex and every admitted input. Predictable actions reveal no further hidden-state information once their past is conditioned upon, as in Proposition A.3.

Lemma G.2 (Uniform contraction). *For any two simplex beliefs and the same (u, y) ,*

$$\|F(p, u, y) - F(\tilde{p}, u, y)\|_1 \leq \rho \|p - \tilde{p}\|_1, \quad \rho = 2 \frac{1 + \epsilon}{1 - \epsilon} (1 - \gamma) \leq \frac{2}{3}. \quad (\text{G.10})$$

Proof. For fixed positive likelihood vector r , write $B_r(q) = \text{diag}(r)q/(q^T r)$. Subtract the two normalized vectors by adding and subtracting $\text{diag}(r)\tilde{q}/(q^T r)$. The first term has norm at most $(r_{\max}/r_{\min}) \|q - \tilde{q}\|_1$. The second has norm $|q^T r - \tilde{q}^T r|/(q^T r)$, at most the same bound. Thus $\text{Lip}_1(B_r) \leq 2r_{\max}/r_{\min}$. Equation (G.8) multiplies differences by $1 - \gamma$. Use $r_{\max} \leq 1 + \epsilon$, $r_{\min} \geq 1 - \epsilon$, and (G.1) to obtain the result. $\square$

Lemma G.3 (An invariant shrinking domain). *Let*

$$a_\epsilon = \frac{2\epsilon}{\gamma(1 - \epsilon)}, \quad D_\epsilon = \{p \in \Delta_d : \|p - \nu\|_1 \leq a_\epsilon\}. \quad (\text{G.11})$$

Then $\nu \in D_\epsilon$, $F(D_\epsilon, u, y) \subset D_\epsilon$, and $a_\epsilon \leq (10/3)\epsilon$. Every attainable posterior belongs to D_ϵ , for every time and every admitted history.

Proof. For any pre-report belief q ,

$$\|B_r(q) - q\|_1 = \sum_i q_i \frac{|r_i - q^T r|}{q^T r} \leq \frac{2\epsilon}{1 - \epsilon}.$$

Consequently

$$\|F(p, u, y) - \nu\|_1 \leq (1 - \gamma) \|p - \nu\|_1 + \frac{2\epsilon}{1 - \epsilon}.$$

The right side is at most a_ϵ when $p \in D_\epsilon$, by its definition. The numerical bound follows from $\gamma \geq 4/5$ and $1 - \epsilon \geq 3/4$. Start the induction at ν . $\square$

The set D_ϵ is a compact convex polytope for each fixed calibration. Its representatives need not all be attainable at a particular time; they are normalized beliefs with valid updates and belong to a proved invariant domain. No extrapolation outside the probability simplex is used. At $\epsilon = 0$ the domain is the singleton $\{\nu\}$.

G.3. The actual acquisition density. Fix a preceding history, so its pre-report belief q is fixed with (G.8). For $\epsilon > 0$ and a fixed sign y , the command map

$$\Phi_{q,y} : u \mapsto F(p, u, y)_+ \in \mathbb{R}^d$$

is one-to-one. Indeed its odds satisfy

$$\frac{p'_i}{p'_0} = \frac{q_i}{q_0} (1 + y\epsilon \tanh u_i), \quad (\text{G.12})$$

so u_i is recovered by applying artanh to $\{p'_i q_0 / (p'_0 q_i) - 1\} / (y\epsilon)$. The image of the command interior has a smooth inverse. Boundary commands have zero probability under the uniform acquisition law.

Lemma G.4 (Posterior Jacobian and unconditional density). *For $\epsilon > 0$,*

$$|\det D\Phi_{q,y}(u)| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \text{sech}^2 u_i. \quad (\text{G.13})$$

Under independent uniform commands and actual reports, the law of $p_{t,+}$ has a Lebesgue density bounded by $B_d \epsilon^{-d}$ at every $t \geq 1$, where one possible uniform constant is

$$B_d = 2^{-d} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1). \quad (\text{G.14})$$

The bound is conditional on every preceding history as well as unconditional.

Proof. Write $r'_i = y \epsilon \operatorname{sech}^2 u_i$. In the d nonzero-state coordinates,

$$D\Phi_{q,y} = \operatorname{diag} \left(\frac{q_i r'_i}{D_y} \right) - \left(\frac{q_i r_i}{D_y} \right)_i \left(\frac{q_j r'_j}{D_y} \right)_j^\top.$$

The determinant lemma gives the determinant of the diagonal matrix times

$$1 - \sum_{i=1}^d \frac{q_i r_i}{D_y} = \frac{q_0}{D_y}.$$

Taking absolute values gives (G.13).

Before any conditioning on the sign, the joint density of the uniform command and that sign is $2^{-d} D_y / 2$. Change variables through the injective command map. Its contribution to the posterior density is

$$\frac{2^{-(d+1)} D_y^{d+2}}{q_0 \prod_{i=1}^d q_i \epsilon^d \prod_{i=1}^d \operatorname{sech}^2 u_i} \leq 2^{-(d+1)} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1) \epsilon^{-d}.$$

Sum the two signs, allowing their images to overlap, to obtain (G.14). Finally integrate the conditional bounds against the actual preceding-history law. The same bound survives because it is independent of that history. Thus no event probability or complementary command volume is discarded. $\square$

This is a full-density upper bound for the actual law, which is sufficient for a quantization lower bound. It is stronger than merely exhibiting a nonzero differential at one command, but it does not assert that every point of the invariant polytope is attained. Those are different statements.

G.4. A fixed codebook and the proof of the theorem. Project D_ϵ to its d coordinates p_+ . It lies in the cube $\nu_+ + [-a_\epsilon, a_\epsilon]^d$. Partition this cube into k^d cells, $k = \lfloor M^{1/d} \rfloor$, and select one belief in D_ϵ from each occupied cell. Fix those representatives once and for all, along with a first-minimum tie convention. Empty cells need no label. The discarded coordinate satisfies $p_0 = 1 - \sum_{i=1}^d p_i$. Hence within a cell the full-simplex ℓ^1 distance is at most twice the ℓ^1 distance of its d coordinates. The resulting codebook has at most M points and covering radius

$$r_M \leq \frac{4da_\epsilon}{k} \leq \frac{80d}{3} \epsilon M^{-1/d}. \quad (\text{G.15})$$

Choose the nearest representative in full ℓ^1 distance. This preserves the covering bound and is Borel. The initial state is exactly ν ; thereafter the same codebook and the same update apply at every step.

Proof of Theorem G.1. For $\epsilon > 0$, use (G.9) at the retained belief and then quantize with (G.15). Lemmas G.2 and G.3 justify the whole procedure on every history. By Corollary F.2,

$$\sup_{t \geq 1} \|p_t - \hat{p}_t\|_1 \leq 3r_M \leq 80d\epsilon M^{-1/d}. \quad (\text{G.16})$$

Equation (G.5) bounds the excess by $(\tau^2/d)(80d)^2 \epsilon^2 M^{-2/d}$. This proves a pathwise upper bound for one machine, hence all the required mean and worst-history upper bounds. A checkpoint encoder can use the same final nearest-centre procedure.

For the lower bound, Lemma G.4 and Lemma E.2 give

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_j \|p_{t,+} - c_j\|_2^2 \geq \frac{1}{2} (2B_d v_d)^{-2/d} \epsilon^2 M^{-2/d}.$$

Multiply by τ^2/d and use Theorem E.1. The bound holds for every $t \geq 1$ under the one fixed exploration law, with coding randomness allowed. Every causal encoder is a checkpoint encoder at each time, proving the causal lower bound for finite or infinite horizons. The inequality between mean and worst-history risks is immediate. When $\epsilon = 0$, both reports have probability $1/2$ independent of the state and command. Refreshing preserves the uniform preparation, so $p_t = \nu$ on every history. A single label predicts the terminal probe exactly relative to the full-history oracle. This proves the zero case and completes the uniform statement. $\square$

Corollary G.5 (Persistent bits and double attenuation). *For b persistent bits, take $M = 2^b$ in (G.7). With the fixed terminal probe the squared excess has order $\epsilon^2 2^{-2b/d}$. If the actual terminal probe is instead*

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau \epsilon \mathbf{1}_{\{i=j\}},$$

the order is $\epsilon^4 2^{-2b/d}$, with the same uniformity and the same filter. At $\epsilon = 0$, zero bits suffice in both models.

Proof. The first assertion substitutes the number of labels. In the second experiment the physical readout difference is multiplied by ϵ , while the acquisition process, posterior law, and filter are unchanged. Its squared excess is therefore multiplied by ϵ^2 . For the lower bound the same scaling applies to arbitrary decoder centres after clipping to the physical mean interval. The zero case was established above. $\square$

G.5. Feedback-uniform emulation in this model. For fixed command u , the probability of the positive report under belief p is

$$g_+(p, u) = \frac{1}{2} \left(1 + \epsilon \sum_{i=1}^d ((1 - \gamma)p_i + \gamma \nu_i) \tanh u_i \right).$$

For binary laws their total variation is the difference of their positive probabilities. Thus

$$\|g(\cdot \mid p, u) - g(\cdot \mid \tilde{p}, u)\|_{\text{TV}} \leq \frac{1}{2} \epsilon (1 - \gamma) \|p - \tilde{p}\|_1. \quad (\text{G.17})$$

By Theorem F.1, the actual report-path law and the finite-state reference report-path law obey

$$\|P_N^\alpha - \hat{P}_N^\alpha\|_{\text{TV}} \leq \min\{1, 40dN\epsilon^2(1 - \gamma)M^{-1/d}\} \quad (\text{G.18})$$

for every common feedback controller. This statement is uniform over controllers but is finite-horizon; it does not identify infinite path laws. If a terminal probe is included in the compared observed path, its additional conditional error is at most $\tau \|p_t - \hat{p}_t\|_1$, averaged over its random index. This term is to be added to (G.18). Prediction regret for that probe instead uses its squared error, as in (G.5); the two claims are different.

The bound vanishes for the acquisition report law when $\gamma = 1$: each new state is independently uniform, so previous belief errors have no effect on the next acquisition report. The current posterior can still have nontrivial geometry and require finite prediction memory for the immediate terminal probe. This limiting case is a useful check on the meaning of each estimate.

G.6. Calibration and finite precision. The theorem is uniform over (G.1), but the codebook and update may use the known values of γ and ϵ . It is not a calibration-blind theorem. Numerically approximating each belief update by an admissible belief within η in ℓ^1 changes (G.16) to $3(r_M + \eta)$. The physical excess is at most $9\tau^2(r_M + \eta)^2/d$. Thus choosing η proportional to $\epsilon M^{-1/d}$ preserves the order. This is an accuracy requirement; it does not assert zero workspace or a universal bit-complexity bound for computing the update.

For rational calibration one can specify the invariant polytope by rational linear inequalities. Representatives can be selected in its occupied grid cells by a finite feasibility procedure; centres and a membership/tie rule form a finite read-only program. With nonrational calibration the theorem uses the declared calibrated real-number model; a numerical implementation must supply its own certified approximation. The only persistent acquired information after a report is its representative label.

Finally, the refresh lower bound and the fixed positive state floor are substantive. If γ tends to zero, the constant $2/3$ need not remain valid and the argument does not give a horizon-uniform theorem. That case is not hidden in the calibration interval.

APPENDIX H. NORMALIZED RESPONSE AND OBSERVATION MODULES

H.1. The covariance pairing. Let π be a probability measure and f, g bounded measurable functions. For u near zero define

$$\pi_u(dx) = \frac{e^{ug(x)}\pi(dx)}{\pi(e^{ug})}.$$

Proposition H.1 (Normalized response). *The derivative of a normalized prediction is*

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \pi(fg) - \pi(f)\pi(g) = \text{Cov}_\pi(f, g). \quad (\text{H.1})$$

More generally, if $L_u > 0$ is differentiable with the dominated derivatives needed for the quotient rule, then the derivative at u_0 of $\pi(fL_u)/\pi(L_u)$ is the covariance of f with $\partial_u \log L_u|_{u_0}$ under the normalized law at u_0 .

Proof. Differentiate numerator and denominator. At zero their values are $\pi(f)$ and one, and their derivatives are $\pi(fg)$ and $\pi(g)$. This proves (H.1). Apply the same quotient rule at u_0 and divide the derivative of L_u by L_{u_0} to obtain the general statement. Boundedness, or the stated domination in the general case, justifies integration of the derivatives. $\square$

For vector tests and several commands this gives a cross-covariance matrix. One index may represent an acquisition direction and the other a future test. That matrix is not automatically the Fisher information for an unknown physical parameter. In (G.9), command changes are selected by the observer; in Theorem D.3, parameter changes vary the family of physical laws. The similar algebra does not equate the two statistical problems.

H.2. Finite positive functionals and exact product fibres. Suppose a real algebra of functions contains finite-dimensional spaces $U_1, \dots, U_L$, and let V contain all their products. A parameter is a linear functional $\ell \in \Omega \subset V^*$, where Ω is open in V^* . Positivity of measured blocks may define this open set. The observed bilinear forms are

$$B_\nu(f, g) = \ell(fg), \quad f, g \in U_\nu.$$

There are no cross-contact products unless the experiment actually measures them. Set

$$W = \sum_{\nu=1}^L \text{span}\{fg : f, g \in U_\nu\} \subset V.$$

Theorem H.2 (Product-observation fibre). *The observation fibre through ℓ is*

$$(\ell + W^\perp) \cap \Omega. \quad (\text{H.2})$$

Its local dimension is $\dim V - \dim W$. Recovery of every functional in Ω is equivalent to $W = V$. If the parameter domain is relatively open in an affine subspace with direction $L \subset V^$, the local invisible space is instead $L \cap W^\perp$.*

Proof. Two observations agree precisely when the difference functional vanishes on every generating product, equivalently on W . This gives (H.2). Openness allows all sufficiently small perturbations in the annihilator, so the local dimension is its dimension. The affine-domain version restricts those perturbations to L . $\square$

This is linear duality. The difficult model-specific work can lie in proving that these forms are actually observed, determining W , describing the locus where its dimension drops, and controlling real positive realizability. For the calibrated Hankel family of [16], V is a polynomial space and the observation is a normal compression of an inverse information matrix. The theorem supplies its abstract fibre mechanism, not the paper's complete degeneracy classification.

H.3. The contact calculation. For clarity, the local geometric interface can also be derived directly. Let $M \subset \mathbb{R}^k$ have an embedded smooth germ at x_0 , and let $R > 0$. Assume the local minimizing projection onto that germ computes the global distance in a neighborhood of x_0 . Define

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b).$$

Let Z have orthonormal columns spanning the Euclidean normal space at x_0 , and put $B = Z^\top R Z$.

Proposition H.3 (Normal second jet). *Under these local projection assumptions,*

$$C_R(x_0 + h) = h^\top Z B^{-1} Z^\top h + o(\|h\|^2), \quad D^2 C_R(x_0) = 2 Z B^{-1} Z^\top.$$

Consequently the full second jet determines exactly the positive normal block B and has the tangent space as its kernel.

Proof. Replace the germ by its tangent space T to compute the quadratic term. The first-order minimizing condition for $\min_{v \in T} (h - v)^\top R^{-1} (h - v)$ says $R^{-1} (h - v) = Z n$ for some n . Hence $h - v = R Z n$, and normal projection gives $Z^\top h = B n$. The minimum is $(Z^\top h)^\top B^{-1} (Z^\top h)$. A smooth local parametrization of the germ has derivative spanning T ; its tangent Gram matrix in the positive R^{-1} metric is invertible. The implicit function theorem gives a differentiable minimizing parameter with the stated linear term. The quadratic expansion and the projection assumption identify it with C_R near x_0 . Positive block inversion is one-to-one, proving the observation assertion. $\square$

If $R = J H(\theta) J^\top$ with Hankel entries $H(\theta)_{ab} = \theta_{a+b}$, then $v^\top R w$ evaluates the moment functional on the product of the two corresponding coefficient polynomials. Theorem H.2 applies after this explicit identification. A higher-order distance jet has additional curvature and projection terms; replacing Sym^2 by Sym^m is not a derivation of a higher-order experiment.

H.4. Why general dynamic tests need not form a finite algebra.

Proposition H.4 (Finite continuous algebras have finite quotients). *Let X be compact Hausdorff and let $W \subset C(X, \mathbb{R})$ be a finite-dimensional unital subalgebra. Then W is the algebra of all functions constant on a finite clopen partition of X . The number of nonempty cells is $\dim W$.*

Proof. For $f \in W$, sufficiently many powers of f are linearly dependent. A nonzero polynomial therefore vanishes on $f(X)$, so that image is finite. A finite basis of W has only finitely many joint values; its fibres form a finite clopen partition. Every member of W is constant on these fibres. Conversely, for a joint value a , choose for every distinct joint value b a basis function $f_{i(a,b)}$ separating them. The product

$$\prod_{b \neq a} \frac{f_{i(a,b)} - f_{i(a,b)}(b)}{f_{i(a,b)}(a) - f_{i(a,b)}(b)}$$

is in W and is the indicator of the fibre at a . These indicators are linearly independent and span all functions on the partition, which proves the result. $\square$

Thus a finite multiplication-closed continuous algebra is a useful model, but it cannot by itself represent arbitrary continuous predictive information. Growing test spaces and positive operator words are the more general primitive. Neither the algebra dimension nor one tangent rank replaces the actual predictive law or its quantization geometry.

APPENDIX I. EXPONENTIAL TRANSFORMS AND CONDITIONAL LIMIT INTERFACES

I.1. Random channels have nonlinear exponential pullbacks. For a Markov kernel $G : X \rightsquigarrow Y$, bounded measurable $f : Y \rightarrow \mathbb{R}$, and a fixed $\beta > 0$, define

$$\mathcal{T}_{G,\beta}f(x) = \beta^{-1} \log \int e^{\beta f(y)} G(dy | x). \quad (\text{I.1})$$

For a probability P on X put $\mathcal{P}_{P,\beta}(h) = \beta^{-1} \log P(e^{\beta h})$. These expressions concern finite experiments and bounded sources; no long-time limit is presumed.

Theorem I.1 (Exponential composition and entropy duality). *For $H : Y \rightsquigarrow Z$ and bounded f ,*

$$\mathcal{T}_{GH,\beta}f = \mathcal{T}_{G,\beta}(\mathcal{T}_{H,\beta}f), \quad (\text{I.2})$$

$$\mathcal{P}_{PG,\beta}(f) = \mathcal{P}_{P,\beta}(\mathcal{T}_{G,\beta}f), \quad (\text{I.3})$$

$$\mathcal{P}_{P,\beta}(h) = \sup_{Q \ll P} \{Qh - \beta^{-1} D(Q \| P)\}. \quad (\text{I.4})$$

The supremum is attained at $dP^h/dP = e^{\beta h}/P(e^{\beta h})$. The source derivatives are

$$D\mathcal{P}_{P,\beta}(h)[u] = P^h u, \quad D^2\mathcal{P}_{P,\beta}(h)[u, v] = \beta \text{Cov}_{P^h}(u, v). \quad (\text{I.5})$$

Proof. Exponentiate (I.2). Its right side becomes $G(H(e^{\beta f}))$, which is the definition of the composed kernel's exponential integral. Integrating the same identity against P proves (I.3). For any $Q \ll P$ of finite entropy,

$$D(Q \| P^h) = D(Q \| P) - \beta Qh + \log P(e^{\beta h}) \geq 0.$$

Rearrangement gives the upper inequality in (I.4), and $Q = P^h$ attains equality. Measures of infinite entropy do not improve the supremum for bounded h . Finally boundedness permits differentiation of the exponential integral in the directions u, v ; one quotient rule gives the mean and the second gives the covariance with the factor β . $\square$

If G is the deterministic map g , the exponential pullback is exactly $f \circ g$. If G is random, it is generally not Gf . Jensen gives $\mathcal{T}_{G,\beta}f \geq Gf$, with equality precisely when f is constant $G(\cdot | x)$ -almost surely at each relevant x . The entropy variational principle is classical [13]; the proof fixes its conventions for this experiment category rather than asserting a new variational principle.

Proposition I.2 (Second-order chain rule for a random channel). *Put $h = \mathcal{T}_{G,\beta}f$ and*

$$G_f(dy | x) = \frac{e^{\beta f(y)} G(dy | x)}{G(e^{\beta f})(x)}.$$

Then the tilted joint law is $P^h(dx)G_f(dy | x)$ and

$$D\mathcal{T}_{G,\beta}(f)[u] = G_f u,$$

$$D^2\mathcal{T}_{G,\beta}(f)[u, v] = \beta \text{Cov}_{G_f}(u, v),$$

$$D^2\mathcal{P}_{PG,\beta}(f)[u, v] = \beta \text{Cov}_{P^h}(G_f u, G_f v) + \beta \mathbb{E}_{P^h} \text{Cov}_{G_f}(u, v).$$

The last expression is $\beta \text{Cov}_{P^h G_f}(u, v)$.

Proof. Multiplying the two tilted densities cancels $G(e^{\beta f})(x)$ and produces the original joint law multiplied by $e^{\beta f(y)}$ and its single total normalizer. Differentiation of the conditional normalizer gives the first two formulas just as in (I.5). Apply the ordinary second-order chain rule to (I.3). Its first term is the covariance of conditional means and its second is the mean of conditional covariances. The total covariance formula gives the final identity. $\square$

The conditional covariance term is essential. Discarding it would treat a random observation as a deterministic projection. This connects the positive-kernel and source-response layers exactly, while keeping them distinct from Fisher information for a model parameter.

I.2. A finite risk-sensitive recursion. Consider a known prepared predictive model with finite action sets, state update $F_t(s, a, y)$, report kernel $g_t(dy \mid s, a)$, bounded observed stage cost $c_t(s, a, y)$, and bounded terminal cost v_N . The cost here is an optimization objective, not an undeclared memory resource. For $\beta > 0$ define

$$v_t(s) = \min_{a \in A_t} \frac{1}{\beta} \log \int \exp\{\beta[c_t(s, a, y) + v_{t+1}(F_t(s, a, y))]\} g_t(dy \mid s, a). \quad (\text{I.6})$$

Proposition I.3 (Finite-horizon exponential dynamic programming). *The value (I.6) is the least value of $\beta^{-1} \log \mathbb{E} \exp(\beta[\sum_{j=t}^{N-1} c_j + v_N])$ over all causal strategies from s . A deterministic Borel minimizing policy exists with a fixed tie rule.*

Proof. Write $w_t = e^{\beta v_t} > 0$. Conditional expectation over the next report and then the future gives the multiplicative recursion $w_t(s) = \min_a \int e^{\beta c_t(s, a, y)} w_{t+1}(F_t(s, a, y)) g_t(dy \mid s, a)$. At the last step this is the exact conditional objective. Induct backwards; future randomized strategies cannot improve the already minimized continuation, and mixing current actions produces a convex combination of their positive objective values, no less than their minimum. Finite Borel action comparisons give a Borel first minimizer. Taking logarithms recovers (I.6). $\square$

This recursion needs a state sufficient for the future experiment and its cost. It cannot be applied to an arbitrary lossy statistic without proving closure. Compact infinite action sets would require appropriate existence and selection hypotheses; they are not needed for this proposition.

I.3. Deterministic LDP contractions. We record a conditional interface for later work. Suppose a sequence P_n on a Polish space has a good large-deviation principle with speed $a_n \rightarrow \infty$ and rate I . If T is continuous into another Polish space, its pushforward has rate

$$I_T(y) = \inf_{T(x)=y} I(x). \quad (\text{I.7})$$

For an open or closed target set its inverse image has the same property, so the LDP lower and upper bounds give (I.7). Goodness follows because each finite-rate infimum is attained on compact sublevels and the sublevels of I_T are continuous images of those of I . This is the standard contraction principle [14]. It is not a consequence of the raw experiment axioms.

For a random channel one needs the joint law, a conditional rate, or a separate exponential approximation theorem. A small total-variation simulation defect does not supply these ingredients; an explicit example is given in Section J.

I.4. Regular quadratic and labelled-phase interfaces.

Proposition I.4 (Quadratic constrained reduction). *If $H > 0$ and A has full row rank, then*

$$\inf_{Ah=z} \frac{1}{2} h^\top H h = \frac{1}{2} z^\top (A H^{-1} A^\top)^{-1} z.$$

Proof. The Lagrange equations give $h = H^{-1} A^\top (A H^{-1} A^\top)^{-1} z$. The constraint matrix in parentheses is positive definite. Substitution gives the value, and strict convexity gives the unique minimizer. $\square$

Only if a separate probabilistic theorem identifies H^{-1} with the appropriate Gaussian covariance does this become a covariance contraction statement. Existence of an LDP alone is not that theorem.

For finitely many labelled components with common speed a_n , suppose component rates are I_j and their positive weights satisfy $-a_n^{-1} \log w_{n,j} \rightarrow c_j$, with $\min_j c_j = 0$. The joint labelled rate is $c_j + I_j$ and the unlabelled rate is $\min_j (c_j + I_j(x))$. To verify the bounds, bound a finite sum above by the number of summands times its largest summand and below by each summand; then apply the component LDPs. Compactness follows from the finite union of component compact sublevels. Dynamic prediction may still require the posterior over labels. A marginal log-sum is not automatically a Markov semigroup.

I.5. Closure and finite-dimensional memory. For a Markov kernel P and state map q , an exact reduced kernel $\bar{P}$ satisfies

$$P(f \circ q) = (\bar{P}f) \circ q.$$

Iterating this identity proves every finite-time identity, including the corresponding exponential transforms. If it fails, the projected state need not be Markov. For a finite-dimensional linear evolution split as

$$\dot{u} = A_{PP}u + A_{PQ}v, \quad \dot{v} = A_{QP}u + A_{QQ}v,$$

variation of constants in the second equation gives exactly

$$\begin{aligned} \dot{u}(t) &= A_{PP}u(t) + A_{PQ}e^{tA_{QQ}}v(0) \\ &\quad + \int_0^t A_{PQ}e^{(t-s)A_{QQ}}A_{QP}u(s) ds. \end{aligned} \tag{I.8}$$

The unresolved initial-state term is not optional. Unbounded generators, common form domains, and differentiable operator transports require separate analytic results; the finite-dimensional identity does not prove them. This is the precise boundary at which the history and memory program of [17] can be connected to the present foundations.

APPENDIX J. BOUNDARY EXAMPLES AND NON-IMPLICATIONS

The following examples specify the distinctions used in the theorems. They are part of the mathematical scope, not objections to studying stronger results under stronger hypotheses.

Example J.1 (Less statistical information, more exact prediction labels). Let $Z \sim \text{Bernoulli}(1/2)$ and initially observe $X = Z$. The future target is again Z . Two labels retain the posterior mean exactly. Garble X into $Y = X + \xi$ with independent $\xi \sim N(0, 1)$. Bayes' formula and the ratio of the two Gaussian densities give

$$p(Y) = \mathbb{P}(Z = 1 \mid Y) = \frac{1}{1 + e^{1/2 - Y}}.$$

This is a smooth strictly increasing bijection from $\mathbb{R}$ to $(0, 1)$. The density of Y is positive everywhere, so the posterior law is continuous and has a positive density on every compact subinterval of $(0, 1)$. No finite set of centres can equal this posterior almost surely. More quantitatively, restrict to a compact interval where the density has a positive lower bound and apply Lemma E.3 in one dimension. Every fixed M has strictly positive optimal excess squared error. Both oracles are different: absolute decision risk obeys statistical data processing, but excess coding risk relative to each oracle need not.

Example J.2 (The same mean map with two noise geometries). Let $Z \sim N(\lambda, 1)$. For known $\epsilon > 0$, observing $Y_\epsilon = \epsilon Z$ is equivalent to observing Z by division, and its Fisher information for λ is one. Observing $\tilde{Y}_\epsilon = \epsilon\lambda + \xi$ with independent $\xi \sim N(0, 1)$ has Fisher information ϵ^2 . Both

means equal $\epsilon\lambda$. Only the first covariance is scaled with the mean. Consequently a small singular value of a mean map does not determine its statistical information without the noise law.

Example J.3 (Rank does not record global fibres or contact order). The derivative of $x \mapsto x^2$ is nonzero at 1 and -1 , yet their images coincide. For another distinction, let $Y_i = \lambda^r + \xi_i$, $\lambda \geq 0$, with independent unit Gaussian noise and $r \geq 2$. Every mean map has zero derivative at zero, but its local amplitude scale is $n^{-1/(2r)}$. Indeed the divergence between $\lambda = 0$ and a positive alternative is $n\lambda^{2r}/2$. Taking this bounded gives the lower resolution scale by two-point testing. For an upper bound use $\hat{\lambda} = (\bar{Y}_+)^{1/r}$ and $|x_+^{1/r} - b^{1/r}| \leq |x - b|^{1/r}$ for $b \geq 0$; its mean squared error is $O(n^{-1/r})$. The order of contact, not just the rank, matters.

Example J.4 (Reachability is not attained mass). A protocol may permit all commands in an interval while the exploration law always chooses a single command. The set of legally attainable predictions can be continuous even if that particular exploration law visits only one prediction. Another exploration on the same command interface can spread positive mass over a full interval. Their average quantization risks differ. Theorem G.1 avoids this ambiguity by specifying independent uniform commands and deriving their density.

Example J.5 (Equal first two moments do not give equal laws). The two laws $\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$ and $\frac{1}{2}\delta_0 + \frac{1}{4}\delta_{-\sqrt{2}} + \frac{1}{4}\delta_{\sqrt{2}}$ have mean zero and variance one, but are different. Equality of a finite jet or covariance needs a model-specific identifiability theorem to imply anything stronger.

Example J.6 (Fisher preservation need not be sufficiency). The Kagan–Shepp phenomenon, analyzed in [11], gives an insufficient statistic preserving Fisher information. We do not reproduce that external construction here or claim it as new. Its precise logical role is to prevent promotion of the equality case of Theorem D.3 into the parameter-uniform recovery assertion of Theorem D.2.

Example J.7 (Small variation error can change the exponential rate). For $a_n \rightarrow \infty$ and $c > 0$, let

$$P_n = \delta_0, \quad Q_n = (1 - e^{-a_n c})\delta_0 + e^{-a_n c}\delta_1.$$

Their total variation is $e^{-a_n c} \rightarrow 0$. At speed a_n , however, P_n gives rate $+\infty$ to 1, while Q_n gives it rate c . Thus even exponentially small variation error need not preserve a complete rate function. Appropriate superexponential approximation hypotheses are substantially stronger than an ordinary vanishing simulation defect.

Example J.8 (Linear-source pressure can convexify a nonconvex rate). Let P_n put probability $1/2$ at each of -1 and 1 , for all n . At speed n its rate vanishes at those two points and is infinite elsewhere. Its limiting pressure for linear source sx is $|s|$. The Legendre transform of $|s|$ vanishes on the entire interval $[-1, 1]$. The full bounded-continuous-source Laplace functional and the restricted linear-source transform are not interchangeable.

Example J.9 (Observation does not ensure Markov closure). Consider the deterministic chain $a \mapsto a$, $b \mapsto c$, $c \mapsto c$, with report $q(a) = q(b) = 0$, $q(c) = 1$. The current report 0 does not specify the next report: the two compatible hidden states give different answers. Thus no single reduced kernel works for every preparation. Future-test equivalence distinguishes a and b ; merely keeping their current report does not.

Example J.10 (Zero evidence has no canonical normalized limit). On a two-point space the positive measures $(\delta, 0)$ and $(0, \delta)$ both tend to the zero measure as $\delta \downarrow 0$. Their normalized probabilities tend to different point masses. A continuous normalized extension at zero cannot be assigned from the zero measure alone.

Example J.11 (A simulator may need delay memory). Let the unknown parameter be $\lambda \in \{0, 1\}$. A source with no control actions reveals λ at time one and constants thereafter. A target reveals

a constant at time one and λ at time two. Their terminal information is equivalent. However, a parameter-independent simulator with only one retained state after time one, and no other retained tape, has the same internal state and the same new inputs under both parameters. Its second output therefore has the same law under both parameters and cannot equal the required two different point masses. Two states suffice by retaining the bit. If a target controller also retains its own state, the direct implementation is a product. Terminal sufficiency does not erase this state cost.

Example J.12 (A finite command alphabet is a different experiment). At a fixed finite time, finite actions and finite reports produce only finitely many histories. With enough labels they can be retained exactly. The positive lower bound of Theorem G.1 for all finite M uses the stipulated continuously distributed commands. Rounding the physical command alphabet itself changes that experiment. Approximating only the filter's arithmetic while observing the original experiment is instead the perturbation problem in Section F.

These examples rule out four unqualified shortcuts: deriving global identifiability from local rank, deriving memory from information order, deriving statistical rates from a noiseless feature map, and deriving exponential asymptotics from ordinary variation approximation. The positive results in this paper specify the extra structure needed in each case.

APPENDIX K. INTERFACES WITH SINGULAR GEOMETRY AND DYNAMICS

The foundations established here do not require all experiments to be contractions of one physical path law or one rate function. They give a common language in which an actual map or causal simulator must be exhibited before a comparison is asserted.

K.1. What is established in this installment. The exact results are path construction, future-test minimality with stated realization conditions, charged simulation composition, posterior and entropy identities, the finite/dominated score identity, normalized response, product-observation fibres, checkpoint quantization, and exponential composition. The quantitative results are the feedback- and stopping-compatible finite-state emulation theorem and the uniform dynamic memory law for the refreshing model. Their proofs occur in this article. The LDP and operator discussions are conditional interfaces, with a standard contraction principle and a finite-dimensional memory identity; no general Sinai or kinetic limit theorem is asserted here.

The principal dynamic model verifies a compact shrinking domain, actual probability mass, and stable updates directly. It gives one time-uniform isotropic degeneration law. It does not classify all anisotropic collisions, all hidden Markov filters, all priors, or weak-refreshing limits. The bounded comparison constants are not exact optimizer transition constants.

K.2. The role of the two geometric manuscripts. The finite-horizon manuscript [15] provides finer anisotropic profiles for its positive monomial experiments and a separate bounded observation-algebra law. Those hypotheses should remain attached to their conclusions. The present dynamic result is not a replacement for their collision geometry, and their fixed-horizon constants are not assumed uniform as the horizon grows.

The contact manuscript [16] provides a different geometric realization of information loss. Theorem H.2 explains its linear-functional interface. A complete extension to singular statistical resolution still requires the actual noise metric, calibration, parameter nuisance, and admissible real domain. In particular independent measurement noise and correlated raw-data score noise can react differently to a small multiplication singular value. A complex determinantal component is not automatically a physically attainable real statistical submodel.

K.3. A precise next target. Let a parameter target be $\tau(\lambda)$ and let an actual feature map be $O(\lambda)$. Its inverse modulus is

$$\omega(\delta) = \sup\{d(\tau(\lambda), \tau(\lambda')) : d_Z(O(\lambda), O(\lambda')) \leq \delta\}.$$

If a finite-memory procedure estimates that feature with a proved common error bound δ , fitting the model on a compact parameter set gives an error at most $\omega(2\delta)$: the true parameter is an admissible fit, and the triangle inequality bounds the discrepancy of the two model features. A matched statistical lower bound requires actual distributional closeness. If $\|P_{n,\lambda_0} - P_{n,\lambda_1}\|_{TV} \leq a < 1$, any estimator has maximum squared target risk at least

$$\frac{1-a}{8}d(\tau(\lambda_0), \tau(\lambda_1))^2.$$

Indeed choose the nearer of the two target points as a test; on an incorrect choice the estimation error is at least half their separation. The two error probabilities sum to at least $1 - a$.

These two elementary bounds specify a route to a resource–statistical singularity theorem, but do not prove a universal matched modulus law. Covering, testing, attainable alternatives, and tail control must be supplied. The geometry-of-rates literature, including [12], is the appropriate comparison, not a claim that all minimax theory follows from one rank locus.

K.4. The physical realization obligations remain. For deterministic hyperbolic and particle systems, one must still establish the true experiment kernels, physical clock handling, tightness and recovery at exponential scale, and the domains of any response or memory operators. A pathwise finite-state prediction error does not provide a Fourier local limit theorem. A static covariance factorization does not construct a collision process. A projected state does not acquire a Markov semigroup merely because it is named a predictive coordinate.

The historical typed-contraction program [17] can therefore be reorganized as a family of verified realization theorems attached to these interfaces. Its model-specific proofs are not assumed completed by the present definitions. In particular memory response in a varying Hilbert or form domain requires its own transport and domain arguments.

K.5. Conclusion. A common foundation is obtained by keeping the entire experiment ahead of its geometric representations. Positive preparation and future tests give prediction; a real acquisition law gives a distortion problem; stable updates give an online implementation; a causal simulator gives a comparison, with its state and error charged. In the refreshing model these steps yield a uniform finite-memory law through loss of the acquisition channel. The separate entropy, score, and exponential identities explain which forms of information transport are valid. Stronger singular and long-time results must enter through corresponding theorems, not through additional meanings assigned to the word “contraction”.

APPENDIX L. MEASURABILITY, FINITE COVERS, AND A SCALAR BENCHMARK

L.1. Finite-label implementation. A finite codebook $C = \{c_1, \dots, c_M\}$ in a metric space has a Borel nearest-centre selector whenever each distance $s \mapsto d(s, c_j)$ is Borel. Define its label by the least index attaining the minimum. The set on which label j is selected is the intersection of the finitely many sets $\{d(s, c_j) < d(s, c_i)\}$ for $i < j$ and $\{d(s, c_j) \leq d(s, c_i)\}$ for $i > j$. These sets are Borel. Thus the finite-label recursions used in this paper do not require an unproved measurable choice of a nearest point in an infinite set.

Codebook construction and codebook execution are different questions. Existence of finitely many representatives in nonempty cells is an ordinary finite choice. When an algorithmic construction is claimed, one must also supply a representation and a feasibility procedure for those cells. In Section G, rational calibration gives a rational polytope; finite rational linear feasibility supplies

this additional construction. This does not charge an unbounded list of history-dependent centres to a read-only program.

L.2. Conditional distributions and common versions. The existence of regular conditional distributions on standard Borel spaces is a classical input [4]. Its relevant conclusion is existence of a kernel under a fixed joint law. It does not provide one canonical version on every null history, nor does it alone give a version continuous in a parameter or uniformly valid on an uncountable family of null sets. The statements in the main text accordingly take one of three forms: an explicit positive finite-report formula; a compact continuous realization as in Theorem B.3; or an already supplied jointly measurable predictive realization with its applicable quantifiers.

There is no hidden claim that all quotient images of a Borel space are standard Borel. The evaluation image into a countable product can be given its induced measurable structure, but to use ordinary kernel calculus on it one must verify that the image and descended maps have the required measurability. The compact theorem verifies precisely that fact in one useful class. Keeping the full posterior remains a valid alternative when compression of the state is not yet justified.

L.3. Quantization after a readout change.

Proposition L.1 (Bi-Lipschitz readout comparison). *Let S be a metric space with a probability ν , and let $p : S \rightarrow \mathbb{R}^q$ be an injective readout satisfying*

$$a d(s, t) \leq \|p(s) - p(t)\|_2 \leq b d(s, t), \quad a, b > 0.$$

If S has an M -point cover of radius r with centres in S , the checkpoint prediction risk is at most $b^2 r^2$. If some measurable map $T : \mathbb{R}^q \rightarrow \mathbb{R}^k$ has Lipschitz constant $L > 0$, and the law of $Tp(s)$ has density at most A , that risk is at least

$$\frac{1}{2L^2} (2Av_k M)^{-2/k}.$$

The second assertion does not require decoder centres to belong to $p(S)$.

Proof. For the upper bound map the cover centres through p and use the upper Lipschitz estimate. For any decoder centres $z_j \in \mathbb{R}^q$, the inequality $\|p(s) - z_j\|_2 \geq L^{-1} \|Tp(s) - Tz_j\|_2$ reduces the lower bound to Lemma E.2. Take the infimum over the original centres. The constant a is useful for comparing intrinsic covers in both directions, but it is not needed in this particular density-based lower bound. In particular, the argument does not require inversion of a near-singular statistical covariance. $\square$

A linear projection or bounded coordinate map can be used for T . A coordinate change whose Lipschitz constant tends to infinity changes the bound; it cannot be called a harmless normalization at a degeneracy.

Proposition L.2 (Exact scalar uniform quantization). *For U uniform on an interval of length $a > 0$, the optimal squared M -centre quantization error is $a^2/(12M^2)$, for every integer $M \geq 1$.*

Proof. Order the centres. Their nearest-centre cells intersect the interval in intervals, allowing empty cells. If a cell has length l_j , its contribution is minimized by its midpoint and equals $l_j^3/(12a)$. Since $\sum_j l_j = a$, convexity gives $\sum_j l_j^3 \geq a^3/M^2$. Hence the total error is at least $a^2/(12M^2)$. Equal-length cells with their midpoints attain that value. Arbitrary randomized encoders cannot improve it by Theorem E.1. $\square$

This exact formula is a benchmark for the meaning of label cardinality, not a claim that the dynamic posterior density in Section G is uniform. The latter is bounded by its actual Jacobian and can depend on the preceding history. Its two-sided order, rather than an exact optimizer constant, is what we prove.

L.4. Predictable error budgets and random horizons. The stopped version of Lemma C.2 is often more useful than the deterministic sum. In a comparison with common visible stopping rule σ , append an absorbing terminal symbol. The two conditional kernels coincide once the symbol has appeared. Their discrepancy at time t is therefore bounded by $\mathbf{1}_{\{t < \sigma\}} e_t(H_t)$, a measurable function of the common prefix. The telescoping proof integrates it against the actual prefix law. This gives the expectation of the active-step sum, without replacing a random horizon by an uncharged successful count.

If the active-step sum is bounded by a deterministic budget B on every path, its expectation is at most B . If only an expected budget is known, that is the exact kind of budget furnished by the theorem. A tail-budget statement requires an additional probability estimate. These are distinct resource certificates even though the same positive summands occur in all three.

APPENDIX M. EXPLICIT FINITE CHECKS AND THE LIMITS OF COMPUTATION

M.1. A two-state posterior calculation. The smallest dynamic instance has $d = 1$, $\gamma = 4/5$, and $\epsilon = 1/4$. Put $v = \tanh u$ and select $v = 1/2$, which corresponds to the legal command $u = \operatorname{artanh}(1/2) < 1$. Starting from the uniform preparation, the pre-report belief is $q = (1/2, 1/2)$, and

$$\mathbb{P}(Y = +1) = \frac{17}{32}, \quad \mathbb{P}(Y = -1) = \frac{15}{32}.$$

The posterior probabilities of state 1 are respectively

$$p_1^+ = \frac{9}{17}, \quad p_1^- = \frac{7}{15}.$$

Thus the same current command produces two different posteriors. Discarding the negative report or assigning its conditional law mass one would alter the acquisition law used in the proof. With the fixed terminal probe the corresponding success probabilities are

$$\frac{1}{2} + \frac{1}{4} \frac{9}{17} = \frac{43}{68}, \quad \frac{1}{2} + \frac{1}{4} \frac{7}{15} = \frac{37}{60}.$$

The mean posterior agrees with the pre-report probability:

$$\frac{17}{32} \frac{9}{17} + \frac{15}{32} \frac{7}{15} = \frac{1}{2}.$$

This is the posterior martingale property for the report step; the refresh step has its own affine prediction and is not being omitted.

The derivative with respect to v , rather than u , at this point for the positive report is

$$\frac{q_0 q_1 \epsilon}{D_+^2} = \frac{16}{289}.$$

Multiplying by $dv/du = 1 - v^2 = 3/4$ gives $12/289$, exactly (G.13) in one dimension. These distinct derivatives are recorded because replacing a physical command by a nonlinear coordinate without its Jacobian would change the preparation density.

M.2. A direct resource calculation. Suppose a report converter has two persistent states (for example, it retains the previous binary report), the target predictor has M labels, and a feedback controller has D labels. The implemented composite uses at most $2MD$ triples. On the next step it reads the old triple, the current input, and fresh independent randomness, and writes the next triple. The source history may be longer, but it is not consulted by this implementation. Any reduction below $2MD$ needs a proved state identification; it is not obtained by calling the converter a sufficient statistic.

Conversely, if a converter is memoryless and deterministic, its state set has one element. In that special case it need not enlarge the target label count. This is the correct context for a nonincrease

assertion. It does not imply the false general statement about exact predictive memory refuted in Example J.1.

M.3. A finite precision requirement rather than a free real tape. For $d = 1$ the invariant domain is an interval inside the probability simplex. A codebook can be constructed using interval intersections, and its labels are ordinary finite integers. One evaluates the two rational Bayes fractions, with the command likelihood evaluated to a specified accuracy, then selects the nearest admissible representative. The perturbation form of Theorem F.1 accounts for the incurred state error. Since denominators are at least $3/4$, no division by a vanishing report probability occurs.

For general fixed d , the same procedure uses a finite polytope feasibility construction. It may be computationally expensive. Our cardinality theorem does not state an optimal running time or workspace bound. The program may store its predetermined codebook, as permitted by the read-only model; it may not put acquired commands or reports in that program. The current command must be discarded after updating the label, and the terminal decoder may not reread it as free side information.

M.4. Role of the accompanying diagnostics. The source package includes exact-rational checks of Bayes normalization, the odds inverse, the rank-one determinant identity, the refresh contraction inequality at a finite family of inputs, and the scalar quantization identity. It also checks a finite conditional-covariance identity, a feedback path-law comparison, and specified deliberately incorrect formulas. These computations are reproducibility aids for signs, constants, and implemented examples.

The universal statements in this article are proved analytically above. A finite test does not establish measurability, a global differential bound, a minimax lower bound, or uniformity over all times. In particular, no number of successful finite diagnostics certifies that every possible hidden Markov model has the displayed memory law. Build receipts, source hashes, and diagnostic output belong to the accompanying delivery record rather than to the logical assumptions of any theorem.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] T. Fritz, A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics, *Adv. Math.* **370** (2020), 107239. arXiv:1908.07021.
- [3] T. Fritz, T. Gonda, P. Perrone, and E. F. Rischel, Representable Markov categories and comparison of statistical experiments in categorical probability, *Theoret. Comput. Sci.* **961** (2023), 113896. arXiv:2010.07416.
- [4] O. Kallenberg, *Foundations of Modern Probability*, third ed., Probability Theory and Stochastic Modelling, vol. 99, Springer, 2021. doi:10.1007/978-3-030-61871-1.
- [5] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *J. Statist. Phys.* **104** (2001), 817–879. arXiv:cond-mat/9907176.
- [6] M. L. Littman, R. S. Sutton, and S. Singh, Predictive representations of state, in *Advances in Neural Information Processing Systems 14*, MIT Press, 2002, 1555–1561 (2001 conference). Publisher proceedings entry.
- [7] N. Barnett and J. P. Crutchfield, Computational mechanics of input-output processes: structured transformations and the ϵ -transducer, *J. Statist. Phys.* **161** (2015), 404–451. arXiv:1412.2690.
- [8] G. Pagès and H. Pham, Optimal quantization methods for nonlinear filtering with discrete-time observations, *Bernoulli* **11** (2005), 893–932. doi:10.3150/bj/1130077599.
- [9] D. S. Kalogerias and A. P. Petropulu, Grid based nonlinear filtering revisited: recursive estimation and asymptotic optimality, preprint, 2016. arXiv:1604.02631.
- [10] N. Saldi, S. Yüksel, and T. Linder, On the asymptotic optimality of finite approximations to Markov decision processes with Borel spaces, *Math. Oper. Res.* (2017). doi:10.1287/moor.2016.0832.
- [11] D. Pollard, A note on insufficiency and the preservation of Fisher information, in *From Probability to Statistics and Back: High-Dimensional Models and Processes*, IMS Collections, vol. 9, 2013. doi:10.1214/12-IMSCOLL919; arXiv:1107.3797.

- [12] D. L. Donoho and R. C. Liu, Geometrizing rates of convergence, II, *Ann. Statist.* **19** (1991), 633–667. doi:10.1214/aos/1176348114.
- [13] P. Dupuis and H. Wang, Subsolutions of an Isaacs equation and efficient schemes for importance sampling, *Math. Oper. Res.* **32** (2007), 723–757. doi:10.1287/moor.1070.0266.
- [14] S. R. S. Varadhan, *Large Deviations*, lecture notes, Courant Institute, Spring 2010, Sections 1–2. Author’s lecture notes.
- [15] Q. Qi, Attainable information and causal compression at exponent collisions, A1 v37, manuscript and companion, September 2026. Immutable repository edition, commit 90465076589f.
- [16] Q. Qi, Recovery of information metrics: multiplication failure schemes, A2 v112, manuscript, September 2026. Immutable mathematical edition, commit 0c696736e6ec.
- [17] Q. Qi, Path-space cotangent rigidity, universal contractions, and tangent representations, historical C2 manuscript, Round 17. Immutable historical snapshot, commit c04845b66132.
- [18] P. J. Bushell, On the projective contraction ratio for positive linear mappings, *J. London Math. Soc.* (2) **6** (1973), 256–258. doi:10.1112/jlms/s2-6.2.256.
- [19] A. D. Kara and S. Yüksel, Near optimality of finite memory feedback policies in partially observed Markov decision processes, *J. Mach. Learn. Res.* **23** (2022), no. 11, 1–46. Publisher edition.
- [20] J. Subramanian, A. Sinha, R. Seraj, and A. Mahajan, Approximate information state for approximate planning and reinforcement learning in partially observed systems, *J. Mach. Learn. Res.* **23** (2022), no. 12, 1–83. Publisher edition.
- [21] A. D. Kara, E. Bayraktar, and S. Yüksel, Near optimal approximations and finite memory policies for POMPDs with continuous spaces, preprint, 2024. arXiv:2410.02895.
- [22] Y. E. Demirci, A. D. Kara, and S. Yüksel, Refined bounds on near optimality finite window policies in POMDPs and their reinforcement learning, preprint, 2024. arXiv:2409.04351.
- [23] M. Kim, Finite memory belief approximation for optimal control in partially observable Markov decision processes, preprint, 2026. arXiv:2601.03132.
- [24] S. Marzen and J. P. Crutchfield, Predictive rate-distortion for infinite-order Markov processes, *J. Statist. Phys.* **163** (2016), 1312–1338. Preprint arXiv:1412.2859, under the title “Circumventing the curse of dimensionality in prediction”.
- [25] S. P. Loomis and J. P. Crutchfield, Exploring predictive states via Cantor embeddings and Wasserstein distance, *Chaos* **32** (2022), 123115. doi:10.1063/5.0102603.
- [26] G. N. Nair and R. J. Evans, Stabilization with data-rate-limited feedback: tightest attainable bounds, *Systems Control Lett.* **41** (2000), 49–56. doi:10.1016/S0167-6911(00)00037-2.
- [27] S. Tatikonda and S. Mitter, Control under communication constraints, *IEEE Trans. Automat. Control* **49** (2004), 1056–1068. doi:10.1109/TAC.2004.831187.
- [28] A. S. Matveev and A. V. Savkin, *Estimation and Control over Communication Networks*, Control Engineering, Birkhäuser, Boston, 2009. doi:10.1007/978-0-8176-4607-3.
- [29] E. Lindenstrauss and M. Tsukamoto, From rate distortion theory to metric mean dimension: variational principle, *IEEE Trans. Inform. Theory* **64** (2018), 3590–3609. doi:10.1109/TIT.2018.2806219. arXiv:1702.05722.
- [30] J. V. Bondar and P. Milnes, Amenability: a survey for statistical applications of Hunt–Stein and related conditions on groups, *Z. Wahrscheinlichkeitstheorie verw. Gebiete* **57** (1981), 103–128. doi:10.1007/BF00533716.
- [31] P. Heinrich and J. Kahn, Minimax rates for finite mixture estimation, preprint (2015), arXiv:1504.03506v1. The comparison in the text uses the explicitly numbered theorems of this fixed preprint edition.